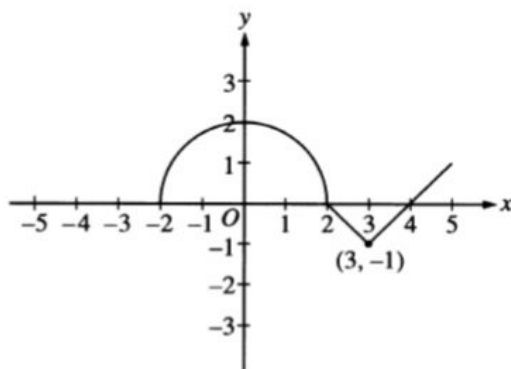


2-2-

1.



The graph of the function f consists of a semicircle and two line segments as shown above. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

- Find $g(3)$.
- Find all the values of x on the open interval $(-2, 5)$ at which g has a relative maximum. Justify your answer.
- Write an equation for the line tangent to the graph of g at $x = 3$.
- Find the x -coordinate of each point of inflection of the graph of g on the open interval $(-2, 5)$. Justify your answer.

Part A

2 points are earned for the correct answer

<-1> each incorrect area

<-1> error in summing

$$\begin{aligned}
 g(3) &= \int_0^3 f(t) dt \\
 &= \frac{1}{4} \cdot \pi \cdot 2^2 - \frac{1}{2} = \pi - \frac{1}{2}
 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for the correct answer

<-1> each incorrect area

2-2-

<-1> error in summing

$$g(3) = \int_0^3 f(t) dt$$

$$= \frac{1}{4} \cdot \pi \cdot 2^2 - \frac{1}{2} = \pi - \frac{1}{2}$$

Part B1 point is earned for correctly answering that relative maximum at $x = 2$ only1 point is earned for correctly answering $g'(x) = f(x)$ or interprets $g(x)$ as area accumulator1 point is earned for the correct justification (ignore discussion at $x = 5$) $g(x)$ has relative maximum at $x = 2$ because $g'(x) = f(x)$ changes from positive to negative at $x = 2$ 

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly answering that relative maximum at $x = 2$ only1 point is earned for correctly answering $g'(x) = f(x)$ or interprets $g(x)$ as area accumulator1 point is earned for the correct justification (ignore discussion at $x = 5$) $g(x)$ has relative maximum at $x = 2$ because $g'(x) = f(x)$ changes from positive to negative at $x = 2$ **Part C**1 point is earned for the correct answer $g'(3) = -1$

$$g(3) = \pi - \frac{1}{2}$$

$$g'(3) = f(3) = -1$$

1 point is earned for the correct equation using $g(3)$ and $g'(3)$

$$y - \left(\pi - \frac{1}{2} \right) = -1(x - 3)$$

2-2-



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $g'(3) = -1$

$$g(3) = \pi - \frac{1}{2}$$

$$g'(3) = f(3) = -1$$

1 point is earned for the correct equation using $g(3)$ and $g'(3)$

$$y - \left(\pi - \frac{1}{2} \right) = -1(x - 3)$$

Part D

1 point is earned for the correct points of inflection with x -coordinates 0 and 3 only

1 point is earned for the correct justification (ignore discussion at $x = 2$)

1/2 if $x = 0, 3$ selected as candidates and $x = 3$ discarded because $g''(3)$ does not exist

1/2 if $x = 0, 2, 3$ selected as candidates and $x = 2$ and $x = 3$ discarded because $g''(2)$ and $g''(3)$ do not exist

graph of g has points of inflection with x -coordinates $x = 0$ and $x = 3$

because $g''(x) = f(x)$ changes from positive to negative at $x = 0$ and from negative to positive at $x = 3$

or

because $g'(x) = f(x)$ changes from increasing to decreasing at $x = 0$ and from decreasing to increasing at $x = 3$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct points of inflection with x -coordinates 0 and 3 only

1 point is earned for the correct justification (ignore discussion at $x = 2$)

1/2 if $x = 0, 3$ selected as candidates and $x = 3$ discarded because $g''(3)$ does not exist

2-2-

1/2 if $x = 0, 2, 3$ selected as candidates and $x = 2$ and $x = 3$ discarded because $g''(2)$ and $g''(3)$ do not exist

graph of g has points of inflection with x -coordinates $x = 0$ and $x = 3$

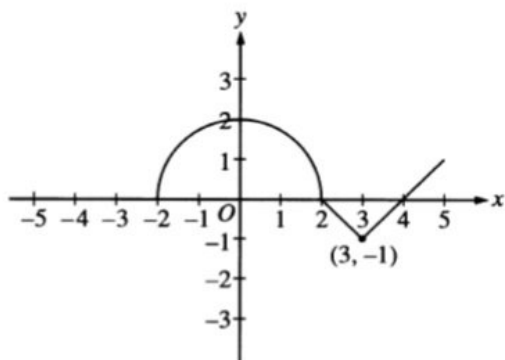
because $g''(x) = f(x)$ changes from positive to negative at $x = 0$ and from negative to positive at $x = 3$

or

because $g'(x) = f(x)$ changes from increasing to decreasing at $x = 0$ and from decreasing to increasing at $x = 3$

2-2-

2.



The graph of the function f consists of a semicircle and two line segments as shown above. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

Write an equation for the line tangent to the graph of g at $x = 3$.

Part C

1 point is earned for the correct answer $g'(3) = -1$

$$g(3) = \pi - \frac{1}{2}$$

$$g'(3) = f(3) = -1$$

1 point is earned for the correct equation using $g(3)$ and $g'(3)$

$$y - \left(\pi - \frac{1}{2}\right) = -1(x - 3)$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct answer $g'(3) = -1$

$$g(3) = \pi - \frac{1}{2}$$

$$g'(3) = f(3) = -1$$

1 point is earned for the correct equation using $g(3)$ and $g'(3)$

2-2-

$$y - \left(\pi - \frac{1}{2}\right) = -1(x - 3)$$

2-2-

3. Suppose that the function f has a continuous second derivative for all x , and that $f(0) = 2$, $f'(0) = -3$, and $f''(0) = 0$. Let g be a function whose derivative is given by $g'(x) = e^{-2x}(3f(x) + 2f'(x))$ for all x .
- (a) Write an equation of the line tangent to the graph of f at the point where $x = 0$.
- (b) Is there sufficient information to determine whether or not the graph of f has a point of inflection when $x = 0$? Explain your answer.
- (c) Given that $g(0) = 4$, write an equation of the line tangent to the graph of g at the point where $x = 0$.
- (d) Show that $g''(x) = e^{-2x}(-6f(x) - f'(x) + 2f''(x))$. Does g have a local maximum at $x = 0$? Justify your answer.

Part A

1 point is earned for the equation

Slope at $x = 0$ is $f'(0) = -3$

At $x = 0$, $y = 2$

$$y - 2 = -3(x - 0)$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the equation

Slope at $x = 0$ is $f'(0) = -3$

At $x = 0$, $y = 2$

$$y - 2 = -3(x - 0)$$

Part B

1 point is earned for the answer

1 point is earned for the explanation

No, Whether $f''(x)$ changes sign at $x = 0$ is unknown. The only given value of $f''(x)$ is $f''(0) = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

2-2-

1 point is earned for the answer

1 point is earned for the explanation

No, Whether $f''(x)$ changes sign at $x = 0$ is unknown. The only given value of $f''(x)$ is $f''(0) = 0$.**Part C**1 point is earned for: $g'(0)$

1 point is earned for the equation

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$\begin{aligned} g'(0) &= e^0 (3f(0) + 2f'(0)) \\ &= 3(2) + 2(-3) = 0 \end{aligned}$$

$$y - 4 = 0(x - 0)$$

$$y = 4$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $g'(0)$

1 point is earned for the equation

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$\begin{aligned} g'(0) &= e^0 (3f(0) + 2f'(0)) \\ &= 3(2) + 2(-3) = 0 \end{aligned}$$

$$y - 4 = 0(x - 0)$$

$$y = 4$$

Part D

2 points are earned for verify derivative 0/2 product or chain rule error

<-1> algebra errors

1 point is earned for: $g'(0) = 0$ and $g''(0)$

1 point is earned for the answer and reasoning

2-2-

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g''(x) = (-2e^{-2x})(3f(x) + 2f'(x)) + e^{-2x} (3f'(x) + 2f''(x))$$

$$= e^{-2x} (-6f(x) + f'(x) + 2f''(x))$$

$$g''(0) = e^0 [(-6)(2) - (-3) + 2(0)] = -9$$

Since $g'(0) = 0$ and $g''(0) < 0$, g does have a local maximum at $x = 0$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for verify derivative 0/2 product or chain rule error

<-1> algebra errors

1 point is earned for: $g'(0) = 0$ and $g''(0)$

1 point is earned for the answer and reasoning

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g''(x) = (-2e^{-2x})(3f(x) + 2f'(x)) + e^{-2x} (3f'(x) + 2f''(x))$$

$$= e^{-2x} (-6f(x) + f'(x) + 2f''(x))$$

$$g''(0) = e^0 [(-6)(2) - (-3) + 2(0)] = -9$$

Since $g'(0) = 0$ and $g''(0) < 0$, g does have a local maximum at $x = 0$.

2-2-

Suppose that the function f has a continuous second derivative for all x , and that $f(0) = 2$, $f'(0) = -3$, and $f''(0) = 0$. Let g be a function whose derivative is given by $g'(x) = e^{-2x} (3f(x) + 2f'(x))$ for all x .

4. Write an equation of the line tangent to the graph of f at the point where $x = 0$.

Part A

1 point is earned for the equation

Slope at $x = 0$ is $f'(0) = -3$

At $x = 0$, $y = 2$

$$y - 2 = -3(x - 0)$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the equation

Slope at $x = 0$ is $f'(0) = -3$

At $x = 0$, $y = 2$

$$y - 2 = -3(x - 0)$$

2-2-

5. Given that $g(0) = 4$, write an equation of the line tangent to the graph of g at the point where $x = 0$.
-

Part C

1 point is earned for: $g'(0)$

1 point is earned for the equation

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g'(0) = e^0 (3f(0) + 2f'(0))$$

$$= 3(2) + 2(-3) = 0$$

$$y - 4 = 0(x - 0)$$

$$y = 4$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $g'(0)$

1 point is earned for the equation

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g'(0) = e^0 (3f(0) + 2f'(0))$$

$$= 3(2) + 2(-3) = 0$$

$$y - 4 = 0(x - 0)$$

$$y = 4$$

2-2-

6. Let h be a function defined for all $x \neq 0$ such that $h(4) = -3$ and the derivative of h is given by $h'(x) = \frac{x^2 - 2}{x}$ for all $x \neq 0$.

- (a) Find all values of x for which the graph of h has a horizontal tangent, and determine whether h has a local maximum, a local minimum, or neither at each of these values. Justify your answers.
- (b) On what intervals, if any, is the graph of h concave up? Justify your answer.
- (c) Write an equation for the line tangent to the graph of h at $x = 4$.
- (d) Does the line tangent to the graph of h at $x = 4$ lie above or below the graph of h for $x > 4$? Why?

Part A

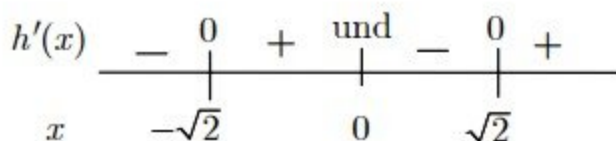
1 point is earned for: $x = \pm\sqrt{2}$

1 point is earned for the analysis

2 points are earned for the conclusions

<-1> not dealing with discontinuity at 0

$$h'(x) = 0 \text{ at } x = \pm\sqrt{2}$$



Local minima at $x = -\sqrt{2}$ and at $x = \sqrt{2}$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $x = \pm\sqrt{2}$

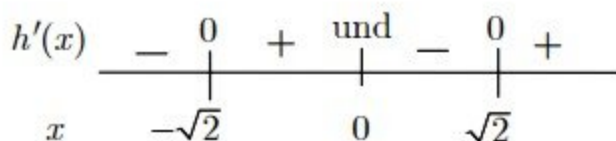
1 point is earned for the analysis

2 points are earned for the conclusions

2-2-

<-1> not dealing with discontinuity at 0

$$h'(x) = 0 \text{ at } x = \pm\sqrt{2}$$

Local minima at $x = -\sqrt{2}$ and at $x = \sqrt{2}$ **Part B**1 point is earned for: $h''(x)$ 1 point is earned for: $h''(x) > 0$

1 point is earned for the answer

$$h''(x) = 1 + \frac{2}{x^2} > 0 \text{ for all } x \neq 0. \text{ Therefore, the graph of } h \text{ is concave up for all } x \neq 0.$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $h''(x)$ 1 point is earned for: $h''(x) > 0$

1 point is earned for the answer

$$h''(x) = 1 + \frac{2}{x^2} > 0 \text{ for all } x \neq 0. \text{ Therefore, the graph of } h \text{ is concave up for all } x \neq 0.$$

Part C

1 point is earned for the tangent line equation

2-2-

$$h'(4) = \frac{16-2}{4} = \frac{7}{2}$$

$$y + 3 = \frac{7}{2}(x - 4)$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the tangent line equation

$$h'(4) = \frac{16-2}{4} = \frac{7}{2}$$

$$y + 3 = \frac{7}{2}(x - 4)$$

Part D

1 point is earned for the answer with reason

The tangent line is below the graph because the graph of h is concave up for $x > 4$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with reason

The tangent line is below the graph because the graph of h is concave up for $x > 4$.

2-2-

7. Let h be a function defined for all $x \neq 0$ such that $h(4) = -3$ and the derivative of h is given by $h'(x) = \frac{x^2 - 2}{x}$ for all $x \neq 0$.

Write an equation for the line tangent to the graph of h at $x = 4$.

Part C

1 point is earned for the tangent line equation

$$h'(4) = \frac{16-2}{4} = \frac{7}{2}$$

$$y + 3 = \frac{7}{2}(x - 4)$$



0

1

The student response earns all of the following points:

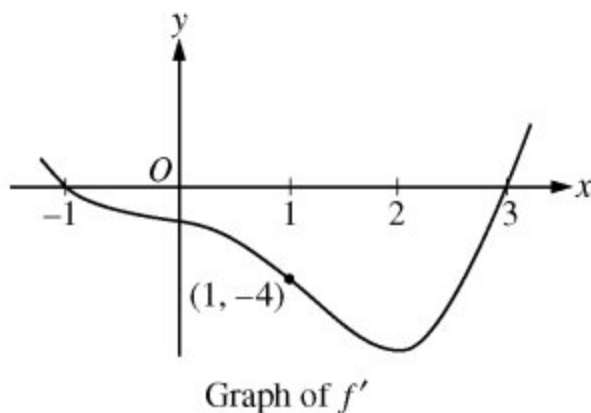
1 point is earned for the tangent line equation

$$h'(4) = \frac{16-2}{4} = \frac{7}{2}$$

$$y + 3 = \frac{7}{2}(x - 4)$$

2-2-

8.



Let f be a twice-differentiable function defined on the interval $-1.2 < x < 3.2$ with $f(1) = 2$. The graph of f' , the derivative of f , is shown above. The graph of f' crosses the x -axis at $x = -1$ and $x = 3$ and has a horizontal tangent at $x = 2$. Let g be the function given by $g(x) = e^{f(x)}$.

Write an equation for the line tangent to the graph of g at $x = 1$.

Part A

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by $y = e^2 - 4e^2(x - 1)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x)$

1 point is earned for $g(1)$ and $g'(1)$

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by $y = e^2 - 4e^2(x - 1)$.

2-2-

2-2-

9. Let f be the function given by $f(x) = 3x^4 + x^3 - 21x^2$.
- (a) Write an equation of the line tangent to the graph of f at the point $(2, -28)$.
- (b) Find the absolute minimum *value* of f . Show the analysis that leads to your conclusion.
- (c) Find the x -coordinate of each point of inflection on the graph of f . Show the analysis that leads to your conclusion.

Part A

1 point is earned for the correct $f'(x)$

$$f'(x) = 12x^3 + 3x^2 - 42x$$

1 point is earned for the correct $f'(2)$

$$f'(2) = 24$$

1 point is earned for the correct equation of tangent

$$y + 28 = 24(x - 2)$$

$$\text{or } y = 24x - 76$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct $f'(x)$

$$f'(x) = 12x^3 + 3x^2 - 42x$$

1 point is earned for the correct $f'(2)$

$$f'(2) = 24$$

1 point is earned for the correct equation of tangent

$$y + 28 = 24(x - 2)$$

$$\text{or } y = 24x - 76$$

Part B

2-2-

1 point is earned for the correct zeros of $f'(x)$ 0/1: if f' is quadratic

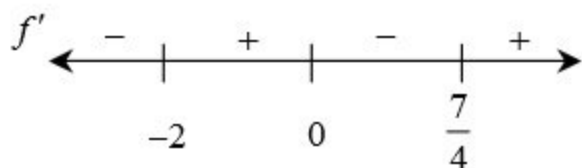
$$12x^3 + 3x^2 - 42x = 0$$

$$3x(4x^2 + x - 14) = 0$$

$$3x(4x - 7)(x + 2) = 0$$

$$x = 0, x = \frac{7}{4}, x = -2$$

1 point is earned for the correct analysis -- must indicate behavior of function over entire real line

Min must be at -2 or $7/4$

$$f(-2) = -44, f\left(\frac{7}{4}\right) = -30.816$$

1 point is earned for the correct answer

Absolute min is -44



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct zeros of $f'(x)$ 0/1: if f' is quadratic

$$12x^3 + 3x^2 - 42x = 0$$

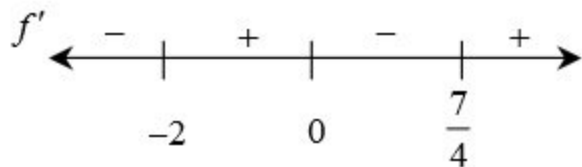
$$3x(4x^2 + x - 14) = 0$$

$$3x(4x - 7)(x + 2) = 0$$

$$x = 0, x = \frac{7}{4}, x = -2$$

1 point is earned for the correct analysis -- must indicate behavior of function over entire real line

2-2-



Min must be at -2 or $7/4$

$$f(-2) = -44, f\left(\frac{7}{4}\right) = -30.816$$

1 point is earned for the correct answer

Absolute min is -44

Part C

1 point is earned for the correct $f''(x)$

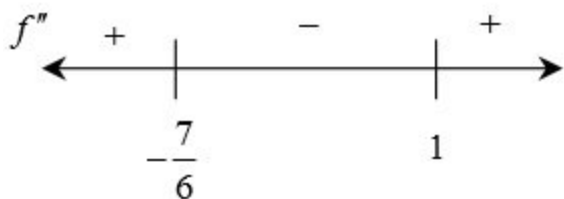
$$\begin{aligned} f''(x) &= 36x^2 + 6x - 42 \\ &= 6(6x^2 + x - 7) \\ &= 6(6x + 7)(x - 1) \end{aligned}$$

1 point is earned for the correct zeros of $f''(x)$

0/1: if f'' is linear

$$\text{zeros at } x = -\frac{7}{6}, x = 1$$

1 point is earned for the correct answer including analysis of sign of f'' or of concavity



The x -coordinates of the points of inflection are $x = -7/6$ and $x = 1$



0	1	2	3
---	---	---	---

2-2-

The student response earns all of the following points:

1 point is earned for the correct $f''(x)$

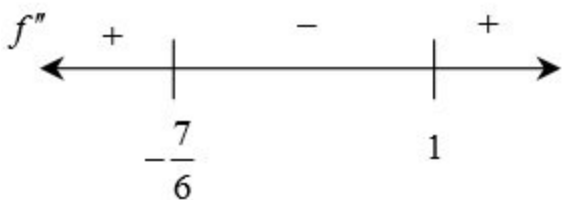
$$\begin{aligned}f''(x) &= 36x^2 + 6x - 42 \\&= 6(6x^2 + x - 7) \\&= 6(6x + 7)(x - 1)\end{aligned}$$

1 point is earned for the correct zeros of $f''(x)$

0/1: if f'' is linear

zeros at $x = -\frac{7}{6}, x = 1$

1 point is earned for the correct answer including analysis of sign of f'' or of concavity



The x -coordinates of the points of inflection are $x = -7/6$ and $x = 1$

2-2-

10. Let f be the function given by $f(x) = 3x^4 + x^3 - 21x^2$.

Write an equation of the line tangent to the graph of f at the point $(2, -28)$.

Part A

1 point is earned for the correct $f'(x)$

$$f'(x) = 12x^3 + 3x^2 - 42x$$

1 point is earned for the correct $f'(2)$

$$f'(2) = 24$$

1 point is earned for the correct equation of tangent

$$y + 28 = 24(x - 2) \text{ or } y = 24x - 76$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the correct $f'(x)$

$$f'(x) = 12x^3 + 3x^2 - 42x$$

1 point is earned for the correct $f'(2)$

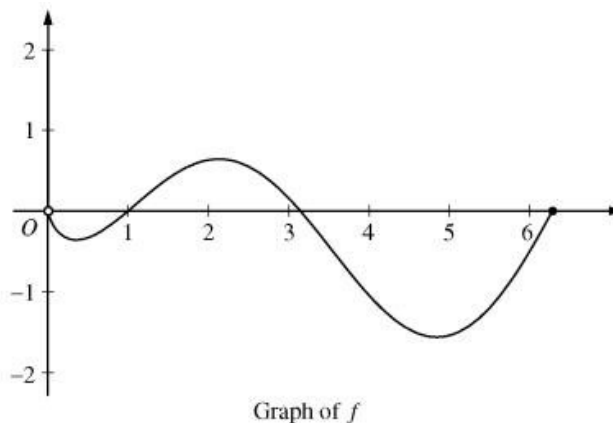
$$f'(2) = 24$$

1 point is earned for the correct equation of tangent

$$y + 28 = 24(x - 2) \text{ or } y = 24x - 76$$

2-2-

11.



Let f be the function given by $f(x) = (\ln x)(\sin x)$. The figure above shows the graph of f for $0 < x \leq 2\pi$. The function g is defined by $g(x) = \int_1^x f(t) dt$ for $0 < x \leq 2\pi$.

- Find $g(1)$ and $g'(1)$.
- On what intervals, if any, is g increasing? Justify your answer.
- For $0 < x \leq 2\pi$, find the value of x at which g has an absolute minimum. Justify your answer.
- For $0 < x < 2\pi$, is there a value of x at which the graph of g is tangent to the x -axis? Explain why or why not.

Part A

1 point is earned for $g(1)$

1 point is earned for $g'(1)$

$$g(1) = \int_1^1 f(t) dt = 0 \text{ and } g'(1) = f(1) = 0$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(1)$

1 point is earned for $g'(1)$

$$g(1) = \int_1^1 f(t) dt = 0 \text{ and } g'(1) = f(1) = 0$$

2-2-

Part B

1 point is earned for the interval

1 point is earned for the reason

Since $g'(x) = f(x)$, g is increasing on the interval $1 \leq x \leq \pi$ because $f(x) > 0$ for $1 < x < \pi$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the interval

1 point is earned for the reason

Since $g'(x) = f(x)$, g is increasing on the interval $1 \leq x \leq \pi$ because $f(x) > 0$ for $1 < x < \pi$.

Part C

1 point is earned if identifies 1 and 2π as candidates or indicates that the graph of g decreases, increases, then decreases

1 point is earned if justifies $g(2\pi) < g(1)$

1 point is earned for the answer

For $0 < x < 2\pi$, $g'(x) = f(x) = 0$ when $x = 1, \pi$. $g' = f$ changes from negative to positive only at $x = 1$. The absolute minimum must occur at $x = 1$ or at the right endpoint. Since $g(1) = 0$ and

$$g(2\pi) = \int_1^{2\pi} f(t) dt = \int_1^{\pi} f(t) dt + \int_{\pi}^{2\pi} f(t) dt < 0$$

by comparison of the two areas, the absolute minimum occurs at $x = 2\pi$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2-2-

1 point is earned if identifies 1 and 2π as candidates or indicates that the graph of g decreases, increases, then decreases

1 point is earned if justifies $g(2\pi) < g(1)$

1 point is earned for the answer

For $0 < x < 2\pi$, $g'(x) = f(x) = 0$ when $x = 1, \pi$. $g' = f$ changes from negative to positive only at $x = 1$. The absolute minimum must occur at $x = 1$ or at the right endpoint. Since $g(1) = 0$ and

$$g(2\pi) = \int_1^{2\pi} f(t) dt = \int_1^{\pi} f(t) dt + \int_{\pi}^{2\pi} f(t) dt < 0$$

by comparison of the two areas, the absolute minimum occurs at $x = 2\pi$.

Part D

1 point is earned for the answer of "yes" with $x = 1$

1 point is earned for the explanation

Yes, the graph g is tangent to the x -axis at $x = 1$ since $g(1) = 0$ and $g'(1) = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

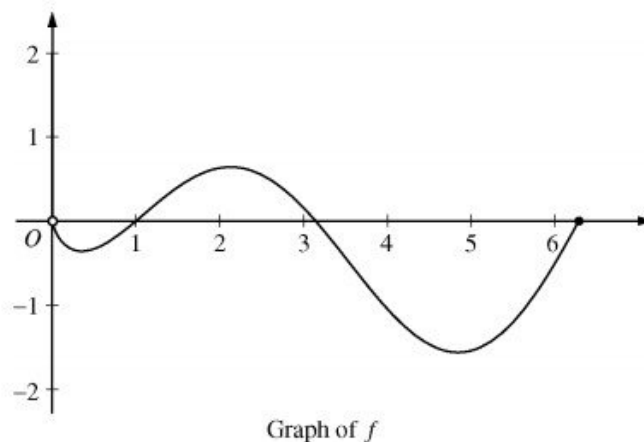
1 point is earned for the answer of "yes" with $x = 1$

1 point is earned for the explanation

Yes, the graph g is tangent to the x -axis at $x = 1$ since $g(1) = 0$ and $g'(1) = 0$.

2-2-

12.



Let f be the function given by $f(x) = (\ln x)(\sin x)$. The figure below shows the graph of f for $0 < x \leq 2\pi$.

The function g is defined by $g(x) = \int_1^x f(t) dt$ for $0 < x \leq 2\pi$.

For $0 < x \leq 2\pi$, is there a value of x at which the graph of g is tangent to the x -axis? Explain why or why not.

Part D

One point is earned for answer of "yes" with $x = 1$

One point is earned for explanation

Yes, the graph of g is tangent to the x -axis at $x = 1$

since $g(1) = 0$ and $g'(1) = 0$



0	1	2
---	---	---

The student earns all of the following points:

One point is earned for answer of "yes" with $x = 1$

One point is earned for explanation

Yes, the graph of g is tangent to the x -axis at $x = 1$

2-2-

since $g(1) = 0$ and $g'(1) = 0$

2-2-

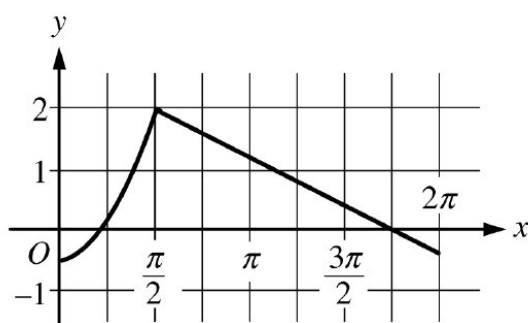
13. Let f be the function defined by $f(x) = e^x \cos(x)$.

(a) Find the average rate of change of f on the interval $0 \leq x \leq \pi$.

(b) What is the slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$?

(c) Find the absolute minimum value of f on the interval $0 \leq x \leq 2\pi$. Justify your answer.

(d) Let g be a differentiable function such that $g\left(\frac{\pi}{2}\right) = 0$. The graph of g' , the derivative of g , is shown below. Find the value of $\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)}$ or state that it does not exist. Justify your answer.



Graph of g'

Part A

1 point is earned for answer

The average rate of change of f on the interval $0 \leq x \leq \pi$ is $\frac{f(\pi) - f(0)}{\pi - 0} = \frac{-e^\pi - 1}{\pi}$.



0

1

The student response earns one of the following points:

1 point is earned for answer

The average rate of change of f on the interval $0 \leq x \leq \pi$ is $\frac{f(\pi) - f(0)}{\pi - 0} = \frac{-e^\pi - 1}{\pi}$.

Part B

1 point is earned for $f'(x)$

1 point is earned for slope

$$f'(x) = e^x \cos x - e^x \sin x$$

$$f'\left(\frac{3\pi}{2}\right) = e^{3\pi/2} \cos\left(\frac{3\pi}{2}\right) - e^{3\pi/2} \sin\left(\frac{3\pi}{2}\right) = e^{3\pi/2}$$

2-2-

The slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$ is $e^{3\pi/2}$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for slope

$$f'(x) = e^x \cos x - e^x \sin x$$

$$f'\left(\frac{3\pi}{2}\right) = e^{3\pi/2} \cos\left(\frac{3\pi}{2}\right) - e^{3\pi/2} \sin\left(\frac{3\pi}{2}\right) = e^{3\pi/2}$$

The slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$ is $e^{3\pi/2}$.

Part C

1 point is earned for sets $f(x) = 0$

1 point is earned for identifies $x = \frac{\pi}{4}$, $x = \frac{5\pi}{4}$ as candidates

1 point is earned for answer with justification

$$f'(x) = 0 \Rightarrow \cos x - \sin x = 0 \Rightarrow x = \frac{\pi}{4}, x = \frac{5\pi}{4}$$

2-2-

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{1}{\sqrt{2}}e^{5\pi/4}$
2π	$e^{2\pi}$

The absolute minimum value of f on $0 \leq x \leq 2\pi$ is $-\frac{1}{\sqrt{2}}e^{5\pi/4}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for sets $f'(x) = 0$

1 point is earned for identifies as $x = \frac{\pi}{4}$, $x = \frac{5\pi}{4}$ candidates

1 point is earned for answer with justification

$$f'(x) = 0 \Rightarrow \cos x - \sin x = 0 \Rightarrow x = \frac{\pi}{4}, x = \frac{5\pi}{4}$$

2-2-

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{1}{\sqrt{2}}e^{5\pi/4}$
2π	$e^{2\pi}$

The absolute minimum value of f on $0 \leq x \leq 2\pi$ is $-\frac{1}{\sqrt{2}}e^{5\pi/4}$.

Part D

1 point is earned for g is continuous at $x = \pi/2$ and limits equal 0

1 point is earned for applies L'Hospital's Rule

1 point is earned for answer

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

$$\lim_{x \rightarrow \pi/2} f(x) = 0$$

Because g is differentiable, g is continuous.

$$\lim_{x \rightarrow \pi/2} g(x) = g\left(\frac{\pi}{2}\right) = 0$$

By L'Hospital's Rule,

$$\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \pi/2} \frac{f'(x)}{g'(x)} = \frac{-e^{\pi/2}}{2}.$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2-2-

1 point is earned for g is continuous at $x = \pi/2$ and limits equal 0

1 point is earned for applies L'Hospital's Rule

1 point is earned for answer

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

$$\lim_{x \rightarrow \pi/2} f(x) = 0$$

Because g is differentiable, g is continuous.

$$\lim_{x \rightarrow \pi/2} g(x) = g\left(\frac{\pi}{2}\right) = 0$$

By L'Hospital's Rule,

$$\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \pi/2} \frac{f'(x)}{g'(x)} = \frac{-e^{\pi/2}}{2}.$$

2-2-

14.

x	-1.5	-1.0	-0.5	0	0.5	1.0	1.5
$f(x)$	-1	-4	-6	-7	-6	-4	-1
$f'(x)$	-7	-5	-3	0	3	5	7

Let f be a function that is differentiable for all real numbers. The table above gives the values of f and its derivative f' for selected points x in the closed interval $-1.5 \leq x \leq 1.5$. The second derivative of f has the property that $f''(x) > 0$ for $-1.5 \leq x \leq 1.5$.

Write an equation of the line tangent to the graph of f at the point where $x = 1$. Use this line to approximate the value of $f(1.2)$. Is this approximation greater than or less than the actual value of $f(1.2)$? Give a reason for your answer.

Part B

1 point is earned for the tangent line

1 point is earned for computes y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$$y = 5(x - 1) - 4$$

$$f(1.2) \approx 5(0.2) - 4 = -3$$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval $1 < x < 1.2$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for computes y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

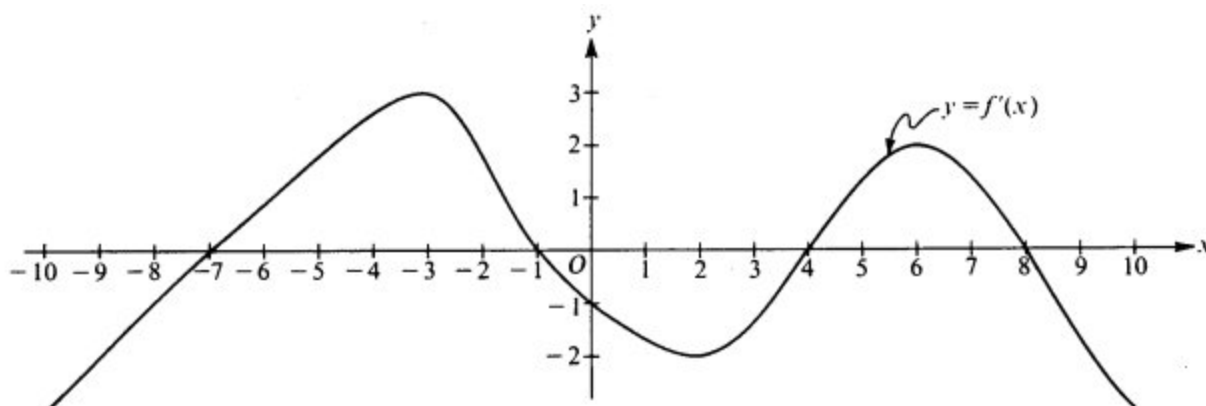
$$y = 5(x - 1) - 4$$

$$f(1.2) \approx 5(0.2) - 4 = -3$$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval $1 < x < 1.2$.

2-2-

15.



Note: This is the graph of the derivative of f , not the graph of f .

The figure above shows the graph of f' , the derivative of a function f . The domain of f is the set of all real numbers x such that $-10 \leq x \leq 10$.

- For what values of x does the graph of f have a horizontal tangent?
- For what values of x in the interval $(-10, 10)$ does f have a relative maximum? Justify your answer
- For what values of x is the graph of f concave downwards?

Part A

3 point is earned for solving $f'(x) = 0$

1 point is **deducted** for each incorrect point

1 point is **deducted** for each missing point

horizontal tangent $\Leftrightarrow f'(x) = 0$

$x = -7, -1, 4, 8$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

3 point is earned for solving $f'(x) = 0$

1 point is **deducted** for each incorrect point

1 point is **deducted** for each missing point

horizontal tangent $\Leftrightarrow f'(x) = 0$

$x = -7, -1, 4, 8$

2-2-

Part B

2 points are earned for answer

1 point is earned for justification

1 point is **deducted** for each incorrect or missing pointRelative maxima at $x = -1, 8$ because f' changes from positive to negative at these points

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for answer

1 point is earned for justification

1 point is **deducted** for each incorrect or missing pointRelative maxima at $x = -1, 8$ because f' changes from positive to negative at these points**Part C**

1 of 3 points are earned for correct sub intervals

2 of 3 points are earned for 1 correct interval with analysis

3 of 3 points are earned for 2 correct intervals

 f concave downward $\Leftrightarrow f'$ decreasing $(-3, 2), (6, 10)$ 

0	1	2	3
---	---	---	---

The student response earns all points:

1 of 3 points are earned for correct sub intervals

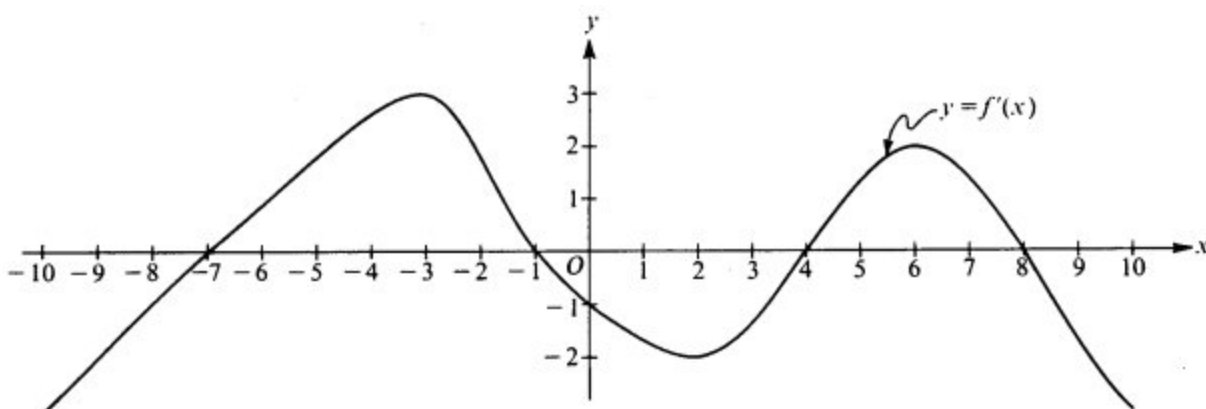
2 of 3 points are earned for 1 correct interval with analysis

3 of 3 points are earned for 2 correct intervals

 f concave downward $\Leftrightarrow f'$ decreasing $(-3, 2), (6, 10)$

2-2-

16.



Note: This is the graph of the derivative of f , not the graph of f .

The figure above shows the graph of f' , the derivative of a function f . The domain of f is the set of all real numbers x such that $-10 \leq x \leq 10$.

For what values of x does the graph of f have a horizontal tangent?

Part A

3 point is earned for solving $f'(x) = 0$

1 point is deducted for each incorrect point

1 point is deducted for each missing point

horizontal tangent $\Leftrightarrow f'(x) = 0$

$x = -7, -1, 4, 8$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

3 point is earned for solving $f'(x) = 0$

1 point is deducted for each incorrect point

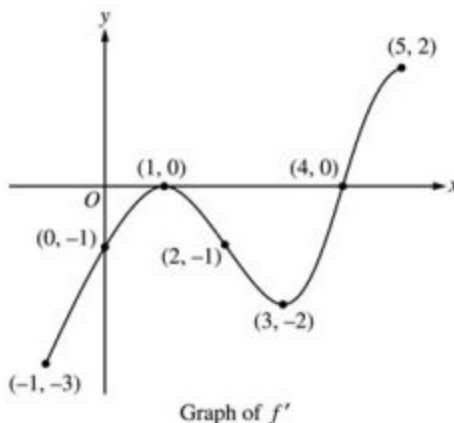
1 point is deducted for each missing point

horizontal tangent $\Leftrightarrow f'(x) = 0$

$x = -7, -1, 4, 8$

2-2-

17.



The figure above shows the graph of f' , the derivative of the function f , on the closed interval $-1 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x = 1$ and $x = 3$. The function f is twice differentiable with $f(2) = 6$.

- (a) Find the x -coordinate of each of the points of inflection of the graph of f . Give a reason for your answer.
- (b) At what value of x does f attain its absolute minimum value on the closed interval $-1 \leq x \leq 5$? At what value of x does f attain its absolute maximum value on the closed interval $-1 \leq x \leq 5$? Show the analysis that leads to your answers.
- (c) Let g be the function defined by $g(x) = xf(x)$. Find an equation for the line tangent to the graph of g at $x = 2$.

Part A

1 point is earned for $x = 1, x = 3$

1 point is earned for the reason

$x = 1$ and $x = 3$ because the graph of f' changes from increasing to decreasing at $x = 1$, and changes from decreasing to increasing at $x = 3$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x = 1, x = 3$

1 point is earned for the reason

$x = 1$ and $x = 3$ because the graph of f' changes from increasing to decreasing at $x = 1$, and changes from decreasing to increasing at $x = 3$.

Part B

2-2-

1 point is earned for indicating f decreases then increases

1 point is earned for eliminating $x = 5$ for maximum

1 point is earned for absolute minimum at $x = 4$

1 point is earned for absolute maximum at $x = -1$

The function f decreases from $x = -1$ to $x = 4$, then increases from $x = 4$ to $x = 5$.

Therefore, the absolute minimum value for f is at $x = 4$. The absolute maximum value must occur at $x = -1$ or at $x = 5$.

$$f(5) - f(-1) = \int_{-1}^5 f'(t) dt < 0$$

Since $f(5) < f(-1)$, the absolute maximum value occurs at $x = -1$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for indicating f decreases then increases

1 point is earned for eliminating $x = 5$ for maximum

1 point is earned for absolute minimum at $x = 4$

1 point is earned for absolute maximum at $x = -1$

The function f decreases from $x = -1$ to $x = 4$, then increases from $x = 4$ to $x = 5$.

Therefore, the absolute minimum value for f is at $x = 4$. The absolute maximum value must occur at $x = -1$ or at $x = 5$.

$$f(5) - f(-1) = \int_{-1}^5 f'(t) dt < 0$$

Since $f(5) < f(-1)$, the absolute maximum value occurs at $x = -1$.

Part C

2 points are earned for $g'(x)$

1 point is earned for the tangent line

2-2-

$$g'(x) = f(x) + xf'(x)$$

$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

Tangent line is $y = 4(x - 2) + 12$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for the tangent line

$$g'(x) = f(x) + xf'(x)$$

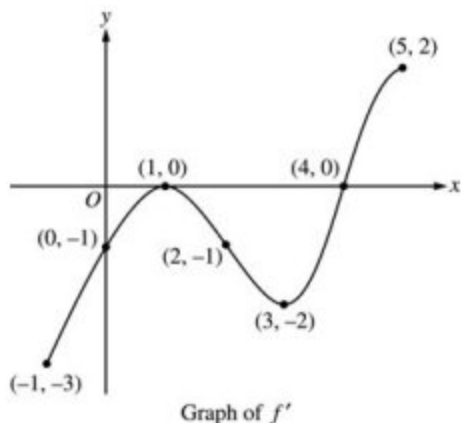
$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

Tangent line is $y = 4(x - 2) + 12$

2-2-

18.



The figure above shows the graph of f' , the derivative of the function f , on the closed interval $-1 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x=1$ and $x=3$. The function f is twice differentiable with $f(2)=6$.

Let g be the function defined by $g(x) = xf(x)$. Find an equation for the line tangent to the graph of g at $x=2$.

Part C

2 point is earned for $g'(x)$

1 point is earned for the tangent line

$$g'(x) = f(x) + xf'(x)$$

$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

Tangent line is $y = 4(x-2) + 12$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 point is earned for $g'(x)$

1 point is earned for the tangent line

$$g'(x) = f(x) + xf'(x)$$

$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

2-2-

Tangent line is $y=4(x-2)+12$

2-2-

19. Let f be the function given by $f(x) = \frac{\ln x}{x}$ **mfrac** for all $x > 0$. The derivative of f is given by $f'(x) = \frac{1 - \ln x}{x^2}$.

- (a) Write an equation for the line tangent to the graph of f at $x = e^2$.
- (b) Find the x -coordinate of the critical point of f . Determine whether this point is a relative minimum, a relative maximum, or neither for the function f . Justify your answer.
- (c) The graph of the function f has exactly one point of inflection. Find the x -coordinate of this point.
- (d) Find $\lim_{x \rightarrow 0^+} f(x)$.

Part A

1 point is earned for the $f(e^2)$ and $f'(e^2)$

1 point is earned for the answer

$$f(e^2) = \frac{\ln e^2}{e^2} = \frac{2}{e^2}, \quad f'(e^2) = \frac{1 - \ln e^2}{(e^2)^2} = -\frac{1}{e^4}$$

An equation for the tangent line is $y = \frac{2}{e^2} - \frac{1}{e^4}(x - e^2)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the $f(e^2)$ and $f'(e^2)$

1 point is earned for the answer

$$f(e^2) = \frac{\ln e^2}{e^2} = \frac{2}{e^2}, \quad f'(e^2) = \frac{1 - \ln e^2}{(e^2)^2} = -\frac{1}{e^4}$$

An equation for the tangent line is $y = \frac{2}{e^2} - \frac{1}{e^4}(x - e^2)$.

2-2-

Part B

3 points can be earned.

1 point for $x = e$

1 point for relative maximum

1 point for justification

$f'(x) = 0$ when $x = e$. The function f has a relative maximum at $x = e$ because $f'(x)$ changes from positive to negative at $x = e$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

3 points can be earned.

1 point for $x = e$

1 point for relative maximum

1 point for justification

$f'(x) = 0$ when $x = e$. The function f has a relative maximum at $x = e$ because $f'(x)$ changes from positive to negative at $x = e$.

Part C2 points are earned for $f''(x)$

1 point is earned for answer

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2 \ln x}{x^3} \text{ for all } x > 0$$

$$f''(x) = 0 \text{ when } -3 + 2 \ln x = 0$$

$$x = e^{3/2}$$

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$.

2-2-



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $f''(x)$

1 point is earned for answer

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2 \ln x}{x^3} \text{ for all } x > 0$$

$$f''(x) = 0 \text{ when } -3 + 2 \ln x = 0$$

$$x = e^{3/2}$$

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$.

Part D

1 point is earned for answer

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x} = -\infty \text{ or Does Not Exist}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x} = -\infty \text{ or Does Not Exist}$$

2-2-

20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(x) = \begin{cases} \sqrt{9-x^2} & \text{for } -3 \leq x \leq 0 \\ -x + 3 \cos\left(\frac{\pi x}{2}\right) & \text{for } 0 < x \leq 4 \end{cases}$$

Let f be the function defined above.

- (a) Find the average rate of change of f on the interval $-3 \leq x \leq 4$.
- (b) Write an equation for the line tangent to the graph of f at $x = 3$.
- (c) Find the average value of f on the interval $-3 \leq x \leq 4$.
- (d) Must there be a value of x at which $f(x)$ attains an absolute maximum on the closed interval $-3 \leq x \leq 4$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point does not require a simplified answer. Substitution of function values is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ answer

Solution:

$$\text{Average rate of change} = \frac{f(4) - f(-3)}{4 - (-3)} = \frac{-1 - 0}{7} = -\frac{1}{7}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-2-

The first point is earned with a correct and labeled value of $f'(3)$.

The second point can be earned with a consistent tangent line equation created from an incorrect value of $f'(3)$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $f'(3)$
- ☐ equation

Solution:

$$f(3) = -3 + 3 \cos\left(\frac{3\pi}{2}\right) = -3$$

$$\text{For } 0 < x < 4, f'(x) = -1 + \left(-3 \sin\left(\frac{\pi x}{2}\right)\right) \cdot \frac{\pi}{2}$$

$$f'(3) = -1 + \left(-3 \sin\left(\frac{3\pi}{2}\right)\right) \cdot \frac{\pi}{2} = -1 + \frac{3\pi}{2}$$

$$\text{An equation for the tangent line is } y = -3 + \left(-1 + \frac{3\pi}{2}\right)(x - 3).$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The constant $\frac{1}{b-a}$ is part of the fourth point.

Simplification is not required for the second point.

The third point is earned with the antiderivative $\frac{-x^2}{2} + \frac{6}{\pi} \sin\left(\frac{\pi x}{2}\right)$. The third point can also be earned with $\frac{-x^2}{2} + \frac{2}{\pi} \sin\left(\frac{\pi x}{2}\right)$, though this response will not be eligible for the fourth point.

Simplification is not required for the fourth point.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

- ☐ integrals of f over $-3 \leq x \leq 0$ and $0 \leq x \leq 4$

2-2-

- ☐ value of $\int_{-3}^0 \sqrt{9-x^2} \, dx$
- ☐ antiderivative of $-x + 3 \cos\left(\frac{\pi x}{2}\right)$
- ☐ answer

Solution:

The average value of f on the interval $-3 \leq x \leq 4$ is $\frac{1}{4-(-3)} \int_{-3}^4 f(x) \, dx$.

$$\int_{-3}^4 f(x) \, dx = \int_{-3}^0 f(x) \, dx + \int_0^4 f(x) \, dx$$

$$\int_{-3}^0 f(x) \, dx = \int_{-3}^0 \sqrt{9-x^2} \, dx = \frac{9\pi}{4}$$

$$\int_0^4 f(x) \, dx = \int_0^4 \left(-x + 3 \cos\left(\frac{\pi x}{2}\right)\right) \, dx = \left[-\frac{1}{2}x^2 + \frac{6}{\pi} \sin\left(\frac{\pi x}{2}\right)\right]_0^4 = -8$$

$$\frac{1}{4-(-3)} \int_{-3}^4 f(x) \, dx = \frac{1}{7} \left(\frac{9\pi}{4} - 8\right)$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned by evaluating each branch of the function at $x = 0$ instead of evaluating the one-sided limits, but the response will not be eligible for the second point.

The second point is earned with correct limit notation for the first point, clear communication that $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0) = 3$, and a statement that the function is continuous on the closed interval, and therefore must have an absolute maximum. Specifically naming the Extreme Value Theorem is not required to earn this point. Responses that indicate that the function is differentiable on the interval $[-3, 4]$ will not earn the point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ continuity at $x = 0$
- ☐ answer with justification

2-2-

Solution:

$\lim_{x \rightarrow 0^-} f(x) = f(0) = 3$ and $\lim_{x \rightarrow 0^+} f(x) = 3$, so f is continuous at $x = 0$.

Because f is continuous on $[-3, 4]$, the Extreme Value Theorem guarantees that f attains an absolute maximum on $[-3, 4]$.

2-2-

21. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The function f is given by $f(x) = \frac{x^3}{e^x}$.

- (a) Write an equation for the line tangent to the graph of f at $x = 5$.
- (b) Write an equation that involves a limit and demonstrates that the x -axis is an asymptote for the graph of f .
- (c) Find the x -coordinate of each critical point of f . Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.
- (d) Find the average value of f' , the derivative of f , on the interval $[-1, 5]$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

$f'(x) = \frac{3x^2 - x^3}{e^x}$ earns the first 2 points.

An unlabeled correct $f'(x)$ earns the first 2 points.

To earn the first point:

- If the quotient rule was used, the presented derivative must include a sum or difference, a factor of e^x , and a polynomial in each term of the numerator.
- If the product rule was used, the presented derivative must include a factor of e^{-x} in each term.

A response that includes $f'(x) = \frac{e^x \cdot 3x^2 - x^3 e^x}{e^{x^2}}$ earns the first point, does not earn the second point, and is eligible for the third point. In this case the response must correctly evaluate this expression at $x = 5$, and use the resulting value as the slope of the tangent line.

The second point cannot be earned without the first. The second point is earned for a correct expression for $f'(x)$ or a correct value of $f'(5)$.

A response that labels the values $f(5)$ and $f'(5)$ correctly with no other supporting work does not earn the first 2 points but is eligible for the third point.

2-2-

A response that presents a correct $f'(x)$ but an incorrect $f'(5)$ earns the first 2 points, but does not earn the third point.

A response that presents a correct equation of the tangent line with no supporting work earns no points in this part.

A response with any arithmetic error in $f'(x)$ does not earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Form of quotient rule (or product rule)
- ☐ $f'(x)$ or $f'(5)$
- ☐ Answer

Solution:

$$f'(x) = \frac{e^x \cdot 3x^2 - x^3 e^x}{(e^x)^2} = \frac{3x^2 - x^3}{e^x}$$

$$f'(5) = \frac{75 - 125}{e^5} = \frac{-50}{e^5}$$

$$f(5) = \frac{125}{e^5}$$

An equation for the tangent line is $y = \frac{125}{e^5} - \frac{50}{e^5}(x - 5)$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

$\lim_{x \rightarrow \infty} f(x) = 0$ or $\lim_{x \rightarrow \infty} \frac{x^3}{e^x} = 0$ earns the point.

A response that presents an expression, rather than an equation, does not earn the point.

A response that presents any additional incorrect limits does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

2-2-

· Equation

Solution:

$$\lim_{x \rightarrow \infty} f(x) = 0$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

$(3x^2 - x^3) = 0$ earns the first point.

A response that presents only one of $x = 0$ or $x = 3$ along with correct answer and justification earns 1 of the last 2 points.

A response that lists additional values of x as critical points does not earn either of the last 2 points.

The second derivative may be used to justify the relative maximum at $x = 3$. Since $f''(x) = 0$ at $x = 0$, it is not sufficient to use the second derivative to conclude that $x = 0$ is neither a maximum or minimum value; more justification is needed.

A sign chart with no explanation is not sufficient justification.

Because the interval is not closed, the Extreme Value Theorem may not be used.

A response that imports e^{x^2} in the denominator of the derivative from part (a) is eligible for all 3 points.

A response that imports either $f'(x) = \frac{e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ or $f'(x) = \frac{-e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ from part (a) is eligible for all 3 points.

· An import of $f'(x) = \frac{e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ should find critical points $x = -3$ and $x = 0$ AND conclude that f has a relative minimum at $x = -3$ and neither a minimum nor a maximum at $x = 0$.

· An import of $f'(x) = \frac{-e^x \cdot 3x^2 + x^3 e^x}{(e^x)^2}$ should find critical points $x = 0$ and $x = 3$ AND conclude that f has neither a minimum nor a maximum at $x = 0$ and a relative minimum at $x = 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $f'(x) = 0$
 $x = 0$ and $x = 3$ only

2-2-

☐☐

Answer with justification

Solution:

$$f'(x) = \frac{e^x \cdot 3x^2 - x^3 e^x}{(e^x)^2} = \frac{x^2 e^x (3 - x)}{e^{2x}} = 0 \Rightarrow x = 0, x = 3$$

$f'(x)$ does not change sign at $x = 0$. Therefore, f has neither a minimum nor a maximum at $x = 0$.

$f'(x)$ changes sign from positive to negative at $x = 3$. Therefore, f has a relative maximum at $x = 3$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A factor of $\frac{1}{6}$ is not needed to earn the first point.

The definite integral can be presented in a single or multiple steps to earn the first point. For example, the following response earns both points: $\int f'(x) dx = f(x)$, so the average value is $\frac{1}{6}(\frac{125}{e^5} + \frac{1}{e^{-1}})$.

$f(5) - f(-1)$ earns the first point and $\frac{1}{6}(\frac{125}{e^5} - (\frac{-1}{e^{-1}}))$ earns both points, provided there are no subsequent errors in simplification.

A response that presents a correct integral along with the correct average value, but provides incorrect or incomplete communication, earns 1 out of 2 points. For example, the following response earns 1 out of 2 points:

$$\int_{-1}^5 f'(x) dx = (\frac{125}{e^5} + e) = \frac{1}{6}(\frac{125}{e^5} + e).$$

$\int_{-1}^5 f'(x) dx$ with an incorrect $f'(x)$ imported from part (a) or part (c) earns the first point.

A response with incorrect limits earns no points.

A response of $\frac{1}{6} \cdot \frac{125}{e^5} - \frac{-1}{e^{-1}}$ or $\frac{1}{6} \cdot f(5) - f(1)$ earns the first point but not the second point.

A response that uses an incorrect value of $f(5)$ or $f(-1)$ does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

2-2-

☐ Integral☐ Answer**Solution:**

$$\frac{1}{5-(-1)} \int_{-1}^5 f'(x) \, dx$$

$$= \frac{1}{6} (f(5) - f(-1)) = \frac{1}{6} \left(\frac{125}{e^5} - \left(\frac{-1}{e^{-1}} \right) \right) = \frac{1}{6} \left(\frac{125}{e^5} + e \right)$$

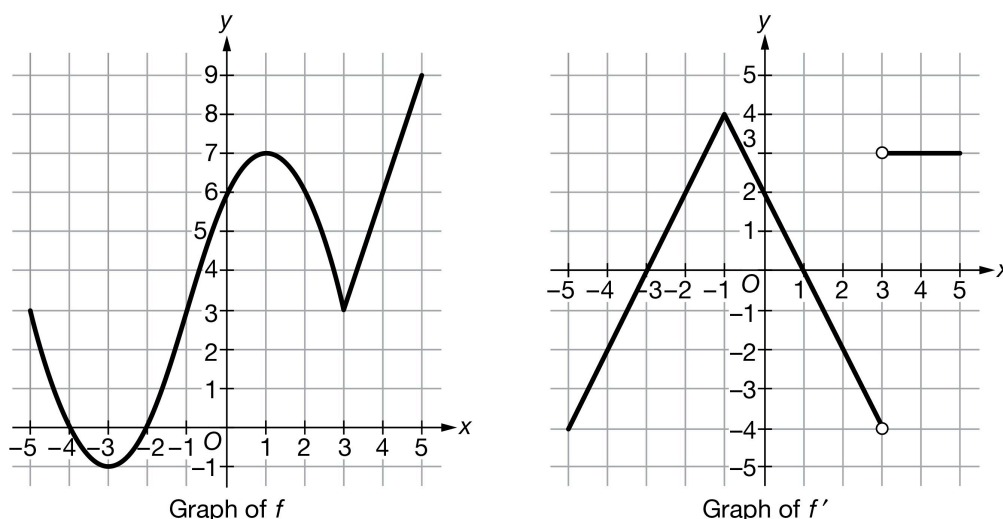
2-2-

22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graphs of the function f and its derivative f' are shown above for $-5 \leq x \leq 5$.

Find the average rate of change of f over the interval $-5 \leq x \leq 5$. For how many values of x in the interval $-5 \leq x \leq 5$ does the instantaneous rate of change of f equal the average rate of change of f over that interval?

Write an equation for the line tangent to the graph of f at $x = 2$.

For each of $\lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x - (-1)}$ and $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$, find the value or give a reason why it does not exist.

Let g be the function defined by $g(x) = f(x) \cdot \ln x$. Find $g'(4)$.

Part A

If the average rate of change has a maximum of one sign error, the response is eligible for the second point based on a correct interpretation of the graph of f' with the presented average rate of change.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

2-2-

The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

$$\frac{f(5)-f(-5)}{5-(-5)} = \frac{9-3}{5-(-5)} = \frac{6}{10} = \frac{3}{5}$$

The average rate of change of f over the interval $-5 \leq x \leq 5$ is $\frac{3}{5}$.

$f'(x)$, the instantaneous rate of change of f at x , equals $\frac{3}{5}$ for two values of x in the interval $-5 \leq x \leq 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct tangent line equation.

Solution:

$$f(2) = 6$$

$$f'(2) = -2$$

An equation of the line tangent to the graph of f at $x = 2$ is $y = 6 - 2(x - 2)$.

Part C

The second point is earned for “does not exist” with a reason that indicates f is not differentiable at that point (e.g., indicates graph has a corner point or sharp point).

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

2-2-

- ☐ $\lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x - (-1)}$
- ☐ $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$ does not exist

Solution:

$$\lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x - (-1)} = f'(-1) = 4$$

$$\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} \text{ does not exist because } f \text{ is not differentiable at } x = 3.$$

Part D

The first point is earned for a correct application of the product rule. The second point is earned for a correct expression that substitutes both correct numerical values.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $g'(x)$
- ☐ answer

Solution:

$$g'(x) = f'(x) \cdot \ln x + f(x) \cdot \frac{1}{x}$$

$$g'(4) = f'(4) \cdot \ln 4 + f(4) \cdot \frac{1}{4} = 3 \ln 4 + 6 \cdot \frac{1}{4} = 3 \ln 4 + \frac{3}{2}$$

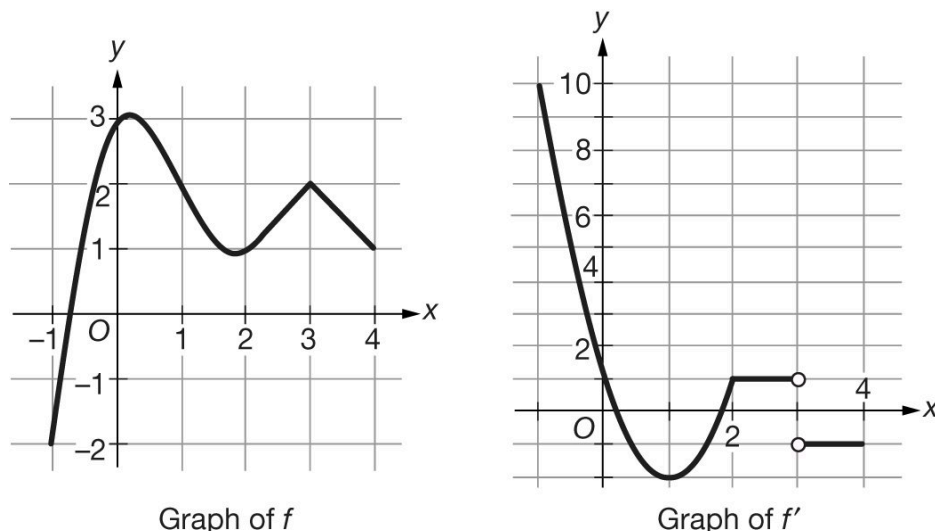
2-2-

23. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graphs of the function f and its derivative f' are shown above for $-1 \leq x \leq 4$.

- Find the average rate of change of f over the interval $-1 \leq x \leq 4$. For how many values of x in the interval $-1 \leq x \leq 4$ does the instantaneous rate of change of f equal the average rate of change of f over that interval?
- Write an equation for the line tangent to the graph of f at $x = 1$.
- For each of $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$ and $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$, find the value or give a reason why it does not exist.
- Let g be the function defined by $g(x) = e^x f(x)$. Find $g'(0)$.

Part A

If the average rate of change has a maximum of one sign error, the response is eligible for the second point based on a correct interpretation of the graph of f' with the presented average rate of change.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

2-2-

The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

$$\frac{f(4)-f(-1)}{4-(-1)} = \frac{1-(-2)}{5} = \frac{3}{5}$$

The average rate of change of f over the interval is $\frac{3}{5}$

$f'(x)$, the instantaneous rate of change of f at x , equals $\frac{3}{5}$ for two values of x in the interval $[-1, 4]$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct tangent line equation.

Solution:

$$f(1) = 2$$

$$f'(1) = -2$$

An equation of the line tangent to the graph of f at $x = 1$ is $y = 2 - 2(x - 1)$.

Part C

The second point is earned for “does not exist” with a reason that indicates f is not differentiable at that point (e.g., indicates graph has a corner point or sharp point).

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

2-2-

The student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$
- ☐ $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$ does not exist

Solution:

$$\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = f'(2) = 1$$

$$\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} \text{ does not exist because } f \text{ is not differentiable at } x = 3.$$

Part D

The first point is earned for a correct application of the product rule. The second point is earned for a correct expression that substitutes both correct numerical values.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $g'(x)$
- ☐ answer

Solution:

$$g'(x) = e^x f(x) + e^x f'(x)$$

$$g'(0) = e^0 f(0) + e^0 f'(0) = 3 + 1 = 4$$

2-2-

24. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Functions f , g , and h are twice-differentiable functions with $g(2) = h(2) = 4$. The line $y = 4 + \frac{2}{3}(x - 2)$ is tangent to both the graph of g at $x = 2$ and the graph of h at $x = 2$.

a) Find $h'(2)$.

b) Let a be the function given by $a(x) = 3x^3h(x)$. Write an expression for $a'(x)$. Find $a'(2)$.

c) The function h satisfies $h(x) = \frac{x^2 - 4}{1 - (f(x))^3}$ for $x \neq 2$. It is known that $\lim_{x \rightarrow 2} h(x)$ can be evaluated using L'Hospital's Rule. Use $\lim_{x \rightarrow 2} h(x)$ to find $f(2)$ and $f'(2)$. Show the work that leads to your answers.

d) It is known that $g(x) \leq h(x)$ for $1 < x < 3$. Let k be a function satisfying $g(x) \leq k(x) \leq h(x)$ for $1 < x < 3$. Is k continuous at $x = 2$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

		✓
0	1	

The student response accurately includes the criteria below.

☐ · answer

Solution:

$$h'(2) = \frac{2}{3}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-2-

The first point is earned for a derivative in the form of the product rule, namely $a'(x) = p(x) \cdot h(x) + q(x) \cdot h'(x)$, where $p(x)$ and $q(x)$ are both polynomials.

Simplification is not required to earn the third point. Function values must be substituted into the expression.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ form of product rule
- ☐ $a'(x)$
- ☐ $a'(2)$

Solution:

$$a'(x) = 9x^2h(x) + 3x^3h'(x)$$

$$a'(2) = 9 \cdot 2^2h(2) + 3 \cdot 2^3h'(2) = 36 \cdot 4 + 24 \cdot \frac{2}{3} = 160$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for showing $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = 4$. This work may appear anywhere in the response.

The second point is earned for calculating $f(2) = 0$ in the context of setting the denominator $1 - (f(x))^3$ equal to 0.

The third point is earned for evidence of applying L'Hospital's Rule, which includes a new limit of a quotient where differentiation was attempted in both the numerator and denominator. Note that incorrect notation such as $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = \frac{0}{0}$ makes a response not eligible to earn the third point.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

☐ $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = 4$

2-2-

- ☐ $f(2)$
- ☐ L'Hospital's Rule
- ☐ $f'(2)$

Solution:

Because h is differentiable, h is continuous, so $\lim_{x \rightarrow 2} h(x) = h(2) = 4$.

Also, $\lim_{x \rightarrow 2} h(x) = \lim_{x \rightarrow 2} \frac{x^2 - 4}{1 - (f(x))^3}$, so $\lim_{x \rightarrow 2} \frac{x^2 - 4}{1 - (f(x))^3} = 4$.

Because $\lim_{x \rightarrow 2} (x^2 - 4) = 0$, we must also have $\lim_{x \rightarrow 2} (1 - (f(x))^3) = 0$.

Thus $\lim_{x \rightarrow 2} f(x) = 1$.

Because f is differentiable, f is continuous, so $f(2) = \lim_{x \rightarrow 2} f(x) = 1$.

Also, because f is twice differentiable, f' is continuous, so $\lim_{x \rightarrow 2} f'(x) = f'(2)$ exists.

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{1 - (f(x))^3} = \lim_{x \rightarrow 2} \frac{2x}{-3(f(x))^2 f'(x)} = \frac{4}{-3(1)^2 \cdot f'(2)} = 4.$$

Thus $f'(2) = -\frac{1}{3}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point does not require naming the squeeze theorem. The justification requires (1) stating g and h are continuous, (2) stating $\lim_{x \rightarrow 2} g(x) = 4$ and $\lim_{x \rightarrow 2} h(x) = 4$ and concluding that $\lim_{x \rightarrow 2} k(x) = 4$, and (3) stating $g(2) = 4 = h(2)$ and concluding $k(2) = 4$.



0

1

The student response accurately includes the criteria below.

- ☐ continuous with justification

Solution:

2-2-

Because g and h are differentiable, g and h are continuous, so $\lim_{x \rightarrow 2} g(x) = g(2) = 4$ and $\lim_{x \rightarrow 2} h(x) = h(2) = 4$.

Because $g(x) \leq k(x) \leq h(x)$ for $1 < x < 3$, it follows from the squeeze theorem that $\lim_{x \rightarrow 2} k(x) = 4$.

Also, $4 = g(2) \leq k(2) \leq h(2) = 4$, so $k(2) = 4$.

Thus k is continuous at $x = 2$.

2-2-

25.  The function g is defined for $x > 0$ with $g(1) = 2$, $g'(x) = \sin(x + 1/x)$, and $g''(x) = (1 - 1/x^2)\cos(x + 1/x)$.
- (a) Find all values of x in the interval $0.12 \leq x \leq 1$ at which the graph of g has a horizontal tangent line.
- (b) On what subintervals of $(0.12, 1)$, if any, is the graph of g concave down? Justify your answer.
- (c) Write an equation for the line tangent to the graph of g at $x = 0.3$.
- (d) Does the line tangent to the graph of g at $x = 0.3$ lie above or below the graph of g for $0.3 < x < 1$? Why?

Part A

1 point is earned for sets $g'(x) = 0$

1 point is earned for the answer

The graph of g has a horizontal tangent line when $g'(x) = 0$. This occurs at $x = 0.163$ and $x = 0.359$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for sets $g'(x) = 0$

1 point is earned for the answer

The graph of g has a horizontal tangent line when $g'(x) = 0$. This occurs at $x = 0.163$ and $x = 0.359$.

Part B

1 point is earned for the answer

1 point is earned for the justification

$g''(x) = 0$ at $x = 0.129458$ and $x = 0.222734$

The graph of g is concave down on $(0.1295, 0.2227)$ because $g''(x) < 0$ on this interval.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

2-2-

1 point is earned for the justification

$$g''(x) = 0 \text{ at } x = 0.129458 \text{ and } x = 0.222734$$

The graph of g is concave down on $(0.1295, 0.2227)$ because $g''(x) < 0$ on this interval.

Part C

1 point is earned for $g'(0.3)$

1 point is earned for the integral expression

1 point is earned for $g(0.3)$

1 point is earned for the equation

$$g'(0.3) = -0.472161$$

$$g(0.3) = 2 + \int_1^{0.3} g'(x) dx = 1.546007$$

An equation for the line tangent to the graph of g is $y = 1.546 - 0.472(x - 0.3)$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(0.3)$

1 point is earned for the integral expression

1 point is earned for $g(0.3)$

1 point is earned for the equation

$$g'(0.3) = -0.472161$$

$$g(0.3) = 2 + \int_1^{0.3} g'(x) dx = 1.546007$$

An equation for the line tangent to the graph of g is $y = 1.546 - 0.472(x - 0.3)$.

Part D

2-2-

1 point is earned for answer with reason

$$g''(x) > 0 \text{ for } 0.3 < x < 1$$

Therefore the line tangent to the graph of g at $x = 0.3$ lies below the graph of g for $0.3 < x < 1$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

$$g''(x) > 0 \text{ for } 0.3 < x < 1$$

Therefore the line tangent to the graph of g at $x = 0.3$ lies below the graph of g for $0.3 < x < 1$.

The function g is defined for $x > 0$ with $g(1)=2$, $g'(x)=\sin(x+1/x)$, and $g''(x)=(1-1/x^2)\cos(x+1/x)$.

26.  Find all values of x in the interval $0.12 \leq x \leq 1$ at which the graph of g has a horizontal tangent line.

Part A

1 point is earned for sets $g'(x)=0$

1 point is earned for the answer

The graph of g has a horizontal tangent line when $g'(x)=0$. This occurs at $x=0.163$ and $x=0.359$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for sets $g'(x)=0$

1 point is earned for the answer

The graph of g has a horizontal tangent line when $g'(x)=0$. This occurs at $x=0.163$ and $x=0.359$.

2-2-

27.  Write an equation for the line tangent to the graph of g at $x=0.3$.
-

Part C

1 point is earned for $g'(0.3)$

1 point is earned for the integral expression

1 point is earned for $g(0.3)$

1 point is earned for the equation

$$g'(0.3) = -0.472161$$

$$g(0.3) = 2 + \int_1^{0.3} g'(x) dx = 1.546007$$

An equation for the line tangent to the graph of g is $y = 1.546 - 0.472(x - 0.3)$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(0.3)$

1 point is earned for the integral expression

1 point is earned for $g(0.3)$

1 point is earned for the equation

$$g'(0.3) = -0.472161$$

$$g(0.3) = 2 + \int_1^{0.3} g'(x) dx = 1.546007$$

An equation for the line tangent to the graph of g is

$$y = 1.546 - 0.472(x - 0.3).$$

2-2-

28. Consider the curve given by the parametric equations

$$x = 2t^3 - 3t^2 \text{ and } y = t^3 - 12t$$

- (a) In terms of t , find dy/dx .
- (b) Write an equation for the line tangent to the curve at the point where $t = -1$.
- (c) Find the x - and y -coordinates for each critical point on the curve and identify each point as having a vertical or horizontal tangent.

Part A

1 point is earned for $\frac{dy}{dt}$

1 point is earned for $\frac{dx}{dt}$

1 point is earned for $\frac{dy}{dx}$

$$\frac{dy}{dt} = 3t^2 - 12$$

$$\frac{dx}{dt} = 6t^2 - 6t$$

$$\frac{dy}{dx} = \frac{3t^2 - 12}{6t^2 - 6t} = \frac{t^2 - 4}{2t^2 - 2t} = \frac{(t+2)(t-2)}{2t(t-1)}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{dy}{dt}$

1 point is earned for $\frac{dx}{dt}$

1 point is earned for $\frac{dy}{dx}$

$$\frac{dy}{dt} = 3t^2 - 12$$

$$\frac{dx}{dt} = 6t^2 - 6t$$

$$\frac{dy}{dx} = \frac{3t^2 - 12}{6t^2 - 6t} = \frac{t^2 - 4}{2t^2 - 2t} = \frac{(t+2)(t-2)}{2t(t-1)}$$

Part B

2-2-1 point is earned for x, y values1 point is earned for $\frac{dy}{dx}$ values

1 point is earned for equation of tangent lines

$$x = -5, y = 11$$

$$\frac{dy}{dx} = -\frac{3}{4}$$

$$y - 11 = -\frac{3}{4}(x + 5)$$

or

$$y = -\frac{3}{4}x + \frac{29}{4}$$

$$4y + 3x = 29$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for x, y values1 point is earned for $\frac{dy}{dx}$ values

1 point is earned for equation of tangent lines

$$x = -5, y = 11$$

$$\frac{dy}{dx} = -\frac{3}{4}$$

$$y - 11 = -\frac{3}{4}(x + 5)$$

or

$$y = -\frac{3}{4}x + \frac{29}{4}$$

$$4y + 3x = 29$$

Part C

2-2-

1 point is earned for finding all t for which his/her $\frac{dy}{dx} = 0$ or is undefined (no points if incorrect values attached)

1 point is earned for finding all (x,y) values for t values

1 point is earned for identifying nature of tangency (type)

t	(x,y)	type
-2	$(-28,16)$	horizontal
0	$(0,0)$	vertical
1	$(-1,-11)$	vertical
2	$(4,-16)$	horizontal



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for finding all t for which his/her $\frac{dy}{dx} = 0$ or is undefined (no points if incorrect values attached)

1 point is earned for finding all (x,y) values for t values

1 point is earned for identifying nature of tangency (type)

t	(x,y)	type
-2	$(-28,16)$	horizontal
0	$(0,0)$	vertical
1	$(-1,-11)$	vertical
2	$(4,-16)$	horizontal

2-2-

29. Consider the curve given by the parametric equations

$$x = 2t^3 - 3t^2 \text{ and } y = t^3 - 12t$$

Write an equation for the line tangent to the curve at the point where $t = -1$.

Part B

1 point is earned for x, y values

1 point is earned for $\frac{dy}{dx}$ values

1 point is earned for equation of tangent lines

$$x = -5, y = 11$$

$$\frac{dy}{dx} = -\frac{3}{4}$$

$$y - 11 = -\frac{3}{4}(x + 5)$$

or

$$y = -\frac{3}{4}x + \frac{29}{4}$$

$$4y + 3x = 29$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for x, y values

1 point is earned for $\frac{dy}{dx}$ values

1 point is earned for equation of tangent lines

$$x = -5, y = 11$$

$$\frac{dy}{dx} = -\frac{3}{4}$$

$$y - 11 = -\frac{3}{4}(x + 5)$$

or

2-2-

$$y = -\frac{3}{4}x + \frac{29}{4}$$

$$4y + 3x = 29$$

2-2-

30. Consider the curve defined by $x^2 + xy + y^2 = 27$.
- (a) Write an expression for the slope of the curve at any point (x, y) .
 - (b) Determine whether the lines tangent to the curve at the x -intercepts of the curve are parallel. Show the analysis that leads to your conclusion.
 - (c) Find the points on the curve where the lines tangent to the curve are vertical.

Part A

2 points are earned for the correct implicit differentiation

<-1> each incorrect derivative of a term

$$2x + xy' + y + 2yy' = 0$$

1 point is earned for correctly solving for y'

0/1 if only one y'

$$y' = \frac{-2x - y}{x + 2y}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct implicit differentiation

<-1> each incorrect derivative of a term

$$2x + xy' + y + 2yy' = 0$$

1 point is earned for correctly solving for y'

0/1 if only one y'

$$y' = \frac{-2x - y}{x + 2y}$$

Part B

2-2-

1 point is earned for correctly using $y = 0$

If $y = 0, x^2 = 27$

$$x = \pm 3\sqrt{3}$$

1 point is earned for correctly finding y' at x -intercepts

at $x = 3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{3\sqrt{3}} = -2$

at $x = -3\sqrt{3}, y' = \frac{2 \cdot 3\sqrt{3}}{-3\sqrt{3}} = -2$

1 point is earned for the correct analysis and conclusion

Tangent lines at x -intercepts are parallel

NOTE: For solution at y -intercepts, eligible for 2nd and 3rd points



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly using $y = 0$

If $y = 0, x^2 = 27$

$$x = \pm 3\sqrt{3}$$

1 point is earned for correctly finding y' at x -intercepts

at $x = 3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{3\sqrt{3}} = -2$

at $x = -3\sqrt{3}, y' = \frac{2 \cdot 3\sqrt{3}}{-3\sqrt{3}} = -2$

1 point is earned for the correct analysis and conclusion

Tangent lines at x -intercepts are parallel

2-2-

NOTE: For solution at y -intercepts, eligible for 2nd and 3rd points**Part C**1 point is earned for correctly setting $x + 2y = 0$ y' undefined if $x + 2y = 0$ 1 point is earned for correctly solving for y (or x) values

$$\begin{aligned}(-2y)^2 + (-2y)y + y^2 &= 27 \\ 3y^2 &= 27 \\ y &= \pm 3\end{aligned}$$

1 point is earned for correctly giving both points

Points are $(-6, 3)$ and $(6, -3)$

NOTE: max 1/3 for horizontal tangents

Max 1/3 if denominator of y' is not of form $ax + by + c$ with $a \cdot b \neq 0$.

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly setting $x + 2y = 0$ y' undefined if $x + 2y = 0$ 1 point is earned for correctly solving for y (or x) values

$$\begin{aligned}(-2y)^2 + (-2y)y + y^2 &= 27 \\ 3y^2 &= 27 \\ y &= \pm 3\end{aligned}$$

1 point is earned for correctly giving both points

Points are $(-6, 3)$ and $(6, -3)$

NOTE: max 1/3 for horizontal tangents

2-2-

Max 1/3 if denominator of y' is not of form $ax + by + c$ with $a \cdot b \neq 0$.

Consider the curve defined by $x^2 + xy + y^2 = 27$.

31. Write an expression for the slope of the curve at any point (x, y) .

Part A

2 points are earned for the correct implicit differentiation

< -1 > each incorrect derivative of a term

$$2x + xy' + y + 2yy' = 0$$

1 point is earned for correctly solving for y'

0/1 if only one y'

$$y' = \frac{-2x - y}{x + 2y}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the correct implicit differentiation

< -1 > each incorrect derivative of a term

$$2x + xy' + y + 2yy' = 0$$

1 point is earned for correctly solving for y'

0/1 if only one y'

$$y' = \frac{-2x - y}{x + 2y}$$

2-2-

32. Determine whether the lines tangent to the curve at the x -intercepts of the curve are parallel. Show the analysis that leads to your conclusion.

Part B

1 point is earned for correctly using $y = 0$

$$\text{If } y = 0, x^2 = 27, x = \pm 3\sqrt{3}$$

1 point is earned for correctly finding y' at x -intercepts

$$\text{at } x = 3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{3\sqrt{3}} = -2$$

$$\text{at } x = -3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{-3\sqrt{3}} = -2$$

1 point is earned for the correct analysis and conclusion

Tangent lines at x -intercepts are parallel

Note: for solution at y -intercepts, eligible for 2nd and 3rd points



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for correctly using $y = 0$

$$\text{If } y = 0, x^2 = 27, x = \pm 3\sqrt{3}$$

1 point is earned for correctly finding y' at x -intercepts

$$\text{at } x = 3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{3\sqrt{3}} = -2$$

$$\text{at } x = -3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{-3\sqrt{3}} = -2$$

1 point is earned for the correct analysis and conclusion

Tangent lines at x -intercepts are parallel

Note: for solution at y -intercepts, eligible for 2nd and 3rd points

2-2-

33. Consider the curve defined by $-8x^2 + 5xy + y^3 = -149$.

(a) Find $\frac{dy}{dx}$.

(b) Write an equation for the line tangent to the curve at the point $(4, -1)$.

(c) There is a number k so that the point $(4.2, k)$ is on the curve. Using the tangent line found in part (b), approximate the value of k .

(d) Write an equation that can be solved to find the actual value of k so that the point $(4.2, k)$ is on the curve.

(e) Solve the equation found in part (d) for the value of k .

Part A

2 points are earned for the correct implicit differentiation

<-1> each error

$$-16x + 5y + 5x \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x - 5y}{5x + 3y^2}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct implicit differentiation

<-1> each error

$$-16x + 5y + 5x \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

2-2-

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x - 5y}{5x + 3y^2}$$

Part B

1 point is earned for correctly evaluating $\frac{dy}{dx}$ at $(4, -1)$

$$\left. \frac{dy}{dx} \right|_{(4,-1)} = \frac{64+5}{20+3} = 3$$

1 point is earned for the correct equation of tangent line

$$y + 1 = 3(x - 4)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly evaluating $\frac{dy}{dx}$ at $(4, -1)$

$$\left. \frac{dy}{dx} \right|_{(4,-1)} = \frac{64+5}{20+3} = 3$$

1 point is earned for the correct equation of tangent line

$$y + 1 = 3(x - 4)$$

Part C

1 point is earned for correctly using $x = 4.2$ in tangent line equation

$$y + 1 = 3(4.2 - 4)$$

1 point is earned for the correct solution

$$y = -1$$

$$y = -0.4$$

$$k \approx -0.4$$

2-2-



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly using $x = 4.2$ in tangent line equation

$$y + 1 = 3(4.2 - 4)$$

1 point is earned for the correct solution

$k = -0.4$

$$y = -0.4$$

$$k \approx -0.4$$

Part D

1 point is earned for correctly using $x = 4.2$ in equation of curve

$$-8(4.2)^2 + 5(4.2)k + k^3 = -149$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for correctly using $x = 4.2$ in equation of curve

$$-8(4.2)^2 + 5(4.2)k + k^3 = -149$$

Part E

1 point is earned for the correct answer

$$k = -0.373$$



0	1
---	---

2-2-

The student response earns all of the following points:

1 point is earned for the correct answer

$$k = -0.373$$

34. Consider the curve defined by $-8x^2 + 5xy + y^3 = -149$.

Write an equation for the line tangent to the curve at the point $(4, -1)$.

Part B

1 point is earned for correctly evaluating $\frac{dy}{dx}$ at $(4, -1)$

$$\left. \frac{dy}{dx} \right|_{(4, -1)} = \frac{64+5}{20+3} = 3$$

1 point is earned for the correct equation of tangent line

$$y + 1 = 3(x - 4)$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly evaluating $\frac{dy}{dx}$ at $(4, -1)$

$$\left. \frac{dy}{dx} \right|_{(4, -1)} = \frac{64+5}{20+3} = 3$$

1 point is earned for the correct equation of tangent line

$$y + 1 = 3(x - 4)$$

2-2-

35. Consider the curve given by $x^2 + 4y^2 = 7 + 3xy$.

(a) Show that $\frac{dy}{dx} = \frac{3y-2x}{8y-3x}$.

(b) Show that there is a point P with x -coordinate 3 at which the line tangent to the curve at P is horizontal. Find the y -coordinate of P .

(c) Find the value of $\frac{d^2y}{dx^2}$ at the point P found in part (b). Does the curve have a local maximum, a local minimum, or neither at the point P ? Justify your answer.

Part A

1 point is earned for implicit differentiation

1 point is earned for solving for y'

$$\begin{aligned} 2x + 8yy' &= 3y + 3xy' \\ (8y - 3x)y' &= 3y - 2x \\ y' &= \frac{3y-2x}{8y-3x} \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for implicit differentiation

1 point is earned for solving for y'

$$\begin{aligned} 2x + 8yy' &= 3y + 3xy' \\ (8y - 3x)y' &= 3y - 2x \\ y' &= \frac{3y-2x}{8y-3x} \end{aligned}$$

Part B

1 point is earned for: $\frac{dy}{dx} = 0$

1 point is earned for showing slope is 0 at (3, 2)

1 point is earned for showing (3, 2) lies on curve

2-2-

$$\frac{3y - 2x}{8y - 3x} = 0; 3y - 2x = 0$$

When $x = 3$, $3y = 6$

$$y = 2$$

$$3^2 + 4 \cdot 2^2 = 25 \text{ and } 7 + 3 \cdot 3 \cdot 2 = 25$$

Therefore, $P = (3, 2)$ is on the curve and the slope is 0 at this point.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{dy}{dx} = 0$

1 point is earned for showing slope is 0 at $(3, 2)$

1 point is earned for showing $(3, 2)$ lies on curve

$$\frac{3y - 2x}{8y - 3x} = 0; 3y - 2x = 0$$

When $x = 3$, $3y = 6$

$$y = 2$$

$$3^2 + 4 \cdot 2^2 = 25 \text{ and } 7 + 3 \cdot 3 \cdot 2 = 25$$

Therefore, $P = (3, 2)$ is on the curve and the slope is 0 at this point.

Part C

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for the value of $\frac{d^2y}{dx^2}$ at $(3, 2)$

1 point is earned for the conclusion with justification

2-2-

$$\frac{d^2y}{dx^2} = \frac{(8y-3x)(3y'-2)-(3y-2x)(8y'-3)}{(8y-3x)^2}$$

$$\text{At } P = (3, 2), \frac{d^2y}{dx^2} = \frac{(16-9)(-2)}{(16-9)^2} = -\frac{2}{7}.$$

Since $y' = 0$ and $y'' < 0$ at P , the curve has a local maximum at P .



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for the value of $\frac{d^2y}{dx^2}$ at $(3, 2)$

1 point is earned for the conclusion with justification

$$\frac{d^2y}{dx^2} = \frac{(8y-3x)(3y'-2)-(3y-2x)(8y'-3)}{(8y-3x)^2}$$

$$\text{At } P = (3, 2), \frac{d^2y}{dx^2} = \frac{(16-9)(-2)}{(16-9)^2} = -\frac{2}{7}.$$

Since $y' = 0$ and $y'' < 0$ at P , the curve has a local maximum at P .

2-2-

36. Consider the curve given by $x^2 + 4y^2 = 7 + 3xy$.

Show that there is a point P with x -coordinates 3 at which the line tangent to the curve at P is horizontal.
Find the y -coordinate of P .

Part B

1 point is earned for: $\frac{dy}{dx} = 0$

1 point is earned for shows slope is 0 at (3, 2)

1 point is earned for shows (3, 2) lies on curve

$$\frac{3y-2x}{8y-3x} = 0; 3y - 2x = 0$$

When $x = 3$, $3y = 6$

$$y = 2$$

$$3^2 + 4 \cdot 2^2 = 25 \text{ and } 7 + 3 \cdot 3 \cdot 2 = 25$$

Therefore, $P = (3, 2)$ is on the curve and the slope is 0 at this point.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{dy}{dx} = 0$

1 point is earned for shows slope is 0 at (3, 2)

1 point is earned for shows (3, 2) lies on curve

$$\frac{3y-2x}{8y-3x} = 0; 3y - 2x = 0$$

When $x = 3$, $3y = 6$

$$y = 2$$

$$3^2 + 4 \cdot 2^2 = 25 \text{ and } 7 + 3 \cdot 3 \cdot 2 = 25$$

Therefore, $P = (3, 2)$ is on the curve and the slope is 0 at this point.

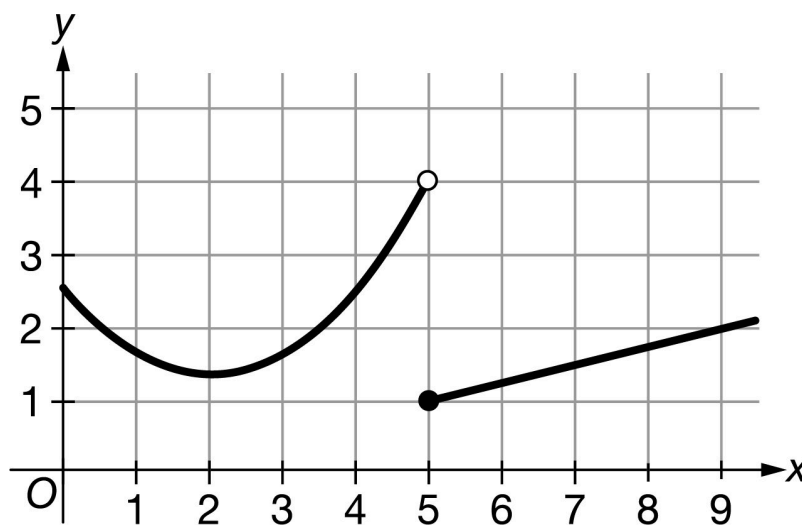
2-2-

37. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

x	3	3.5	4	4.5	5	5.5	6
$f(x)$	1.653	2.02	2.533	3.193	1	1.125	1.25

The graph of the function f and a table of selected values of $f(x)$ are shown above. The graph of f has a horizontal tangent line at $x = 2$, is concave up for $0 < x < 5$, and is linear for $x \geq 5$.

- (a) Approximate the value of $f'(4.5)$ using data from the table. Show the computations that lead to your answer.
- (b) Is there a value of x , for $0 < x < 5$, such that $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} = 0$? Give a reason for your answer.
- (c) For each of the following limits, find the value or explain why it does not exist.
- $\lim_{h \rightarrow 0^+} \frac{f(5+h)-f(5)}{h}$
 - $\lim_{h \rightarrow 0^-} \frac{f(5+h)-f(5)}{h}$
 - $\lim_{h \rightarrow 0} \frac{f(5+h)-f(5)}{h}$

Part A

2-2-

The first point is earned for a correct difference quotient over a subinterval of $3 \leq x < 5$ that contains $x = 4.5$. A response using $(5, 1)$ as the right endpoint earns only 1 point for knowing to use a difference quotient; this response is not eligible for the second point because $x = 5$ is not in the relevant portion of the domain. The second point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ difference quotient
- ☐ approximation

Solution:

$$f'(4.5) \approx \frac{f(4.5) - f(4)}{4.5 - 4} = \frac{3.193 - 2.533}{4.5 - 4} = \frac{0.660}{0.5} = 1.320$$

Part B

The first point is for recognizing that the given expression is the derivative of f . The second point is for the correct answer (there is a value of x , for $0 < x < 5$) and communicating the connection between the derivative and the slope of the tangent line at $x = 2$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$
- ☐ answer with reason

Solution:

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$$

2-2-

$f'(x) = 0$ at a point where the line tangent to the graph of f has slope 0. Since the graph of f has a horizontal tangent at $x = 2$, there is a value of x , for $0 < x < 5$, such that $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = 0$.

Part C

To earn the first point, the response must report the slope of the linear section of the graph. To earn the second point, the response must explain that the difference in the numerator of the expression will approach 3 as h approaches 0.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{h \rightarrow 0^+} \frac{f(5+h) - f(5)}{h}$
- ☐ $\lim_{h \rightarrow 0^-} \frac{f(5+h) - f(5)}{h}$ does not exist with explanation
- ☐ $\lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h}$ does not exist with explanation

Solution:

$$\lim_{h \rightarrow 0^+} \frac{f(5+h) - f(5)}{h} = \frac{1}{4}$$

$$\lim_{h \rightarrow 0^-} \frac{f(5+h) - f(5)}{h} \text{ does not exist because the numerator approaches 3 as the denominator approaches 0.}$$

$$\lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h} \text{ does not exist because the left-hand limit in (ii) does not exist.}$$

—OR—

Because f is not continuous at $x = 5$ due to a jump discontinuity, f is not differentiable at $x = 5$. Since

$$\lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h} \text{ is the definition of the derivative of } f \text{ at } x = 5, \lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h} \text{ does not exist.}$$

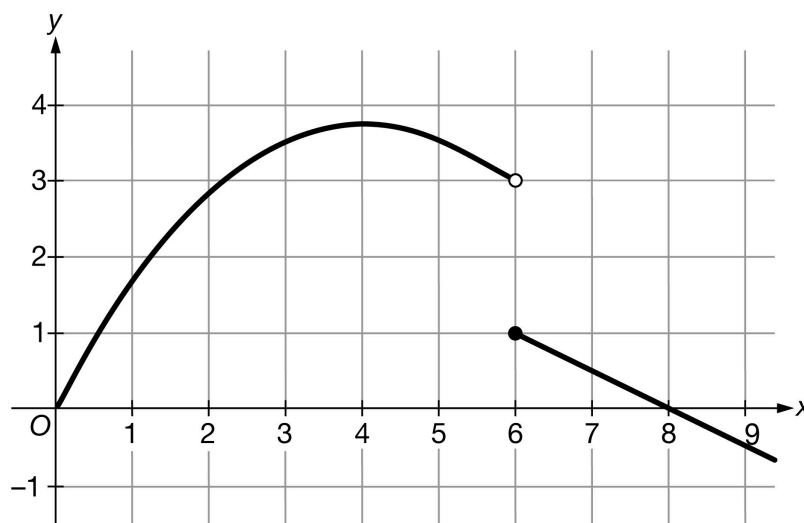
2-2-

38. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

x	4	4.5	5	5.5	6	6.5	7
$f(x)$	3.654	3.600	3.455	3.244	1	0.75	0.5

The graph of the function f and a table of selected values of $f(x)$ are shown above. The graph of f has a horizontal tangent line at $x = 4$, is concave down for $0 < x < 6$, and is linear for $x \geq 6$.

- (a) Approximate the value of $f'(5.5)$ using data from the table. Show the computations that lead to your answer.
- (b) Is there a value of x , for $0 < x < 6$, such that $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} = 0$? Give a reason for your answer.
- (c) For each of the following limits, find the value or explain why it does not exist.
- $\lim_{h \rightarrow 0^+} \frac{f(6+h)-f(6)}{h}$
 - $\lim_{h \rightarrow 0^-} \frac{f(6+h)-f(6)}{h}$
 - $\lim_{h \rightarrow 0} \frac{f(6+h)-f(6)}{h}$

2-2-

Part A

The first point is earned for a correct difference quotient over a subinterval of $4 \leq x < 6$ that contains $x = 5.5$. A response using $(6, 1)$ as the right endpoint earns only 1 point for knowing to use a difference quotient; this response is not eligible for the second point because $x = 6$ is not in the relevant portion of the domain. The second point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ difference quotient
- ☐ approximation

Solution:

$$f'(5.5) \approx \frac{f(5.5) - f(5)}{5.5 - 5} = \frac{3.244 - 3.455}{0.5} = -0.422$$

Part B

The first point is for recognizing that the given expression is the derivative of f . The second point is for the correct answer (there is a value of x , for $0 < x < 6$) and communicating the connection between the derivative and the slope of the tangent line at $x = 4$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$
- ☐ answer with reason

Solution:

2-2-

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$$

$f'(x) = 0$ at a point where the line tangent to the graph of f has slope 0. Since the graph of f has a horizontal tangent at $x = 4$, there is a value of x , for $0 < x < 6$, such that

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = 0.$$

Part C

To earn the first point, the response must report the slope of the linear section of the graph. To earn the second point, the response must explain that the difference in the numerator of the expression will approach 2 as h approaches 0.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{h \rightarrow 0^+} \frac{f(6+h) - f(6)}{h}$
- ☐ $\lim_{h \rightarrow 0^-} \frac{f(6+h) - f(6)}{h}$ does not exist with explanation
- ☐ $\lim_{h \rightarrow 0} \frac{f(6+h) - f(6)}{h}$ does not exist with explanation

Solution:

(i) $\lim_{h \rightarrow 0^+} \frac{f(6+h) - f(6)}{h} = -\frac{1}{2}$

(ii) $\lim_{h \rightarrow 0^-} \frac{f(6+h) - f(6)}{h}$ does not exist because the numerator approaches 2 as the denominator approaches 0.

(iii) $\lim_{h \rightarrow 0} \frac{f(6+h) - f(6)}{h}$ does not exist because the left-hand limit in (ii) does not exist.

—OR—

Because f is not continuous at $x = 6$ due to a jump discontinuity, f is not differentiable at $x = 6$. Since

$\lim_{h \rightarrow 0} \frac{f(6+h) - f(6)}{h}$ is the definition of the derivative of f at $x = 6$, $\lim_{h \rightarrow 0} \frac{f(6+h) - f(6)}{h}$ does not exist.

2-2-

39.

x	1	1.1	1.2	1.3	1.4
$f'(x)$	8	10	12	13	14.5

The function f is twice differentiable for $x > 0$ with $f(1) = 15$ and $f'(1) = 20$. Values of f' , the derivative of f , are given for selected values of x in the table above.

- (a) Write an equation for the line tangent to the graph of f at $x = 1$. Use this line to approximate $f(1.4)$.
- (b) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_1^{1.4} f'(x) dx$. Use the approximation for $\int_1^{1.4} f'(x) dx$ to estimate the value of $f(1.4)$. Show the computations that lead to your answer.
- (c) Use Euler's method, starting at $x = 1$ with two steps of equal size, to approximate $f(1.4)$. Show the computations that lead to your answer.
- (d) Write the second-degree Taylor polynomial for f about $x = 1$. Use the Taylor polynomial to approximate $f(1.4)$.

Part A

1 point is earned for the tangent line

1 point is earned for the approximation

$$f(1) = 15, f'(1) = 8$$

An equation for the tangent line is

$$y = 15 + 8(x - 1).$$

$$f(1.4) \approx 15 + 8(1.4 - 1) = 18.2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for the approximation

$$f(1) = 15, f'(1) = 8$$

An equation for the tangent line is

$$y = 15 + 8(x - 1).$$

2-2-

$$f(1.4) \approx 15 + 8(1.4 - 1) = 18.2$$

Part B

1 point is earned for the midpoint Riemann sum

1 point is earned for the Fundamental Theorem of Calculus

1 point is earned for the answer

$$\int_1^{1.4} f'(x)dx \approx (0.2)(10) + (0.2)(13) = 4.6$$

$$f(1.4) = f(1) + \int_1^{1.4} f'(x)dx$$

$$f(1.4) \approx 15 + 4.6 = 19.6$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the midpoint Riemann sum

1 point is earned for the Fundamental Theorem of Calculus

1 point is earned for the answer

$$\int_1^{1.4} f'(x)dx \approx (0.2)(10) + (0.2)(13) = 4.6$$

$$f(1.4) = f(1) + \int_1^{1.4} f'(x)dx$$

$$f(1.4) \approx 15 + 4.6 = 19.6$$

Part C

1 point is earned for the Euler's method with two steps

1 point is earned for the answer

2-2-

$$f(1.2) \approx f(1) + (0.2)(8) = 16.6$$

$$f(1.4) \approx 16.6 + (0.2)(12) = 19.0$$



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the Euler's method with two steps

1 point is earned for the answer

$$f(1.2) \approx f(1) + (0.2)(8) = 16.6$$

$$f(1.4) \approx 16.6 + (0.2)(12) = 19.0$$

Part D

1 point is earned for the Taylor polynomial

1 point is earned for the approximation

$$T_2(x) = 15 + 8(x - 1) + \frac{20}{2!}(x - 1)^2$$

$$= 15 + 8(x - 1) + 10(x - 1)^2$$

$$f(1.4) \approx 15 + 8(1.4 - 1) + 10(1.4 - 1)^2 = 19.8$$



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the Taylor polynomial

1 point is earned for the approximation

$$T_2(x) = 15 + 8(x - 1) + \frac{20}{2!}(x - 1)^2$$

$$= 15 + 8(x - 1) + 10(x - 1)^2$$

$$f(1.4) \approx 15 + 8(1.4 - 1) + 10(1.4 - 1)^2 = 19.8$$

2-2-

2-2-

40. The function f is defined by $f(x) = \sqrt{25 - x^2}$ for $-5 \leq x \leq 5$.

(a) Find $f'(x)$.

(b) Write an equation for the line tangent to the graph of f at $x = -3$.

(c) Let g be the function defined by $g(x) = \begin{cases} f(x) & \text{for } -5 \leq x \leq -3 \\ x + 7 & \text{for } -3 < x \leq 5. \end{cases}$

Is g continuous at $x = -3$? Use the definition of continuity to explain your answer.

(d) Find the value of $\int_0^5 x\sqrt{25 - x^2} dx$.

Part A

2 points are earned for $f'(x)$

$$f'(x) = \frac{1}{2}(25 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{25 - x^2}}, \quad -5 < x < 5$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for $f'(x)$

$$f'(x) = \frac{1}{2}(25 - x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{25 - x^2}}, \quad -5 < x < 5$$

Part B

1 point is earned for $f'(-3)$

1 point is earned for the answer

$$f'(-3) = \frac{3}{\sqrt{25-9}} = \frac{3}{4}$$

$$f(-3) = \sqrt{25 - 9} = 4$$

An equation for the tangent line is $y = 4 + \frac{3}{4}(x + 3)$.

2-2-



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(-3)$

1 point is earned for the answer

$$f'(-3) = \frac{3}{\sqrt{25-9}} = \frac{3}{4}$$

$$f(-3) = \sqrt{25-9} = 4$$

An equation for the tangent line is $y = 4 + \frac{3}{4}(x + 3)$.

Part C

1 point is earned for considering one-sided limits

1 point is earned for the answer with explanation

$$\lim_{x \rightarrow -3^-} g(x) = \lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^-} \sqrt{25 - x^2} = 4$$

$$\lim_{x \rightarrow -3^+} g(x) = \lim_{x \rightarrow -3^+} (x + 7) = 4$$

Therefore, $\lim_{x \rightarrow -3} g(x) = 4$.

$$g(-3) = f(-3) = 4$$

So, $\lim_{x \rightarrow -3} g(x) = g(-3)$.

Therefore, g is continuous at $x = -3$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering one-sided limits

1 point is earned for the answer with explanation

2-2-

$$\lim_{x \rightarrow -3^-} g(x) = \lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^-} \sqrt{25 - x^2} = 4$$

$$\lim_{x \rightarrow -3^+} g(x) = \lim_{x \rightarrow -3^+} (x + 7) = 4$$

Therefore, $\lim_{x \rightarrow -3} g(x) = 4$.

$$g(-3) = f(-3) = 4$$

So, $\lim_{x \rightarrow -3} g(x) = g(-3)$.

Therefore, g is continuous at $x = -3$.

Part D

2 points are earned for antiderivative

1 point is earned for the answer

$$\text{Let } u = 25 - x^2 \Rightarrow du = -2x dx$$

$$\begin{aligned} \int_0^5 x \sqrt{25 - x^2} dx &= -\frac{1}{2} \int_{25}^0 \sqrt{u} du \\ &= \left[-\frac{1}{2} \cdot \frac{2}{3} u^{3/2} \right]_{u=25}^{u=0} \\ &= -\frac{1}{3} (0 - 125) = \frac{125}{3} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for antiderivative

1 point is earned for the answer

$$\text{Let } u = 25 - x^2 \Rightarrow du = -2x dx$$

$$\begin{aligned} \int_0^5 x \sqrt{25 - x^2} dx &= -\frac{1}{2} \int_{25}^0 \sqrt{u} du \\ &= \left[-\frac{1}{2} \cdot \frac{2}{3} u^{3/2} \right]_{u=25}^{u=0} \\ &= -\frac{1}{3} (0 - 125) = \frac{125}{3} \end{aligned}$$

2-2-

41. The function f is defined by $f(x) = \sqrt{25 - x^2}$ for $-5 \leq x \leq 5$.

Write an equation for the line tangent to the graph of f at $x = -3$.

Part B

1 point is earned for $f'(-3)$

1 point is earned for the answer

$$f'(-3) = \frac{3}{\sqrt{25-9}} = \frac{3}{4}$$

$$f(-3) = \sqrt{25-9} = 4$$

An equation for the tangent line is $y = 4 + \frac{3}{4}(x + 3)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(-3)$

1 point is earned for the answer

$$f'(-3) = \frac{3}{\sqrt{25-9}} = \frac{3}{4}$$

$$f(-3) = \sqrt{25-9} = 4$$

An equation for the tangent line is $y = 4 + \frac{3}{4}(x + 3)$.

42. What is $\lim_{h \rightarrow 0} \frac{\cos\left(\frac{3\pi}{2} + h\right) - \cos\left(\frac{3\pi}{2}\right)}{h}$?

(A) 1

(B) $\frac{\sqrt{2}}{2}$

(C) 0

(D) -1

(E) The limit does not exist.



2-2-

43. $\lim_{x \rightarrow 2} \frac{\ln(x+3) - \ln(5)}{x-2}$ is

(A) 0

(B) $\frac{1}{5}$

(C) $\frac{1}{2}$

(D) 1

(E) nonexistent

44. If $\lim_{h \rightarrow 0} \frac{\arcsin(a+h) - \arcsin(a)}{h} = 2$, which of the following could be the value of a ?

(A) $\frac{\sqrt{2}}{2}$

(B) $\frac{\sqrt{3}}{2}$

(C) $\sqrt{3}$

(D) $\frac{1}{2}$

(E) 2

45. An equation of the line tangent to the graph of $f(x) = x(1 - 2x)^3$ at the point $(1, -1)$ is

(A) $y = -7x + 6$

(B) $y = -6x + 5$

(C) $y = -2x$

(D) $y = 2x - 3$

(E) $y = 7x - 8$

2-2-

46. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be an increasing function with $f(0) = 3$. The derivative of f is given by $f'(x) = \cos(\pi x) + x^4 + 6$.

- (a) Find $f''(-3)$.
- (b) Write an equation for the line tangent to the graph of $y = (f(x))^2$ at $x = 0$.
- (c) Let g be the function defined by $g(x) = f(\sqrt{2x^2 + 7})$. Find $g'(3)$.
- (d) Let h be the inverse function of f . Find $h'(3)$.

Part A

The second point cannot be earned without the first point. The second point does not require a simplified answer; trigonometric function values do not need to be evaluated. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ chain rule
- ☐ answer

Solution:

$$f''(x) = -\pi \sin(\pi x) + 4x^3$$

$$f''(-3) = -\pi \sin(-3\pi) + 4(-3)^3 = -108$$

Part B

2-2-

The second point may be earned based on a derivative expression that includes application of power rule and chain rule with a maximum of one computational error. Trigonometric function values do not need to be evaluated. The third point may be earned if response contains a slope that is consistent with the derivative expression. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response does not accurately include any of the criteria below.

- ☐ $\frac{d}{dx}((f(x))^2)$
- ☐ slope
- ☐ equation for tangent line

Solution:

At $x = 0$, $y = (f(0))^2 = 3^2 = 9$.

$$\frac{d}{dx}((f(x))^2) = 2f(x)f'(x)$$

$$\left. \frac{d}{dx}((f(x))^2) \right|_{x=0} = 2f(0)f'(0) = 2(3)(\cos 0 + 6) = 6(\cos 0 + 6) = 42$$

An equation for the tangent line is $y = 9 + 42x$.

Part C

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. Trigonometric function values do not need to be evaluated. A simplified answer is not required. Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

2-2-

$$g'(x) = f'(\sqrt{2x^2 + 7}) \cdot \frac{4x}{2\sqrt{2x^2 + 7}} = f'(\sqrt{2x^2 + 7}) \cdot \frac{2x}{\sqrt{2x^2 + 7}}$$

$$g'(3) = f'(\sqrt{2 \cdot 3^2 + 7}) \cdot \frac{2 \cdot 3}{\sqrt{2 \cdot 3^2 + 7}} = f'(5) \cdot \frac{6}{5} = (\cos(5\pi) + 631) \cdot \frac{6}{5} = (630) \frac{6}{5} = 756$$

Part D

Substitution of function values is required. Trigonometric function values do not need to be evaluated. Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes the criteria below.

☐ answer

Solution:

Because $f(0) = 3$, $h(3) = 0$.

$$h'(3) = \frac{1}{f'(h(3))} = \frac{1}{f'(0)} = \frac{1}{(\cos 0 + 6)} = \frac{1}{7}$$

2-2-

47. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be an increasing function with $f(0) = 2$. The derivative of f is given by $f'(x) = \sin(\pi x) + x^2 + 3$.

- (a) Find $f''(-2)$.
- (b) Write an equation for the line tangent to the graph of $y = \frac{1}{f(x)}$ at $x = 0$.
- (c) Let g be the function defined by $g(x) = f(\sqrt{3x^2 + 4})$. Find $g'(2)$.
- (d) Let h be the inverse function of f . Find $h'(2)$.

Part A

The second point cannot be earned without the first point. The second point does not require a simplified answer; trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ chain rule
- ☐ answer

Solution:

$$f''(x) = \pi \cos(\pi x) + 2x$$

$$f''(-2) = \pi \cos(-2\pi) + 2(-2) = \pi - 4$$

2-2-

Part B

The second point may be earned based on a derivative expression that includes application of power rule and chain rule with a maximum of one computational error.

The third point may be earned if response contains a slope that is consistent with the derivative expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\frac{d}{dx} \left(\frac{1}{f(x)} \right)$
- ☐ slope
- ☐ equation for tangent line

Solution:

$$\text{At } x = 0, y = \frac{1}{f(0)} = \frac{1}{2}.$$

$$\frac{d}{dx} \left(\frac{1}{f(x)} \right) = \frac{d}{dx} (f(x))^{-1} = -(f(x))^{-2} f'(x) = -\frac{f'(x)}{(f(x))^2}$$

$$\left. \frac{d}{dx} \left(\frac{1}{f(x)} \right) \right|_{x=0} = -\frac{f'(0)}{(f(0))^2} = -\frac{3}{4}$$

An equation for the tangent line is $y = \frac{1}{2} - \frac{3}{4}x$.

Part C

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

2-2-

The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$g'(x) = f'(\sqrt{3x^2 + 4}) \cdot \frac{6x}{2\sqrt{3x^2 + 4}} = f'(\sqrt{3x^2 + 4}) \cdot \frac{3x}{\sqrt{3x^2 + 4}}$$

$$g'(2) = f'(\sqrt{3 \cdot 2^2 + 4}) \cdot \frac{3 \cdot 2}{\sqrt{3 \cdot 2^2 + 4}} = f'(4) \cdot \frac{3}{2} = 19 \cdot \frac{3}{2} = \frac{57}{2}$$

Part D

Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct value of $h'(2)$

Solution:

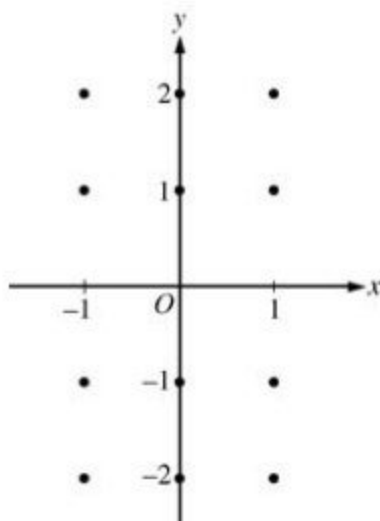
Because $f(0) = 2$, $h(2) = 0$.

$$h'(2) = \frac{1}{f'(h(2))} = \frac{1}{f'(0)} = \frac{1}{3}$$

2-2-

48. Consider the differential equation $\frac{dy}{dx} = -\frac{2x}{y}$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



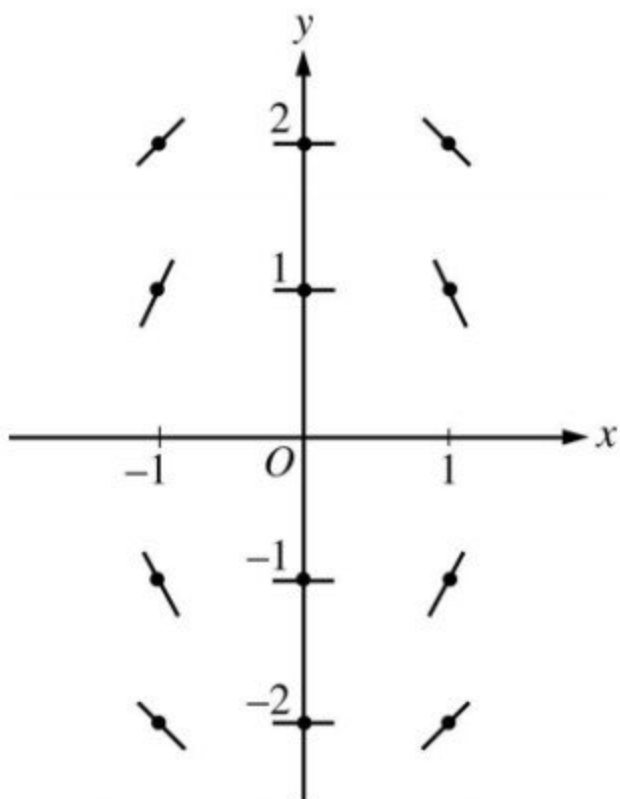
- (b) Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(1) = -1$. Write an equation for the line tangent to the graph of f at $(1, -1)$ and use it to approximate $f(1.1)$.
- (c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(1) = -1$.

Part A

1 point is earned for the zero slopes

1 point is earned for the nonzero slopes

2-2-



✓

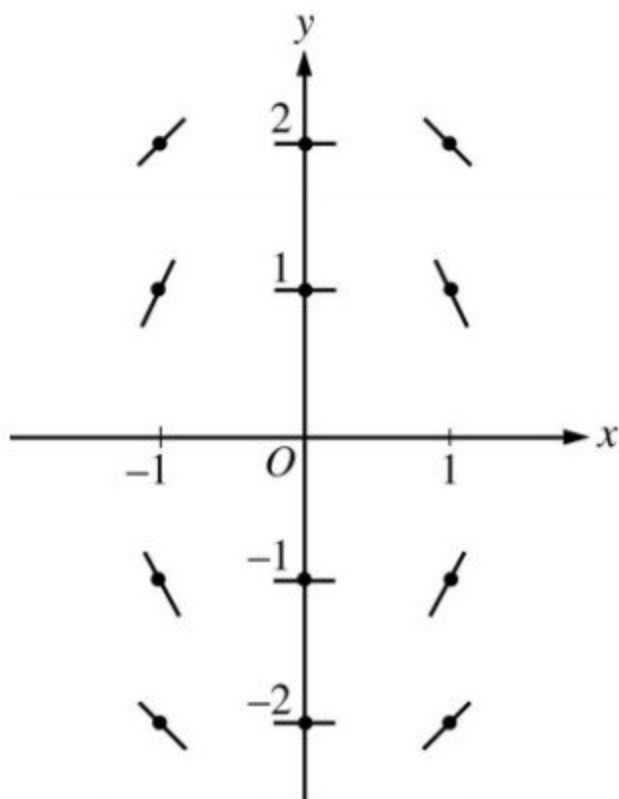
0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the zero slopes

1 point is earned for the nonzero slopes

2-2-

**Part B**

1 point is earned for the equation of the tangent line

1 point is earned for the approximation for $f(1.1)$

The line tangent to f at $(1, -1)$ is $y + 1 = 2(x - 1)$. Thus, $f(1.1)$ is approximately -0.8 .



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the equation of the tangent line

1 point is earned for the approximation for $f(1.1)$

The line tangent to f at $(1, -1)$ is $y + 1 = 2(x - 1)$. Thus, $f(1.1)$ is approximately -0.8 .

Part C

1 point is earned for the separates variables

2-2-

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for y

$$\frac{dy}{dx} = -\frac{2x}{y}$$

$$ydy = -2xdx$$

$$\frac{y^2}{2} = -x^2 + C$$

$$\frac{1}{2} = -1 + C; C = \frac{3}{2}$$

$$y^2 = -2x^2 + 3$$

Since the particular solution goes through (1, -1), y must be negative. Thus the particular solution is

$$y = -\sqrt{3 - 2x^2}.$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for uses initial condition

1 point is earned for solves for y

$$\frac{dy}{dx} = -\frac{2x}{y}$$

$$ydy = -2xdx$$

$$\frac{y^2}{2} = -x^2 + C$$

$$\frac{1}{2} = -1 + C; C = \frac{3}{2}$$

$$y^2 = -2x^2 + 3$$

Since the particular solution goes through (1, -1), y must be negative. Thus the particular solution is

$$y = -\sqrt{3 - 2x^2}.$$

2-2-

49. Consider the differential equation $\frac{dy}{dx} = -\frac{2x}{y}$.

Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(1) = -1$. Write an equation for the line tangent to the graph of f at $(1, -1)$ and use it to approximate $f(1.1)$.

Part B

1 point is earned for the equation of the tangent line

1 point is earned for the approximation for $f(1.1)$

The line tangent to f at $(1, -1)$ is $y + 1 = 2(x - 1)$. Thus, $f(1.1)$ is approximately -0.8 .



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the equation of the tangent line

1 point is earned for the approximation for $f(1.1)$

The line tangent to f at $(1, -1)$ is $y + 1 = 2(x - 1)$. Thus, $f(1.1)$ is approximately -0.8 .

2-2-

50. Consider the differential equation $\frac{dy}{dx} = \frac{3-x}{y}$.

(a) Let $y = f(x)$ be the particular solution to the given differential equation for $1 < x < 5$ such that the line $y = -2$ is tangent to the graph of f . Find the x -coordinate of the point of tangency, and determine whether f has a local maximum, local minimum, or neither at this point. Justify your answer.

(b) Let $y = g(x)$ be the particular solution to the given differential equation for $-2 < x < 8$, with the initial condition $g(6) = -4$. Find $y = g(x)$.

Part A

1 point is earned for: $x = 3$

1 point is earned for the local minimum

1 point is earned for the justification

$$\frac{dy}{dx} = 0 \text{ when } x = 3$$

$$\left. \frac{d^2y}{dx^2} \right|_{(3,-2)} = \left. \frac{-y-y'(3-x)}{y^2} \right|_{(3,-2)} = \frac{1}{2},$$

so f has a local minimum at this point.

or

Because f is continuous for $1 < x < 5$, there is an interval containing $x = 3$ on which $y < 0$. On this interval, $\frac{dy}{dx}$ is negative to the left of $x = 3$ and $\frac{dy}{dx}$ is positive to the right of $x = 3$. Therefore f has a local minimum at $x = 3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $x = 3$

1 point is earned for the local minimum

1 point is earned for the justification

$$\frac{dy}{dx} = 0 \text{ when } x = 3$$

$$\left. \frac{d^2y}{dx^2} \right|_{(3,-2)} = \left. \frac{-y-y'(3-x)}{y^2} \right|_{(3,-2)} = \frac{1}{2},$$

2-2-

so f has a local minimum at this point.

or

Because f is continuous for $1 < x < 5$, there is an interval containing $x = 3$ on which $y < 0$. On this interval, $\frac{dy}{dx}$ is negative to the left of $x = 3$ and $\frac{dy}{dx}$ is positive to the right of $x = 3$. Therefore f has a local minimum at $x = 3$.

Part B

1 point is earned for the separates variables

1 point is earned for the antiderivative of dy term

1 point is earned for the antiderivative of dx term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $g(6) = -4$

1 point is earned for solving for y

$$y \, dy = (3 - x) \, dx$$

$$\frac{1}{2}y^2 = 3x - \frac{1}{2}x^2 + C$$

$$8 = 18 - 18 + C; C = 8$$

$$y^2 = 6x - x^2 + 16$$

$$y = -\sqrt{6x - x^2 + 16}$$



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivative of dy term

1 point is earned for the antiderivative of dx term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $g(6) = -4$

1 point is earned for solving for y

2-2-

$$y \, dy = (3 - x)dx$$

$$\frac{1}{2}y^2 = 3x - \frac{1}{2}x^2 + C$$

$$8 = 18 - 18 + C; C = 8$$

$$y^2 = 6x - x^2 + 16$$

$$y = -\sqrt{6x - x^2 + 16}$$

2-2-

51. Consider the differential equation $\frac{dy}{dx} = \frac{3-x}{y}$.

Let $y = f(x)$ be the particular solution to the given differential equation for $1 < x < 5$ such that the line $y = -2$ is tangent to the graph of f . Find the x -coordinate of the point of tangency, and determine whether f has a local maximum, local minimum, or neither at this point. Justify your answer.

Part A

1 point is earned for: $x = 3$

1 point is earned for the local minimum

1 point is earned for the justification

$$\frac{dy}{dx} = 0 \text{ when } x = 3$$

$$\left. \frac{d^2y}{dx^2} \right|_{(3,-2)} = \left. \frac{-y-y'(3-x)}{y^2} \right|_{(3,-2)} = \frac{1}{2},$$

so f has a local minimum at this point.

or

Because f is continuous for $1 < x < 5$, there is an interval containing $x = 3$ on which $y < 0$. On this interval, $\frac{dy}{dx}$ is negative to the left of $x = 3$ and $\frac{dy}{dx}$ is positive to the right of $x = 3$. Therefore f has a local minimum at $x = 3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $x = 3$

1 point is earned for the local minimum

1 point is earned for the justification

$$\frac{dy}{dx} = 0 \text{ when } x = 3$$

$$\left. \frac{d^2y}{dx^2} \right|_{(3,-2)} = \left. \frac{-y-y'(3-x)}{y^2} \right|_{(3,-2)} = \frac{1}{2},$$

so f has a local minimum at this point.

or

2-2-

Because f is continuous for $1 < x < 5$, there is an interval containing $x = 3$ on which $y < 0$. On this interval, $\frac{dy}{dx}$ is negative to the left of $x = 3$ and $\frac{dy}{dx}$ is positive to the right of $x = 3$. Therefore f has a local minimum at $x = 3$.

2-2-

52. Solutions to the differential equation $\frac{dy}{dx} = xy^3$ also satisfy $\frac{d^2y}{dx^2} = y^3(1 + 3x^2y^2)$. Let $y = f(x)$ be a particular solution to the differential equation $\frac{dy}{dx} = xy^3$ with $f(1) = 2$.

- (a) Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 1$.
- (b) Use the tangent line equation from part (a) to approximate $f(1.1)$. Given that $f(x) > 0$ for $1 < x < 1.1$, is the approximation for $f(1.1)$ greater than or less than $f(1.1)$? Explain your reasoning.
- (c) Find the particular solution $y = f(x)$ with initial condition $f(1) = 2$.

Part A1 point is earned for $f'(1)$

1 point is earned for the answer

$$f'(1) = \left. \frac{dy}{dx} \right|_{(1,2)} = 8$$

An equation of the tangent line is $y = 2 + 8(x - 1)$.

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(1)$

1 point is earned for the answer

$$f'(1) = \left. \frac{dy}{dx} \right|_{(1,2)} = 8$$

An equation of the tangent line is $y = 2 + 8(x - 1)$.**Part B**

1 point is earned for approximation

1 point is earned for conclusion with explanation

$$f(1.1) \approx 2.8$$

Since $y = f(x) > 0$ on the interval $1 \leq x < 1.1$,

2-2-

$$\frac{d^2y}{dx^2} = y^3 (1 + 3x^2y^2) > 0 \text{ on this interval.}$$

Therefore on the interval $1 < x < 1.1$, the line tangent to the graph of $y = f(x)$ at $x = 1$ lies below the curve and the approximation 2.8 is less than $f(1.1)$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for approximation

1 point is earned for conclusion with explanation

$$f(1.1) \approx 2.8$$

Since $y = f(x) > 0$ on the interval $1 \leq x < 1.1$,

$$\frac{d^2y}{dx^2} = y^3 (1 + 3x^2y^2) > 0 \text{ on this interval.}$$

Therefore on the interval $1 < x < 1.1$, the line tangent to the graph of $y = f(x)$ at $x = 1$ lies below the curve and the approximation 2.8 is less than $f(1.1)$

Part C

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

2-2-

$$\frac{dy}{dx} = xy^3$$

$$\int \frac{1}{y^3} dy = \int x dx$$

$$-\frac{1}{2y^2} = \frac{x^2}{2} + C$$

$$-\frac{1}{2 \cdot 2^2} = \frac{1^2}{2} + C \Rightarrow C = -\frac{5}{8}$$

$$y^2 = \frac{1}{\frac{5}{4} - x^2}$$

$$f(x) = \frac{2}{\sqrt{5-4x^2}}, \frac{-\sqrt{5}}{2} < x < \frac{\sqrt{5}}{2}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

$$\frac{dy}{dx} = xy^3$$

$$\int \frac{1}{y^3} dy = \int x dx$$

$$-\frac{1}{2y^2} = \frac{x^2}{2} + C$$

$$-\frac{1}{2 \cdot 2^2} = \frac{1^2}{2} + C \Rightarrow C = -\frac{5}{8}$$

$$y^2 = \frac{1}{\frac{5}{4} - x^2}$$

$$f(x) = \frac{2}{\sqrt{5-4x^2}}, \frac{-\sqrt{5}}{2} < x < \frac{\sqrt{5}}{2}$$

2-2-

53. Solutions to the differential equation $dy/dx = xy^3$ also satisfy $d^2y/dx^2 = y^3(1 + 3x^2y^2)$. Let $y = f(x)$ be a particular solution to the differential equation $dy/dx = xy^3$ with $f(1) = 2$.

Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 1$.

Part A

1 point is earned for $f'(1)$

1 point is earned for the answer

$$f'(1) = \left. \frac{dy}{dx} \right|_{(1,2)} = 8$$

An equation of the tangent line is $y = 2 + 8(x-1)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(1)$

1 point is earned for the answer

$$f'(1) = \left. \frac{dy}{dx} \right|_{(1,2)} = 8$$

An equation of the tangent line is $y = 2 + 8(x-1)$.

2-2-

54. Consider the differential equation $dy/dx = e^y(3x^2 - 6x)$. Let $y = f(x)$ be the particular solution to the differential equation that passes through $(1, 0)$.
- (a) Write an equation for the line tangent to the graph of f at the point $(1, 0)$. Use the tangent line to approximate $f(1.2)$.
- (b) Find $y = f(x)$, the particular solution to the differential equation that passes through $(1, 0)$.

Part A

1 point is earned for $\frac{dy}{dx}$ at the point $(x, y) = (1, 0)$

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,0)} = e^0 (3 \cdot 1^2 - 6 \cdot 1) = -3$$

An equation for the tangent line is $y = -3(x - 1)$.

$$f(1.2) \approx -3(1.2 - 1) = -0.6$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{dy}{dx}$ at the point $(x, y) = (1, 0)$

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,0)} = e^0 (3 \cdot 1^2 - 6 \cdot 1) = -3$$

An equation for the tangent line is $y = -3(x - 1)$.

$$f(1.2) \approx -3(1.2 - 1) = -0.6$$

2-2-

Part B

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables.

$$\frac{dy}{e^y} = (3x^2 - 6x)dx$$

$$\int \frac{dy}{e^y} = \int (3x^2 - 6x)dx$$

$$-e^{-y} = x^3 - 3x^2 + C$$

$$-e^{-0} = 1^3 - 3 \cdot 1^2 + C \Rightarrow C = 1$$

$$-e^{-y} = x^3 - 3x^2 + 1$$

$$e^{-y} = -x^3 + 3x^2 - 1$$

$$-y = \ln(-x^3 + 3x^2 - 1)$$

$$y = -\ln(-x^3 + 3x^2 - 1)$$

Note: This solution is valid on an interval containing $x = 1$ for which $-x^3 + 3x^2 - 1 > 0$.

0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

2-2-

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables.

$$\frac{dy}{e^y} = (3x^2 - 6x)dx$$

$$\int \frac{dy}{e^y} = \int (3x^2 - 6x)dx$$

$$-e^{-y} = x^3 - 3x^2 + C$$

$$-e^{-0} = 1^3 - 3 \cdot 1^2 + C \Rightarrow C = 1$$

$$-e^{-y} = x^3 - 3x^2 + 1$$

$$e^{-y} = -x^3 + 3x^2 - 1$$

$$-y = \ln(-x^3 + 3x^2 - 1)$$

$$y = -\ln(-x^3 + 3x^2 - 1)$$

Note: This solution is valid on an interval containing $x = 1$ for which $-x^3 + 3x^2 - 1 > 0$.

2-2-

55. Consider the differential equation $dy/dx = e^y(3x^2 - 6x)$. Let $y = f(x)$ be the particular solution to the differential equation that passes through $(1, 0)$.

Write an equation for the line tangent to the graph of f at the point $(1, 0)$. Use the tangent line to approximate $f(1.2)$.

Part A

1 point is earned for $\frac{dy}{dx}$ at the point $(x, y) = (1, 0)$

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,0)} = e^0 (3 \cdot 1^2 - 6 \cdot 1) = -3$$

An equation for the tangent line is $y = -3(x - 1)$.

$$f(1.2) \approx -3(1.2 - 1) = -0.6$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{dy}{dx}$ at the point $(x, y) = (1, 0)$

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,0)} = e^0 (3 \cdot 1^2 - 6 \cdot 1) = -3$$

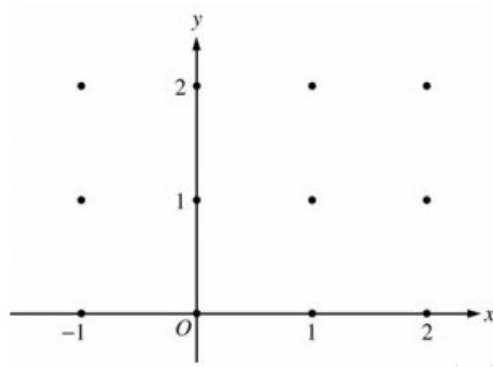
An equation for the tangent line is $y = -3(x - 1)$.

$$f(1.2) \approx -3(1.2 - 1) = -0.6$$

2-2-

56. Consider the differential equation $dy/dx = -xy^2/2$. Let $y = f(x)$ be the particular solution to this differential equation with the initial condition $f(-1) = 2$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



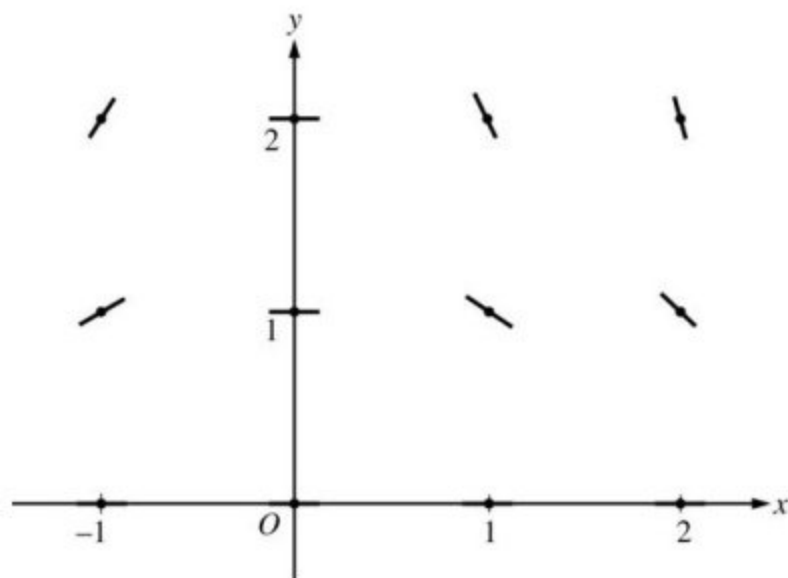
(b) Write an equation for the line tangent to the graph of f at $x = -1$.

(c) Find the solution $y = f(x)$ to the given differential equation with the initial condition $f(-1) = 2$.

Part A

1 point is earned for zero slopes

1 point is earned for nonzero slopes



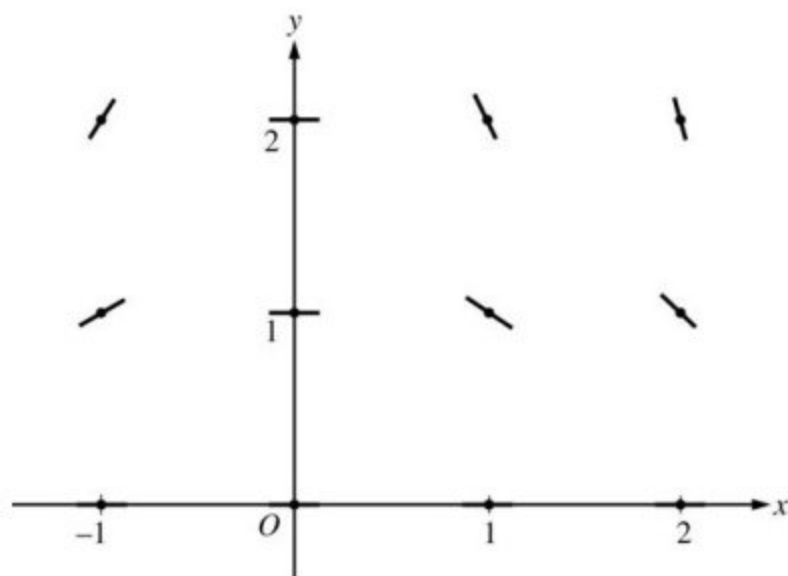
0	1	2
---	---	---

The student response earns all of the following points:

2-2-

1 point is earned for zero slopes

1 point is earned for nonzero slopes

**Part B**

1 point is earned for the equation

$$\text{Slope} = \frac{-(-1)4}{2} = 2$$

$$y - 2 = 2(x + 1)$$



0

1

The student response earns all of the following points:

1 point is earned for the equation

$$\text{Slope} = \frac{-(-1)4}{2} = 2$$

$$y - 2 = 2(x + 1)$$

Part C

1 point is earned if separates variables

2 points are earned for antiderivatives

2-2-

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: max 0/6 if no separation of variables

$$\frac{1}{y^2} dy = -\frac{x}{2} dx$$

$$-\frac{1}{y} = -\frac{x^2}{4} + C$$

$$-\frac{1}{2} = -\frac{1}{4} + C; C = -\frac{1}{4}$$

$$y = \frac{1}{\frac{x^2}{4} + \frac{1}{4}} = \frac{4}{x^2 + 1}$$



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned if separates variables

2 points are earned for antiderivatives

1 point is earned for constant of integration

1 point is earned if uses initial condition

1 point is earned if solves for y

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: max 0/6 if no separation of variables

$$\frac{1}{y^2} dy = -\frac{x}{2} dx$$

$$-\frac{1}{y} = -\frac{x^2}{4} + C$$

$$-\frac{1}{2} = -\frac{1}{4} + C; C = -\frac{1}{4}$$

$$y = \frac{1}{\frac{x^2}{4} + \frac{1}{4}} = \frac{4}{x^2 + 1}$$

2-2-

57. Consider the differential equation $dy/dx = -xy^2/2$. Let $y = f(x)$ be the particular solution to this differential equation with the initial condition $f(-1) = 2$.

Write an equation for the line tangent to the graph of f at $x = -1$.

Part B

1 point is earned for the equation

$$\text{slope} = \frac{-(-1)4}{2} = 2$$

$$y - 2 = 2(x + 1)$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the equation

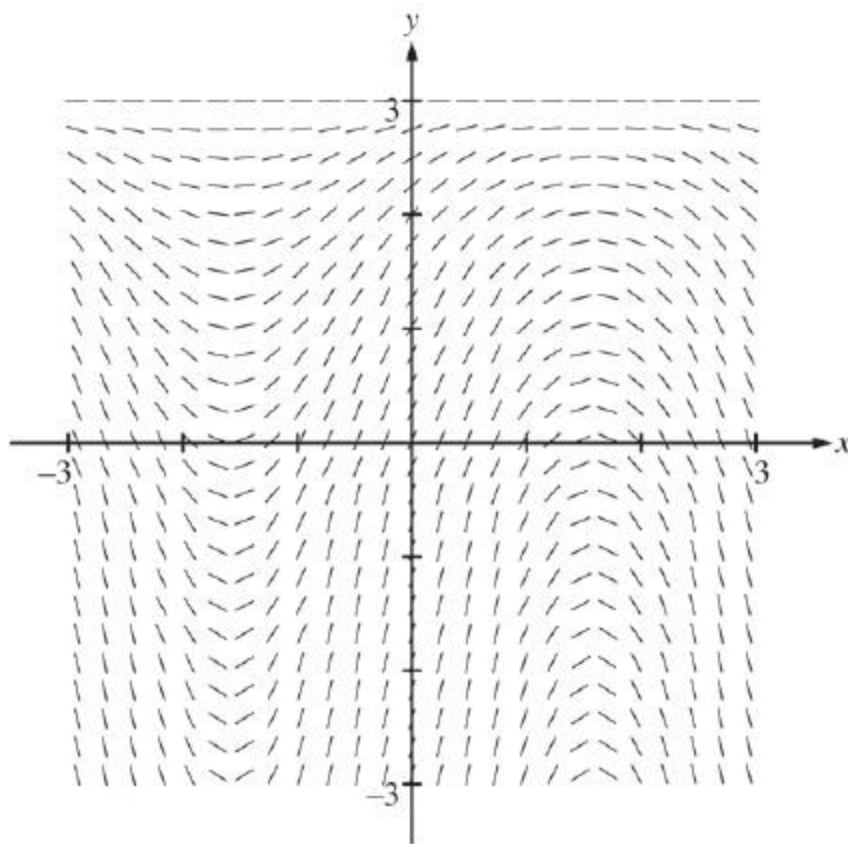
$$\text{slope} = \frac{-(-1)4}{2} = 2$$

$$y - 2 = 2(x + 1)$$

2-2-

58. Consider the differential equation $\frac{dy}{dx} = (3 - y) \cos x$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(0) = 1$. The function f is defined for all real numbers.

(a) A portion of the slope field of the differential equation is given below. Sketch the solution curve through the point $(0, 1)$.

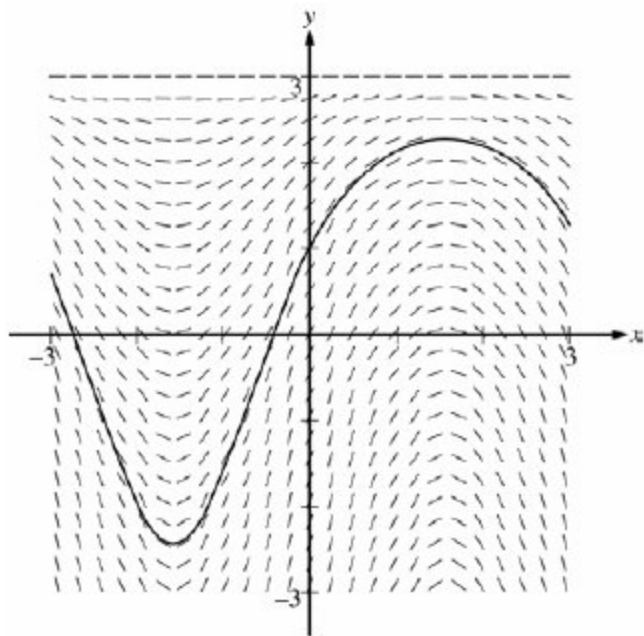


- (b) Write an equation for the line tangent to the solution curve in part (a) at the point $(0, 1)$. Use the equation to approximate $f(0.2)$.
- (c) Find $y = f(x)$, the particular solution to the differential equation with the initial condition $f(0) = 1$.

Part A

1 point is earned for solution curve

2-2-



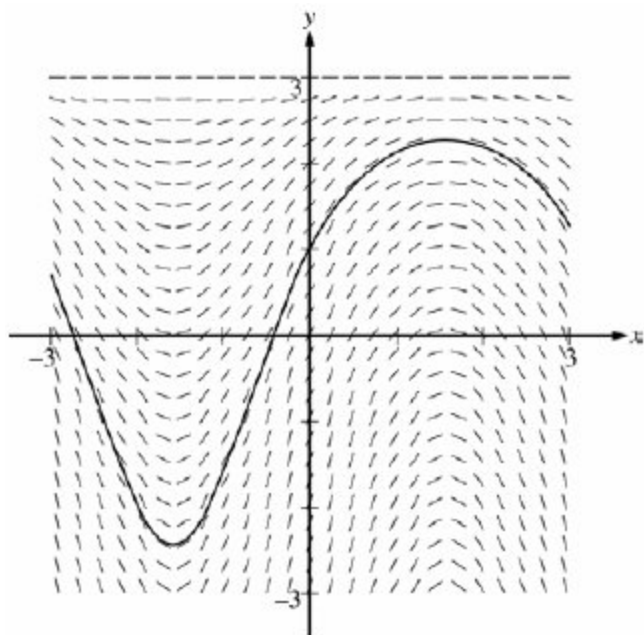
✓

0

1

The student response earns all of the following points:

1 point is earned for solution curve



Part B

1 point is earned for the tangent line equation

2-2-

1 point is earned for the approximation

$$\left. \frac{dy}{dx} \right|_{(x,y)=(0,1)} = 2 \cos 0 = 2$$

An equation for the tangent line is $y = 2x + 1$.

$$f(0.2) \approx 2(0.2) + 1 = 1.4$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left. \frac{dy}{dx} \right|_{(x,y)=(0,1)} = 2 \cos 0 = 2$$

An equation for the tangent line is $y = 2x + 1$.

$$f(0.2) \approx 2(0.2) + 1 = 1.4$$

Part C

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

2-2-

$$\frac{dy}{dx} = (3 - y) \cos x$$

$$\int \frac{dy}{3 - y} = \int \cos x dx$$

$$-\ln |3 - y| = \sin x + C$$

$$-\ln 2 = \sin 0 + C \Rightarrow C = -\ln 2$$

$$-\ln |3 - y| = \sin x - \ln 2$$

Because $y(0) = 1$, $y < 3$, so $|3 - y| = 3 - y$

$$3 - y = 2e^{-\sin x}$$

$$y = 3 - 2e^{-\sin x}$$

Note: this solution is valid for all real numbers.

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the separation of variables

2 points are earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

$$\frac{dy}{dx} = (3 - y) \cos x$$

$$\int \frac{dy}{3 - y} = \int \cos x dx$$

$$-\ln |3 - y| = \sin x + C$$

$$-\ln 2 = \sin 0 + C \Rightarrow C = -\ln 2$$

$$-\ln |3 - y| = \sin x - \ln 2$$

Because $y(0) = 1$, $y < 3$, so $|3 - y| = 3 - y$

2-2-

$$3 - y = 2e^{-\sin x}$$

$$y = 3 - 2e^{-\sin x}$$

Note: this solution is valid for all real numbers.

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

- 59.** Consider the differential equation $\frac{dy}{dx} = (3 - y) \cos x$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(0) = 1$. The function f is defined for all real numbers. Write an equation for the line tangent to the solution curve in part (a) at the point $(0, 1)$. Use the equation to approximate $f(0.2)$.

Part B

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left. \frac{dy}{dx} \right|_{(x,y)=(0,1)} = 2 \cos 0 = 2$$

An equation for the tangent line is $y = 2x + 1$

$$f(0.2) \approx 2(0.2) + 1 = 1.4$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line equation

1 point is earned for the approximation

$$\left. \frac{dy}{dx} \right|_{(x,y)=(0,1)} = 2 \cos 0 = 2$$

An equation for the tangent line is $y = 2x + 1$

$$f(0.2) \approx 2(0.2) + 1 = 1.4$$

2-2-

60. Let f be a function with $f(1) = 4$ such that for all points (x, y) on the graph of f the slope is given by $3x^2 + 1 / 2y$.
- (a) Find the slope of the graph of f at the point where $x = 1$.
 - (b) Write an equation for the line tangent to the graph of f at $x = 1$ and use it to approximate $f(1.2)$.
 - (c) Find $f(x)$ by solving the separable differential equation $dy/dx = 3x^2 + 1 / 2y$ with the initial condition $f(1) = 4$.
 - (d) Use your solution from part (c) to find $f(1.2)$.

Part A

1 point is earned for the answer

$$\frac{dy}{dx} = \frac{3x^2 + 1}{2y}$$

$$\left. \frac{dy}{dx} \right|_{\substack{x=1 \\ y=4}} = \frac{3 + 1}{2 \cdot 4} = \frac{4}{8} = \frac{1}{2}$$



0

1

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{dy}{dx} = \frac{3x^2 + 1}{2y}$$

$$\left. \frac{dy}{dx} \right|_{\substack{x=1 \\ y=4}} = \frac{3 + 1}{2 \cdot 4} = \frac{4}{8} = \frac{1}{2}$$

Part B

1 point is earned for the equation of tangent line

1 point is earned if uses equation to approximate $f(1.2)$

$$y - 4 = \frac{1}{2}(x - 1)$$

2-2-

$$f(1.2) - 4 \approx \frac{1}{2}(1.2 - 1)$$

$$f(1.2) \approx 0.1 + 4 = 4.1$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the equation of tangent line

1 point is earned if uses equation to approximate $f(1.2)$

$$y - 4 = \frac{1}{2}(x - 1)$$

$$f(1.2) - 4 \approx \frac{1}{2}(1.2 - 1)$$

$$f(1.2) \approx 0.1 + 4 = 4.1$$

Part C

1 point is earned if separates variables

1 point is earned for antiderivative of dy term

1 point is earned for antiderivative of dx term

1 point is earned if uses $y = 4$ when $x = 1$ to pick one function out of a family of functions

1 point is earned if solves for y

$$2y \, dy = (3x^2 + 1) \, dx$$

$$\int 2y \, dy = \int (3x^2 + 1) \, dx$$

$$y^2 = x^3 + x + C$$

$$4^2 = 1 + 1 + C$$

$$14 = C$$

2-2-

$$y^2 = x^3 + x + 14$$

$y = \sqrt{x^3 + x + 14}$ is branch with point (1, 4)

$$f(x) = \sqrt{x^3 + x + 14}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned if separates variables

1 point is earned for antiderivative of dy term

1 point is earned for antiderivative of dx term

1 point is earned if uses $y = 4$ when $x = 1$ to pick one function out of a family of functions

1 point is earned if solves for y

$$2y \, dy = (3x^2 + 1) \, dx$$

$$\int 2y \, dy = \int (3x^2 + 1) \, dx$$

$$y^2 = x^3 + x + C$$

$$4^2 = 1 + 1 + C$$

$$14 = C$$

$$y^2 = x^3 + x + 14$$

$y = \sqrt{x^3 + x + 14}$ is branch with point (1, 4)

$$f(x) = \sqrt{x^3 + x + 14}$$

Part D

2-2-

1 point is earned if answer, from student's solution to the given differential equation in (c)

$$f(1.2) = \sqrt{1.2^3 + 1.2 + 14} \approx 4.114$$



0	1
---	---

The student response earns all of the following points:

1 point is earned if answer, from student's solution to the given differential equation in (c)

$$f(1.2) = \sqrt{1.2^3 + 1.2 + 14} \approx 4.114$$

Let f be a function with $f(1) = 4$ such that for all points (x, y) on the graph of f the slope is given by $3x^2 + 1 / 2y$.

61. Find the slope of the graph of f at the point where $x = 1$.

Part A

1 point is earned for the answer

$$\frac{dy}{dx} = \frac{3x^2 + 1}{2y}$$

$$\left. \frac{dy}{dx} \right|_{\substack{x=1 \\ y=4}} = \frac{3 + 1}{2 \cdot 4} = \frac{4}{8} = \frac{1}{2}$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the answer

$$\frac{dy}{dx} = \frac{3x^2 + 1}{2y}$$

$$\left. \frac{dy}{dx} \right|_{\substack{x=1 \\ y=4}} = \frac{3 + 1}{2 \cdot 4} = \frac{4}{8} = \frac{1}{2}$$

2-2-

62. Write an equation for the line tangent to the graph of f at $x = 1$ and use it to approximate $f(1.2)$.
-

Part B

1 point is earned for the equation of tangent line

1 point is earned if uses equation to approximate $f(1.2)$

$$y - 4 = \frac{1}{2}(x - 1)$$

$$f(1.2) - 4 \approx \frac{1}{2}(1.2 - 1)$$

$$f(1.2) \approx 0.1 + 4 = 4.1$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the equation of tangent line

1 point is earned if uses equation to approximate $f(1.2)$

$$y - 4 = \frac{1}{2}(x - 1)$$

$$f(1.2) - 4 \approx \frac{1}{2}(1.2 - 1)$$

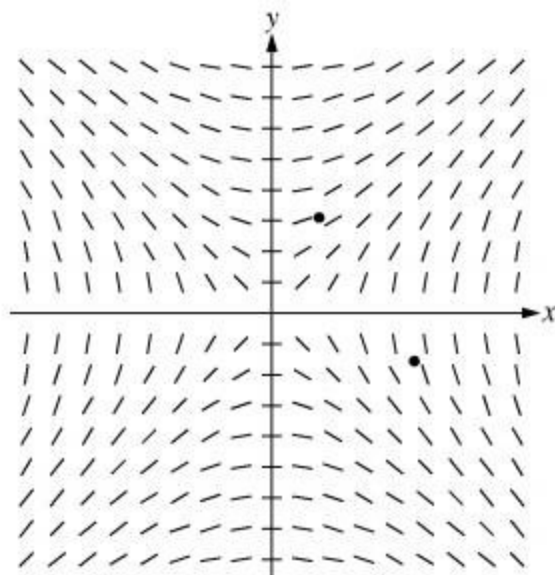
$$f(1.2) \approx 0.1 + 4 = 4.1$$

2-2-

63. Consider the differential equation $\frac{dy}{dx} = \frac{x}{y}$, where $y \neq 0$.

(a) The slope field for the given differential equation is shown below. Sketch the solution curve that passes through the point $(3, -1)$ and sketch the solution curve that passes through the point $(1, 2)$.

(Note: The points $(3, -1)$ and $(1, 2)$ are indicated in the figure.)



(b) Write an equation for the line tangent to the solution curve that passes through the point $(1, 2)$.

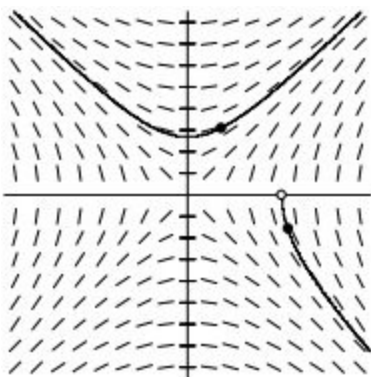
(c) Find the particular solution $y = f(x)$ to the differential equation with the initial condition $f(3) = -1$ and state its domain.

Part A

1 point is earned for the solution curve through $(3, -1)$

1 point is earned for the solution curve through $(1, 2)$

Curves must go through the indicated points, follow the given slope lines, and extend to the boundary of the slope field or the x -axis.



2-2-

✓

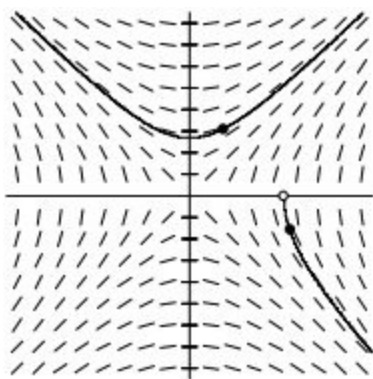
0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the solution curve through (3, -1)

1 point is earned for the solution curve through (1, 2)

Curves must go through the indicated points, follow the given slope lines, and extend to the boundary of the slope field or the x -axis.



Part B

1 point is earned for the equation of tangent line

$$\left. \frac{dy}{dx} \right|_{(1,2)} = \frac{1}{2}$$

An equation for the line tangent to the solution curve is $y - 2 = \frac{1}{2}(x - 1)$.

✓

0	1
---	---

The student response earns one of the following points:

1 point is earned for the equation of tangent line

2-2-

$$\left. \frac{dy}{dx} \right|_{(1,2)} = \frac{1}{2}$$

An equation for the line tangent to the solution curve is $y - 2 = \frac{1}{2}(x - 1)$.

Part C

1 point is earned for separate variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

1 point is earned for domain

$$y \, dy = x \, dx$$

$$\frac{1}{2}y^2 = \frac{1}{2}x^2 + A$$

$$y^2 = x^2 + C$$

$$C = -8$$

Since the particular solution goes through (3, -1), y must be negative.

$$y = -\sqrt{x^2 - 8} \text{ for } x > \sqrt{8}$$



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for separate variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

2-2-

1 point is earned for using initial condition

1 point is earned for solving for y

1 point is earned for domain

$$y \, dy = x \, dx$$

$$\frac{1}{2}y^2 = \frac{1}{2}x^2 + A$$

$$y^2 = x^2 + C$$

$$C = -8$$

Since the particular solution goes through (3, -1), y must be negative.

$$y = -\sqrt{x^2 - 8} \text{ for } x > \sqrt{8}$$

2-2-

64. Consider the differential equation $\frac{dy}{dx} = \frac{x}{y}$, where $y \neq 0$.

Write an equation for the line tangent to the solution curve that passes through the point (1,2).

Part B

1 point is earned for the equation of tangent line

$$\left. \frac{dy}{dx} \right|_{(1,2)} = \frac{1}{2}$$

An equation for the line tangent to the solution curve is $y - 2 = \frac{1}{2}(x - 1)$.



0	1
---	---

The student response earns all of the following points:

$$\left. \frac{dy}{dx} \right|_{(1,2)} = \frac{1}{2}$$

An equation for the line tangent to the solution curve is

$$y - 2 = \frac{1}{2}(x - 1).$$

2-2-

65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

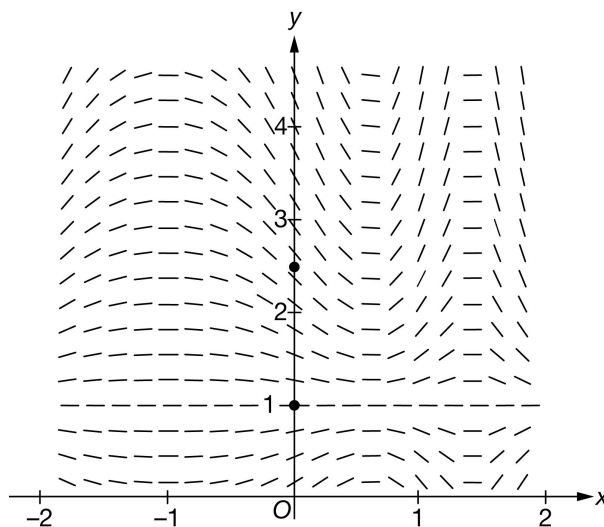
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = (y - 1)(x + 1) \cos(x^2 + 2x + \pi)$.

- (a) A portion of the slope field of the differential equation is given below. Sketch the solution curves through the points $(0, 1)$ and $(0, \frac{5}{2})$.



- (b) Write an equation for the line tangent to the solution curve in part (a) at the point $(0, \frac{5}{2})$.
- (c) Find the particular solution $y = f(x)$ to the differential equation with initial condition $f(0) = \frac{5}{2}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

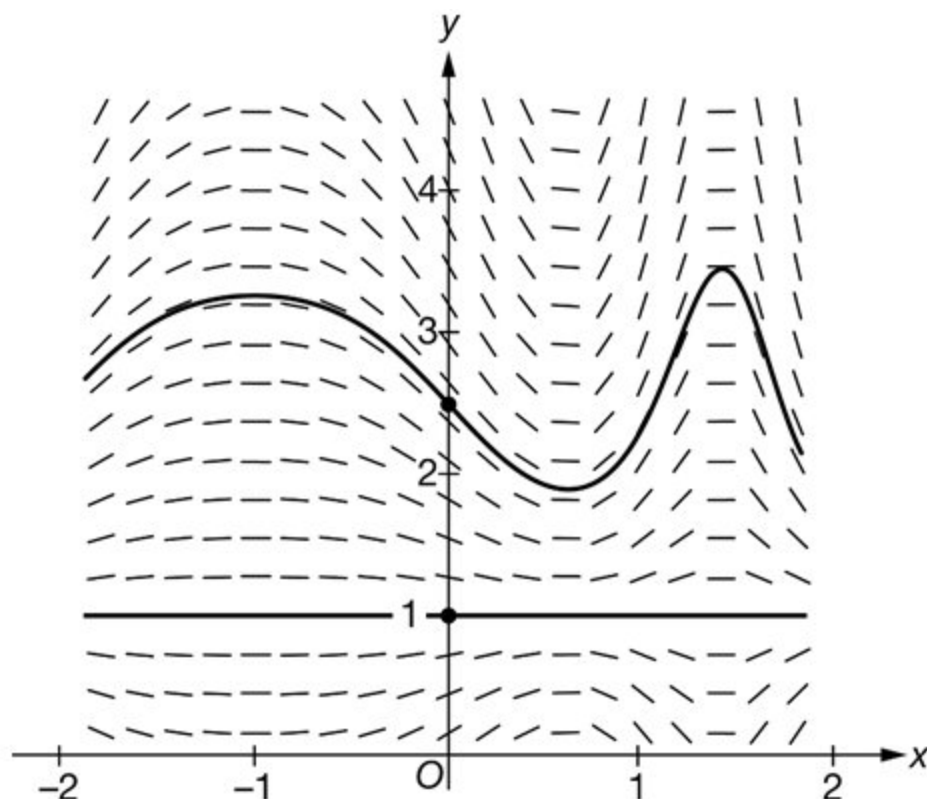


0	1	2
---	---	---

The student response accurately includes both of the criteria below.

2-2-

- ☐ curve through $(0, \frac{5}{2})$
- ☐ horizontal line through $(0, 1)$

Solution:**Part B**

The tangent line equation is not required to be in slope-intercept form.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The student response accurately includes a correct tangent line equation.

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(0, \frac{5}{2})} = \left(\frac{3}{2} \right) (1) (\cos \pi) = -\frac{3}{2}$$

An equation for the tangent line is $y = -\frac{3}{2}x + \frac{5}{2}$.

2-2-

Part C

Zero out of 6 points earned if no separation of variables.

At most 4 out of 6 points earned [1-1-1-1-0-0] if no constant of integration.

The sixth point requires an expression for y .

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response accurately includes all six of the criteria below.

- ☐ separation of variables
- ☐ dy antiderivative
- ☐ substitution
- ☐ dx antiderivative
- ☐ constant of integration and uses initial condition
- ☐ solves for y

Solution:

$$\frac{1}{y-1} dy = (x+1) \cos(x^2 + 2x + \pi) dx$$

$$\int \frac{1}{y-1} dy = \int (x+1) \cos(x^2 + 2x + \pi) dx$$

Let $u = x^2 + 2x + \pi$. Then $du = (2x + 2) dx = 2(x + 1) dx$.

$$\ln|y-1| = \int \frac{1}{2} \cos u du$$

$$\ln|y-1| = \frac{1}{2} \sin u + C = \frac{1}{2} \sin(x^2 + 2x + \pi) + C$$

$$\ln \frac{3}{2} = \frac{1}{2} \sin \pi + C = C$$

$$\ln|y-1| = \frac{1}{2} \sin(x^2 + 2x + \pi) + \ln \frac{3}{2}$$

$$|y-1| = e^{\frac{1}{2} \sin(x^2 + 2x + \pi) + \ln \frac{3}{2}} = \frac{3}{2} e^{\frac{1}{2} \sin(x^2 + 2x + \pi)}$$

2-2-

Because $y(0) = \frac{5}{2}$, $y > 1 \Rightarrow |y - 1| = y - 1$

$$y - 1 = \frac{3}{2} e^{\frac{1}{2} \sin(x^2 + 2x + \pi)}$$

$$y = 1 + \frac{3}{2} e^{\frac{1}{2} \sin(x^2 + 2x + \pi)}$$

2-2-

66. The function f satisfies the equation

$$f'(x) = f(x) + x + 1$$

and $f(0) = 2$. The Taylor series for f about $x = 0$ converges to $f(x)$ for all x .

- (a) Write an equation for the line tangent to the curve $y = f(x)$ at the point where $x = 0$.
 (b) Find $f''(0)$ and find the second-degree Taylor polynomial for f about $x = 0$.
 (c) Find the fourth-degree Taylor polynomial for f about $x = 0$.
 (d) Find $f^{(n)}(0)$, the n th derivative of f at $x = 0$, for $n \geq 2$. Use the Taylor series for f about $x = 0$ and the Taylor series for e^x about $x = 0$ to find a polynomial expression for $f(x) - 4e^x$.

Part A

1 point is earned for the tangent line equation

$$f'(0) = f(0) + 0 + 1 = 3$$

An equation for the tangent line is $y = 3x + 2$.



0	1
---	---

The student earns all of the following points:

1 point is earned for the tangent line equation

$$f'(0) = f(0) + 0 + 1 = 3$$

An equation for the tangent line is $y = 3x + 2$.

Part B

1 point is earned for $f''(0)$

1 point is earned for the second-degree Taylor polynomial

$$f''(x) = f'(x) + 1; f''(0) = f'(0) + 1 = 3 + 1 = 4$$

$$P_2(x) = 2 + 3x + \frac{4}{2!}x^2 = 2 + 3x + 2x^2$$



0	1	2
---	---	---

2-2-

The student earns all of the following points:

1 point is earned for $f''(0)$

1 point is earned for the second-degree Taylor polynomial

$$f''(x) = f'(x) + 1; f''(0) = f'(0) + 1 = 3 + 1 = 4$$

$$P_2(x) = 2 + 3x + \frac{4}{2!}x^2 = 2 + 3x + 2x^2$$

Part C

1 point is earned for $f'''(0)$ and $f^{(4)}(0)$

1 point is earned for the fourth-degree Taylor polynomial

$$f'''(x) = f''(x); f'''(0) = f''(0) = 4$$

$$f^{(4)}(x) = f'''(x); f^{(4)}(0) = f'''(0) = 4$$

$$\begin{aligned} P_4(x) &= 2 + 3x + \frac{4}{2!}x^2 + \frac{4}{3!}x^3 + \frac{4}{4!}x^4 \\ &= 2 + 3x + 2x^2 + \frac{2}{3}x^3 + \frac{1}{6}x^4 \end{aligned}$$



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for $f'''(0)$ and $f^{(4)}(0)$

1 point is earned for the fourth-degree Taylor polynomial

$$f'''(x) = f''(x); f'''(0) = f''(0) = 4$$

$$f^{(4)}(x) = f'''(x); f^{(4)}(0) = f'''(0) = 4$$

$$\begin{aligned} P_4(x) &= 2 + 3x + \frac{4}{2!}x^2 + \frac{4}{3!}x^3 + \frac{4}{4!}x^4 \\ &= 2 + 3x + 2x^2 + \frac{2}{3}x^3 + \frac{1}{6}x^4 \end{aligned}$$

Part D

2-2-

1 point is earned for the $f^{(n)}(0)$ for $n \geq 2$

1 point is earned for the Taylor series for f

1 point is earned for the Taylor series for e^x

1 point is earned for the polynomial expression

$$f^{(n)}(0) = 4 \text{ for } n \geq 2$$

$$f(x) = 2 + 3x + \frac{4}{2!}x^2 + \frac{4}{3!}x^3 + \frac{4}{4!}x^4 + \frac{4}{5!}x^5 + \dots$$

$$4e^x = 4 + 4x + \frac{4}{2!}x^2 + \frac{4}{3!}x^3 + \frac{4}{4!}x^4 + \frac{4}{5!}x^5 + \dots$$

Therefore, $f(x) - 4e^x = -2 - x$.



0	1	2	3	4
---	---	---	---	---

The student earns all of the following points:

1 point is earned for the $f^{(n)}(0)$ for $n \geq 2$

1 point is earned for the Taylor series for f

1 point is earned for the Taylor series for e^x

1 point is earned for the polynomial expression

$$f^{(n)}(0) = 4 \text{ for } n \geq 2$$

$$f(x) = 2 + 3x + \frac{4}{2!}x^2 + \frac{4}{3!}x^3 + \frac{4}{4!}x^4 + \frac{4}{5!}x^5 + \dots$$

$$4e^x = 4 + 4x + \frac{4}{2!}x^2 + \frac{4}{3!}x^3 + \frac{4}{4!}x^4 + \frac{4}{5!}x^5 + \dots$$

Therefore, $f(x) - 4e^x = -2 - x$.

2-2-

67. The function f satisfies the equation $f'(x) = f(x) + x + 1$ and $f(0) = 2$. The Taylor series for f about $x = 0$ converges to $f(x)$ for all x .

Write an equation for the line tangent to the curve $y = f(x)$ at the point where $x = 0$.

Part A

One point is earned for the tangent line equation

$$f'(0) = f(0) + 0 + 1 = 3$$

An equation for the tangent line is $y = 3x + 2$



0	1
---	---

The student earns all of the following points:

One point is earned for the tangent line equation

$$f'(0) = f(0) + 0 + 1 = 3$$

An equation for the tangent line is $y = 3x + 2$

2-2-

68. Consider the curve given by the equation $x^3 - 2xy + y = 22$. It can be shown that $\frac{dy}{dx} = \frac{2y-3x^2}{1-2x}$.

Part A

Find the slope of the line tangent to the curve at the point $(1, -21)$. Show the work that leads to your answer.

Part B

Find the coordinates of all points on the curve at which the line tangent to the curve is vertical, or explain why no such points exist.

Part C

The line tangent to the curve at the point $(-2, 6)$ is horizontal. Use the Second Derivative Test to determine whether the curve has a relative minimum or a relative maximum at $(-2, 6)$. Show the work that leads to your answer.

Part D

For the curve defined by the equation $y + x \cos y = 18$, find $\frac{dy}{dx}$. Show the work that leads to your answer.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response of $\frac{2(-21)-3(1)^2}{1-2(1)}$ earns **P1** (i.e., subsequent errors in simplification will not be considered in scoring for **P1**).
- A response of $\frac{2(-21)-3}{1-2}$ is sufficient to earn **P1**.
- A response of 45 with no supporting work does not earn **P1**.

Model Solution:

$$\text{At } (1, -21), \frac{dy}{dx} = \frac{2(-21)-3(1)^2}{1-2(1)} = 45.$$

The slope of the tangent line is 45.



0

1

2-2-

The response accurately includes the criteria below.

- Answer with supporting work

Part B Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response. Scoring Notes:

- **P2** can be earned with a response of $1 - 2x = 0$, $1 = 2x$, or $x = \frac{1}{2}$.

Model Solution:

A vertical tangent line can only occur when $1 - 2x = 0$ or when $x = \frac{1}{2}$.



0	1
---	---

The response accurately includes the criteria below.

- Considers $1 - 2x = 0$

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To be eligible for **P3**, a response must have earned **P2**.
- To earn **P3**, a response must provide evidence that no points with $x = \frac{1}{2}$ lie on the curve.

Model Solution:

For all points (x, y) on the curve, $x^3 - 2xy + y = 22$.

However, when $x = \frac{1}{2}$, $\left(\frac{1}{2}\right)^3 - 2 \cdot \frac{1}{2} \cdot y + y = \frac{1}{8} - y + y = \frac{1}{8} \neq 22$.

Therefore, there are no points on the curve with $x = \frac{1}{2}$. Thus, there are no points on the curve with a vertical tangent line.



0	1
---	---

The response accurately includes the criteria below.

2-2-

- Answer with explanation

Part C Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P4**, at least four of the five factors in an application of the quotient rule must be correct. The five correct factors are $1 - 2x$, $2\frac{dy}{dx} - 6x$, $2y - 3x^2$, -2 , and $(1 - 2x)^2$.
- Special cases: The following examples earn **P4**.
 - $\frac{d^2y}{dx^2} = \frac{(2y-3x^2)(-2) - (1-2x)\left(2\frac{dy}{dx}-6x\right)}{(1-2x)^2}$ (reversal in the numerator)
 - $\frac{d^2y}{dx^2} = \frac{(1-2x)(2-6x) - (2y-3x^2)(-2)}{(1-2x)^2}$ (chain rule error)

Model Solution:

$$\frac{d^2y}{dx^2} = \frac{(1-2x)\left(2\frac{dy}{dx}-6x\right) - (2y-3x^2)(-2)}{(1-2x)^2}$$



0	1
---	---

The response accurately includes the criteria below.

- Form of quotient rule

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with an incorrect second derivative does not earn **P5**.

Model Solution:

$$\text{At } (-2, 6), \frac{d^2y}{dx^2} = \frac{5(12) - 0(-2)}{5^2} = \frac{12}{5}.$$

2-2-



0	1
---	---

The response accurately includes the criteria below.

- $\frac{d^2y}{dx^2}$ at $(-2, 6)$

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with an incorrect second derivative is eligible to earn **P6** for a consistent answer and reason.
- Special cases: The following examples are eligible to earn **P6** for a consistent answer and reason.

$$\begin{aligned} \circ \frac{d^2y}{dx^2} &= \frac{(2y-3x^2)(-2) - (1-2x)\left(2\frac{dy}{dx} - 6x\right)}{(1-2x)^2} \text{ (reversal in the numerator)} \\ \circ \frac{d^2y}{dx^2} &= \frac{(1-2x)(2-6x) - (2y-3x^2)(-2)}{(1-2x)^2} \text{ (chain rule error)} \end{aligned}$$

Model Solution:

At $(-2, 6)$, $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$, and therefore the curve has a relative minimum there.



0	1
---	---

The response accurately includes the criteria below.

- Answer with reason

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response earns **P7** with an equation that involves either $1 \cdot \cos y + x \cdot (\pm \sin y) \cdot \frac{dy}{dx}$ or $1 \cdot \cos y + x \cdot (\pm \sin y)$.

Model Solution:

2-2-

$$\frac{d}{dx}(y + x \cos y) = \frac{d}{dx}(18)$$

$$\Rightarrow \frac{dy}{dx} + 1 \cdot \cos y + x \cdot (-\sin y) \cdot \frac{dy}{dx} = 0$$



0

1

The response accurately includes the criteria below.

- Product rule

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response earns **P8** with an equation that involves either $\frac{dy}{dx} + 1 \cdot \cos y + x \cdot (\pm \sin y) \cdot \frac{dy}{dx}$ or $\frac{dy}{dx} + 1 \cdot (\pm \sin y) \cdot \frac{dy}{dx}$.

Model Solution:

$$\frac{d}{dx}(y + x \cos y) = \frac{d}{dx}(18)$$

$$\Rightarrow \frac{dy}{dx} + 1 \cdot \cos y + x \cdot (-\sin y) \cdot \frac{dy}{dx} = 0$$



0

1

The response accurately includes the criteria below.

- Implicit differentiation

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned both **P7** and **P8** to be eligible for **P9**.
- **P9** is earned only for a response mathematically equivalent to $\frac{dy}{dx} = \frac{-\cos y}{1-x \sin y}$.

Model Solution:

2-2-

$$\Rightarrow \frac{dy}{dx} \cdot (1 - x \sin y) = -\cos y$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\cos y}{1 - x \sin y}$$



0

1

The response accurately includes the criteria below.

- Answer

69. Let f be the function given by $f(x) = (2x - 1)^5(x + 1)$. Which of the following is an equation for the line tangent to the graph of f at the point where $x = 1$?

(A) $y = 21x + 2$

(B) $y = 21x - 19$



(C) $y = 11x - 9$

(D) $y = 10x + 2$

(E) $y = 10x - 8$

2-2-

70.

x	1	1.1	1.2	1.3	1.4
$f'(x)$	8	10	12	13	14.5

The function f is twice differentiable for $x > 0$ with $f(1) = 15$ and $f'(1) = 20$. Values of f' , the derivative of f , are given for selected values of x in the table above.

Write an equation for the line tangent to the graph of f at $x = 1$. Use this line to approximate $f(1.4)$.

Part A

One point is earned for the tangent line

One point is earned for the approximation

$$f(1) = 15, f'(1) = 8$$

An equation for the tangent line is

$$y = 15 + 8(x - 1)$$

$$f(1.4) \approx 15 + 8(1.4 - 1) = 18.2$$



0	1	2
---	---	---

One point is earned for the tangent line

One point is earned for the approximation

$$f(1) = 15, f'(1) = 8$$

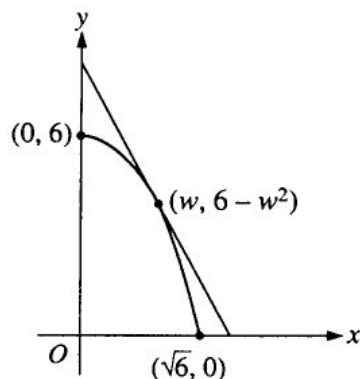
An equation for the tangent line is

$$y = 15 + 8(x - 1)$$

$$f(1.4) \approx 15 + 8(1.4 - 1) = 18.2$$

2-2-

71.



Let $f(x) = 6 - x^2$. For $0 < w < \sqrt{6}$, let $A(w)$ be the area of the triangle formed by the coordinate axes and the line tangent to the graph of f at the point $(w, 6 - w^2)$.

(a) Find $A(1)$.

(b) For what value of w is $A(w)$ a minimum?

Part A

1 point is earned for the correct answer $f'(1) = -2$

$$f(x) = 6 - x^2; f'(x) = -2x$$

$$f'(1) = -2$$

1 point is earned for correctly finding equation of line or x intercept or y intercept

$$y - 5 = -2(x - 1) \text{ or } y = -2x + 7$$

x -intercept: $7/2$ y -intercept: 7

1 point is earned for the correct answer

$$A(1) = \frac{1}{2} \left(\frac{7}{2} \right) (7) = \frac{49}{4}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $f'(1) = -2$

$$f(x) = 6 - x^2; f'(x) = -2x$$

2-2-

$$f'(1) = -2$$

1 point is earned for correctly finding equation of line or x intercept or y intercept

$$y - 5 = -2(x - 1) \text{ or } y = -2x + 7$$

x -intercept: $7/2$ y -intercept: 7

1 point is earned for the correct answer

$$A(1) = \frac{1}{2} \left(\frac{7}{2} \right) (7) = \frac{49}{4}$$

Part B

1 point is earned for the correct equation of line

$$f'(w) = -2w; \quad y - (6 - w^2) = -2w(x - w)$$

1 point is earned for correctly expressing x intercept or y intercept in terms of w

$$x\text{-intercept} : \frac{6+w^2}{2w}; \quad y\text{-intercept} : 6 + w^2$$

1 point is earned for the correct answer for $A(w)$

$$A(w) = \frac{(6+w^2)^2}{4w}$$

1 point is earned for the correct answer for $A'(w)$

$$A'(w) = \frac{4w(2(6+w^2)(2w)) - 4(6+w^2)^2}{16w^2}$$

1 point is earned for correctly solving $A'(w) = 0$

$$A'(w) = 0 \text{ When } (6 + w^2)(3w^2 - 6) = 0$$

$$w = \sqrt{2}$$

1 point is earned for correctly showing solution yields a minimum

2-2-



Note: $A(w)$ must be of the form $\frac{P(w)}{Q(w)}$, neither constant, to be eligible for derivative point

OR

If student is using Theorem that $A(w)$ is minimized at w_1 , where $f(w_1)$ is the average value of the function:

1 point is earned for the correct average value set-up

1 point is earned for the correct antidifferentiation and evaluation

1 point is earned for correctly setting $6 - w^2 = \text{average value}$

1 point is earned for correctly solving for w_1 in domain $(0, \sqrt{6})$

2 points are earned for the clear statement of Theorem

<-1> error in statement

<-2> No statement



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct equation of line

$$f'(w) = -2w; \quad y - (6 - w^2) = -2w(x - w)$$

1 point is earned for correctly expressing x intercept or y intercept in terms of w

$$x\text{-intercept} : \frac{6+w^2}{2w}; \quad y\text{-intercept} : 6 + w^2$$

1 point is earned for the correct answer for $A(w)$

2-2-

$$A(w) = \frac{(6+w^2)^2}{4w}$$

1 point is earned for the correct answer for $A'(w)$

$$A'(w) = \frac{4w(2(6+w^2)(2w)) - 4(6+w^2)^2}{16w^2}$$

1 point is earned for correctly solving $A'(w) = 0$

$$A'(w) = 0 \text{ When } (6 + w^2)(3w^2 - 6) = 0$$

$$w = \sqrt{2}$$

1 point is earned for correctly showing solution yields a minimum



Note: $A(w)$ must be of the form $\frac{P(w)}{Q(w)}$, neither constant, to be eligible for derivative point

OR

If student is using Theorem that $A(w)$ is minimized at w_1 , where $f(w_1)$ is the average value of the function:

1 point is earned for the correct average value set-up

1 point is earned for the correct antidifferentiation and evaluation

1 point is earned for correctly setting $6 - w^2 = \text{average value}$

1 point is earned for correctly solving for w_1 in domain $(0, \sqrt{6})$

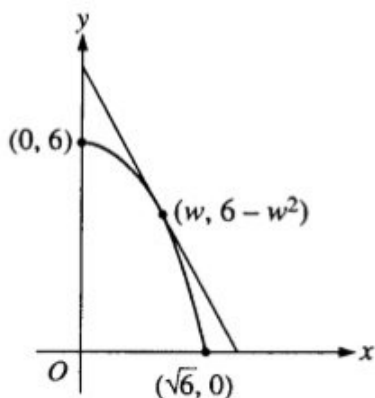
2 points are earned for the clear statement of Theorem

<-1> error in statement

<-2> No statement

2-2-

72.



Let $f(x) = 6 - x^2$. For $0 < w < \sqrt{6}$, let $A(w)$ be the area of the triangle formed by the coordinate axes and the line tangent to the graph of f at the point $(w, 6 - w^2)$.

Find $A(1)$.

Part A

1 point is earned for the correct answer $f'(1) = -2$

$$f(x) = 6 - x^2; f'(x) = -2x$$

$$f'(1) = -2$$

1 point is earned for correctly finding equation of line or x intercept or y intercept

$$y - 5 = -2(x - 1) \text{ or } y = -2x + 7$$

$$x \text{ intercept: } \frac{7}{2} \quad y \text{ intercept: } 7$$

1 point is earned for the correct answer

$$A(1) = \frac{1}{2} \left(\frac{7}{2} \right) (7) = \frac{49}{4}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points

1 point is earned for the correct answer $f'(1) = -2$

2-2-

$$f(x) = 6 - x^2; f'(x) = -2x$$

$$f'(1) = -2$$

1 point is earned for correctly finding equation of line or x intercept or y intercept

$$y - 5 = -2(x - 1) \text{ or } y = -2x + 7$$

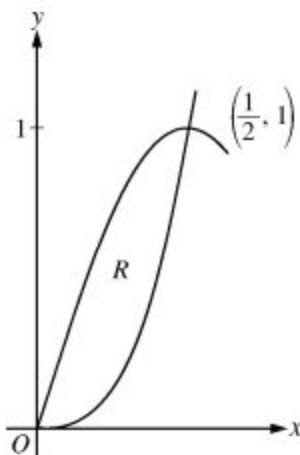
x intercept : $\frac{7}{2}$ y intercept: 7

1 point is earned for the correct answer

$$A(1) = \frac{1}{2}\left(\frac{7}{2}\right)(7) = \frac{49}{4}$$

2-2-

73.



Let R be the region in the first quadrant enclosed by the graphs of $f(x) = 8x^3$ and $g(x) = \sin(\pi x)$, as shown in the figure above.

(a) Write an equation for the line tangent to the graph of f at $x = 1/2$.

(b) Find the area of R .

(c) Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 1$.

Part A

1 point is earned for $f'(\frac{1}{2})$

1 point is earned for the answer

$$f\left(\frac{1}{2}\right) = 1$$

$$f'(x) = 24x^2, \text{ so } f'\left(\frac{1}{2}\right) = 6$$

An equation for the tangent line is $y = 1 + 6\left(x - \frac{1}{2}\right)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(\frac{1}{2})$

1 point is earned for the answer

2-2-

$$f\left(\frac{1}{2}\right) = 1$$

$$f'(x) = 24x^2, \text{ so } f'\left(\frac{1}{2}\right) = 6$$

An equation for the tangent line is $y = 1 + 6\left(x - \frac{1}{2}\right)$.

Part B

1 point is earned for the integrand

2 points are earned the antiderivative

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^{1/2} (g(x) - f(x))dx \\ &= \int_0^{1/2} (\sin(\pi x) - 8x^3)dx \\ &= \left[-\frac{1}{\pi}\cos(\pi x) - 2x^4\right]_{x=0}^{x=1/2} \\ &= -\frac{1}{8} + \frac{1}{\pi} \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

2 points are earned the antiderivative

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^{1/2} (g(x) - f(x))dx \\ &= \int_0^{1/2} (\sin(\pi x) - 8x^3)dx \\ &= \left[-\frac{1}{\pi}\cos(\pi x) - 2x^4\right]_{x=0}^{x=1/2} \\ &= -\frac{1}{8} + \frac{1}{\pi} \end{aligned}$$

2-2-

Part C

1 point is earned for limits and constant

2 points are earned for integrand

$$\pi \int_0^{1/2} \left((1 - f(x))^2 - (1 - g(x))^2 \right) dx$$

$$= \pi \int_0^{1/2} \left((1 - 8x^3)^2 - (1 - \sin(\pi x))^2 \right) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

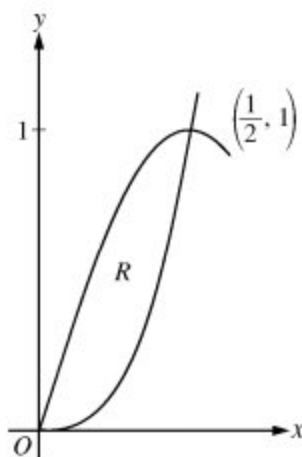
2 points are earned for integrand

$$\pi \int_0^{1/2} \left((1 - f(x))^2 - (1 - g(x))^2 \right) dx$$

$$= \pi \int_0^{1/2} \left((1 - 8x^3)^2 - (1 - \sin(\pi x))^2 \right) dx$$

2-2-

74.



Let R be the region in the first quadrant enclosed by the graphs of $f(x) = 8x^3$ and $g(x) = \sin(\pi x)$, as shown in the figure above.

Write an equation for the line tangent to the graph of f at $x = 1/2$.

Part A

1 point is earned for $f'(\frac{1}{2})$

1 point is earned for the answer

$$f\left(\frac{1}{2}\right) = 1$$

$$f'(x) = 24x^2, \text{ so } f'\left(\frac{1}{2}\right) = 6$$

An equation for the tangent line is $y = 1 + 6\left(x - \frac{1}{2}\right)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(\frac{1}{2})$

1 point is earned for the answer

$$f\left(\frac{1}{2}\right) = 1$$

$$f'(x) = 24x^2, \text{ so } f'\left(\frac{1}{2}\right) = 6$$

An equation for the tangent line is $y = 1 + 6\left(x - \frac{1}{2}\right)$.

2-2-

2-2-

75. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

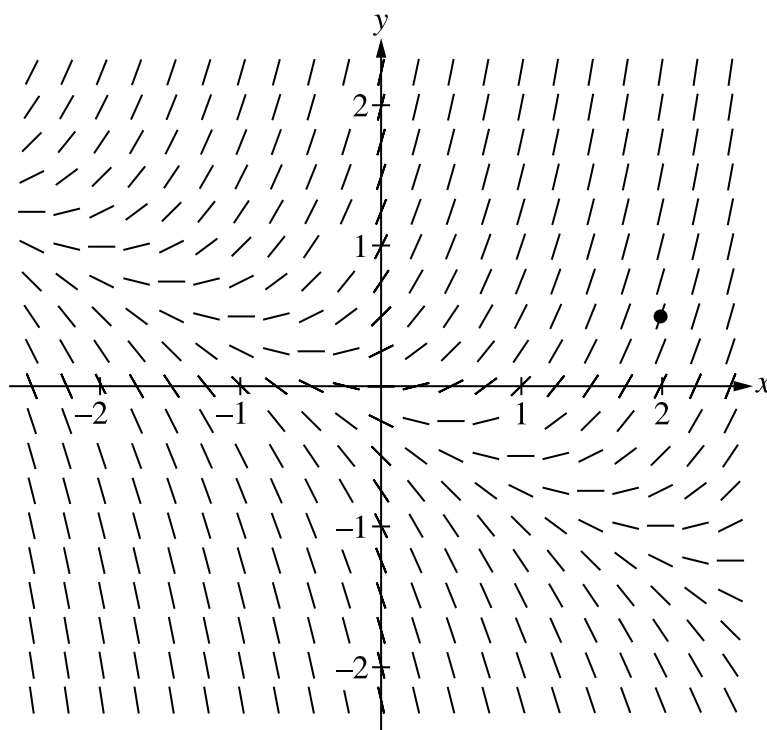
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = x + 2y$ with initial condition $f(2) = 0.5$.

(a) A portion of the slope field for the differential equation $\frac{dy}{dx} = x + 2y$ is given. Sketch the solution curve through the point $(2, 0.5)$.



(b) Write an equation for the line tangent to the graph of f at $x = 2$. Use the equation to approximate $f(2.4)$.

(c) Write the second-degree Taylor polynomial for f about $x = 2$. Use the Taylor polynomial to approximate $f(2.4)$.

(d) It is known that $0 < f'''(x) < 36$ for $2 \leq x \leq 2.4$. Use the Lagrange error bound to show that the approximation from part (c) differs from the actual value of $f(2.4)$ by less than 0.5.

2-2-

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The solution curve must pass through the point $(2, 0.5)$, extend reasonably close (within one row or column of the edge of the slope field) to the left edge and top-right corner of the square, and have no obvious conflicts with the given slope lines.

Do not consider any portion of the solution curve that extends beyond the square.

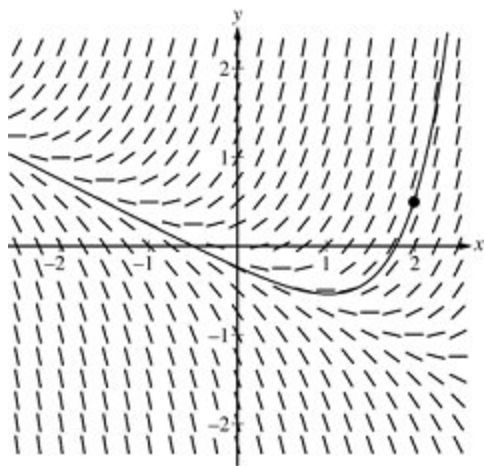


0

1

The student response accurately includes the criteria below.

☐ Solution curve

Solution:**Part B**

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The tangent line equation can be presented in any equivalent form.

A response with incorrect tangent line equation with a slope of 3 will not earn the first point but earns the second point for a consistent approximation.

A response of only $0.5 + 3(2.4 - 2)$ earns the second point but not the first.

2-2-



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Tangent line equation
- ☐ Approximation

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(2,0.5)} = 2 + 2(0.5) = 3$$

An equation for the tangent line is $y = 0.5 + 3(x - 2)$.

$$f(2.4) \approx 0.5 + 3(2.4 - 2) = 1.7$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must present work to support $f''(2) = 7$.

The minimal supporting work is $\frac{d^2y}{dx^2} = 1 + 2\frac{dy}{dx}$.

The expression $\frac{f''(2)}{2!}(x - 2)^2$ will earn the second point even if a value of $f''(2)$ is never presented. If an incorrect value of $f''(2)$ is presented, it must be nonzero in order to earn the second point.

The third point is for adding the tangent line expression and an expression (such as $\frac{f''(2)}{2!}(x - 2)^2$ or $\frac{7}{2}(x - 2)^2$) which has earned the second point.

- An incorrect tangent line expression may be imported from part (b).
- Numeric values of $f(2)$, $f'(2)$, and $f''(2)$ must be included in the polynomial expression in order to earn the third point.
- A correct second-degree polynomial followed by “+ . . .” can earn the third point provided the response uses only the second-degree polynomial in calculating the approximation.
- A response that includes additional terms in a presented polynomial will not earn the third point, but is eligible for the fourth point provided the approximation is found using only the second-degree polynomial.

An answer of just $0.5 + 3(0.4) + \frac{7}{2}(0.4)^2$ earns only the fourth point.

2-2-

A response that presents an incorrect approximation will earn the fourth point if it has earned the third point and correctly evaluates the presented (incorrect) second-degree Taylor polynomial about $x = 2$ at $x = 2.4$.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

- ☐ $f''(2) = 7$
- ☐ $\frac{f''(2)}{2!}(x-2)^2$ -OR- $\frac{7}{2}(x-2)^2$
- ☐ Second-degree Taylor polynomial about $x = 2$
- ☐ Approximation

Solution:

$$f(2) = 0.5$$

$$f'(2) = 3$$

$$\frac{d^2y}{dx^2} = 1 + 2\frac{dy}{dx} = 1 + 2(x + 2y) = 1 + 2x + 4y$$

$$f''(2) = 1 + 2(2) + 4(0.5) = 7$$

$$\begin{aligned} P_2(x) &= f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 \\ &= 0.5 + 3(x-2) + \frac{7}{2}(x-2)^2 \end{aligned}$$

$$f(2.4) \approx P_2(2.4) = 0.5 + 3(0.4) + \frac{7}{2}(0.4)^2 = 2.26$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

· The minimal response to earn the first point is $(\max f'''(x)) \cdot \frac{0.4^3}{3!}$ or $36 \cdot \frac{0.4^3}{3!}$.

· In order to earn the second point, a response must indicate that $|f(2.4) - P_2(2.4)| \leq 36 \cdot \frac{0.4^3}{3!} < 0.5$,

Error bound $= 36 \cdot \frac{0.4^3}{3!} < 0.5$, or Error $\leq 36 \cdot \frac{0.4^3}{3!} < 0.5$.

A response that only shows $36 \cdot \frac{0.4^3}{3!} < 0.5$ does not earn the second point.

A response that equates the error with the bound (e.g., Error $= 36 \cdot \frac{0.4^3}{3!}$) does not earn the second point.

2-2-



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of error bound
- ☐ Answer with supporting work

Solution:

By the Lagrange error bound,

$$|f(2.4) - P_2(2.4)| \leq \left(\max_{2 \leq x \leq 2.4} |f'''(x)| \right) \cdot \frac{|2.4 - 2|^3}{3!}$$

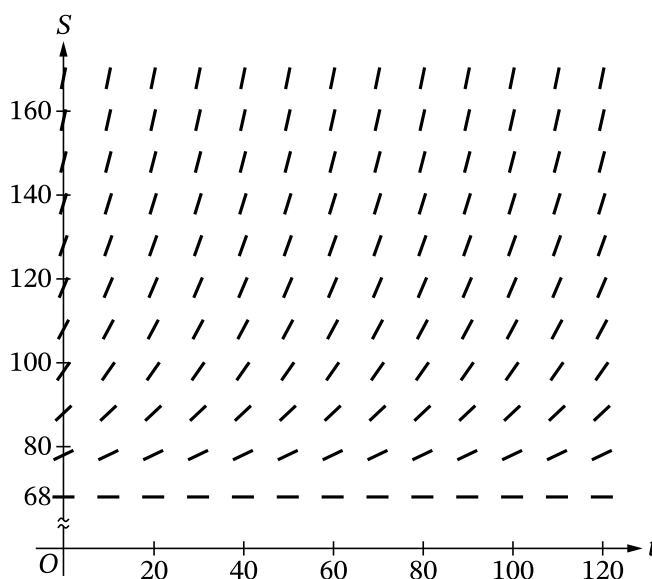
$$< 36 \cdot \frac{0.4^3}{6} = \frac{6 \cdot 64}{1000} < \frac{500}{1000} = 0.5$$

2-2-

76. For $0 \leq t \leq 120$, the temperature of a bowl of soup at time t is modeled by the function S that satisfies the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$, where $S(t)$ is measured in degrees Fahrenheit ($^{\circ}\text{F}$) and t is measured in minutes. The temperature of the bowl of soup is always greater than 68°F and is 148°F at time $t = 0$.

Part A

Explain why the following could not be a slope field for the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$.

**Part B**

Use the line tangent to the graph of S at $t = 0$ to approximate the temperature of the soup at time $t = 5$ minutes.

Part C

According to the model, is the approximation found in part B an overestimate or an underestimate of the temperature of the soup at time $t = 5$ minutes? Give a reason for your answer.

Part D

Using separation of variables, find the particular solution $y = S(t)$ to the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$ with initial condition $S(0) = 148$.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

2-2-

- **P1** can be earned for a response of “because there are no negative slopes” or “because the slopes are positive.”
- **P1** can be earned by using a local argument at any point (t, S) with $0 \leq t \leq 120$ and $S > 68$, rather than a global argument.

Model Solution:

The slopes of the line segments in a slope field for the given differential equation should all be negative because for $S > 68$, $\frac{dS}{dt} < 0$. However, the slopes of the line segments in the slope field given are all nonnegative.



0	1
---	---

The response accurately includes the criteria below.

- Explanation

Part B Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response can earn **P2** with $\frac{dS}{dt} \Big|_{(0, 148)} = -4$, “slope is -4 ,” or equivalent.
- A response that presents a linear approximation with a slope of -4 also earns **P2**. For example, a response of $148 - 4(5)$ earns **P2**.
- A response that declares the slope equal to any nonzero value $k \neq -4$ does not earn **P2**.

Model Solution:

$$\frac{dS}{dt} \Big|_{(0, 148)} = -\frac{1}{20}(148 - 68) = -4$$



0	1
---	---

The response accurately includes the criteria below.

- Slope of tangent line

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

2-2-

Scoring Notes:

- A response that declares the slope equal to any nonzero value $k \neq -4$:
 - Earns **P3** if there is evidence that the given differential equation was used to calculate k and that k was used correctly in a tangent line approximation of the form $148 + k(5)$. Subsequent errors in simplification will not be considered in scoring for **P3**.
 - Does not earn **P3** if k is simply declared to be a slope without a connection to the differential equation.
- **P3** cannot be earned for an approximation at any value of t other than 5.
- A response does not have to present the tangent line equation but must clearly demonstrate its use at $t = 5$ in finding the requested approximation in order to earn **P3**.
- A response of $148 - 4(5)$ earns **P3**. Subsequent errors in simplification will not be considered in scoring for **P3**.

Model Solution:

$$S(5) \approx 148 - 4(5) = 128$$

The temperature of the soup at $t = 5$ minutes is approximately 128°F .



0	1
---	---

The response accurately includes the criteria below.

- Tangent line approximation

Part C Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P4**, a response must include one of the following: $\frac{d^2S}{dt^2}$, $-\frac{1}{20}\frac{dS}{dt}$, $\frac{1}{400}(S - 68)$, “the second derivative,” or equivalent.
 - “ S' is increasing (or decreasing)” or a reference to concavity without a connection to the second derivative is not sufficient to earn **P4**.

Model Solution:

Because $S(t) > 68$, $\frac{d^2S}{dt^2} = -\frac{1}{20}\frac{dS}{dt} = \left(-\frac{1}{20}\right)\left(-\frac{1}{20}\right)(S - 68) = \frac{1}{400}(S - 68) > 0$.

2-2-



0	1
---	---

The response accurately includes the criteria below.

- Considers $\frac{d^2S}{dt^2}$

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To be eligible to earn **P5**, a response must have earned **P4**.
- Responses do not need to say “concave up” to earn **P5**. A response of “underestimate because the second derivative is positive” is sufficient to earn **P5**.

Model Solution:

Because the second derivative is always positive, the tangent line approximation is an underestimate of the temperature predicted by the model.



0	1
---	---

The response accurately includes the criteria below.

- Answer with reason

Part D Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P6**.
- A response earns **P6** if the separation of variables results in an equation of the form $A \frac{dS}{S-68} = B dt$, where A and B are nonzero constants.
- Special case: If the response begins with a separation of the form $A \frac{dS}{S+68} = B dt$, where A and B are correct coefficients, it earns **P6**.

Model Solution:

2-2-

$$\frac{dS}{S-68} = -\frac{1}{20}dt$$



0	1
---	---

The response accurately includes the criteria below.

- Separate variables

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P7**.
- A response is eligible to earn **P7** if the separation of variables resulted in an equation of the form $A \frac{dS}{S-68} = B dt$, where A and B are nonzero constants. A response that presents consistent antiderivatives of the form $A \ln(S - 68)$ (with either parentheses or absolute value) and Bt earns **P7**.
- Special case: If the response began with a separation of the form $A \frac{dS}{S+68} = B dt$, where A and B are correct coefficients, it earns **P7** with two consistent antiderivatives.
- A response with no constant of integration is eligible to earn **P7**.

Model Solution:

$$\ln|S - 68| = -\frac{t}{20} + C$$



0	1
---	---

The response accurately includes the criteria below.

- Finds antiderivatives

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P8**.
- A response with no constant of integration does not earn **P8**.

2-2-

- A response is eligible for **P8** if it has earned P6 and has one of the two antiderivatives correct OR if it has earned **P7**.
- An eligible response earns **P8** by correctly including the constant of integration in an equation and substituting 0 for t and 148 for S .

Model Solution:

$$\ln |148 - 68| = -\frac{0}{20} + C \Rightarrow \ln 80 = C$$

$$\ln (S - 68) = -\frac{t}{20} + \ln 80$$



0	1
---	---

The response accurately includes the criteria below.

- Constant of integration and uses initial condition

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response is eligible for **P9** only if it has earned **P6**, **P7**, and **P8**.
- A response earns **P9** only for an answer of $S(t) = 68 + 80e^{-t/20}$ or equivalent.

Model Solution:

$$S - 68 = e^{-t/20 + \ln 80} = 80e^{-t/20}$$

$$S(t) = 68 + 80e^{-t/20}$$



0	1
---	---

The response accurately includes the criteria below.

- Solves for $S(t)$

2-2-

77. The functions f and g are given by $f(x) = \int_0^{3x} \sqrt{4+t^2} dt$ and $g(x) = f(\sin x)$.

(a) Find $f'(x)$ and $g'(x)$.

(b) Write an equation for the line tangent to the graph of $y = g(x)$ at $x = \pi$.

(c) Write, but do not evaluate, an integral expression that represents the maximum value of g on the interval $0 \leq x \leq \pi$. Justify your answer.

Part A

2 points are earned for $f'(x)$

2 points are earned for $g'(x)$

$$f'(x) = 3\sqrt{4 + (3x)^2}$$

$$g'(x) = f'(\sin x) \cdot \cos x$$

$$= 3\sqrt{4 + (3 \sin x)^2} \cdot \cos x$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for $f'(x)$

2 points are earned for $g'(x)$

$$f'(x) = 3\sqrt{4 + (3x)^2}$$

$$g'(x) = f'(\sin x) \cdot \cos x$$

$$= 3\sqrt{4 + (3 \sin x)^2} \cdot \cos x$$

Part B

1 point is earned for $g(\pi)$ or $g'(\pi)$

1 point is earned for the tangent line equation

$$g(\pi) = 0, g'(\pi) = -6$$

$$\text{Tangent line: } y = -6(x - \pi)$$

2-2-



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(\pi)$ or $g'(\pi)$

1 point is earned for the tangent line equation

$$g(\pi) = 0, g'(\pi) = -6$$

$$\text{Tangent line: } y = -6(x - \pi)$$

Part C

1 point is earned if sets $g'(x) = 0$

1 point is earned if justifies maximum at $\frac{\pi}{2}$

1 point is earned for integral expression at $g\left(\frac{\pi}{2}\right)$

For $0 < x < \pi$, $g'(x) = 0$ only at $x = \frac{\pi}{2}$.

$$g(0) = g(\pi) = 0$$

$$g\left(\frac{\pi}{2}\right) = \int_0^3 \sqrt{4 + t^2} dt > 0$$

The maximum value of g on $[0, \pi]$ is $\int_0^3 \sqrt{4 + t^2} dt$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if sets $g'(x) = 0$

1 point is earned if justifies maximum at $\frac{\pi}{2}$

1 point is earned for integral expression at $g\left(\frac{\pi}{2}\right)$

2-2-

For $0 < x < \pi$, $g'(x) = 0$ only at $x = \frac{\pi}{2}$.

$$g(0) = g(\pi) = 0$$

$$g\left(\frac{\pi}{2}\right) = \int_0^3 \sqrt{4+t^2} dt > 0$$

The maximum value of g on $[0, \pi]$ is $\int_0^3 \sqrt{4+t^2} dt$.

78. The functions f and g are given by $f(x) = \int_0^{3x} \sqrt{4+t^2} dt$ and $g(x) = f(\sin x)$.

Write an equation for the line tangent to the graph of $y = g(x)$ at $x = \pi$.

Part B

1 point is earned for $g(\pi)$ or $g'(\pi)$

1 point is earned for the tangent line equation

$$g(\pi) = 0, g'(\pi) = -6$$

$$\text{Tangent line: } y = -6(x - \pi)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(\pi)$ or $g'(\pi)$

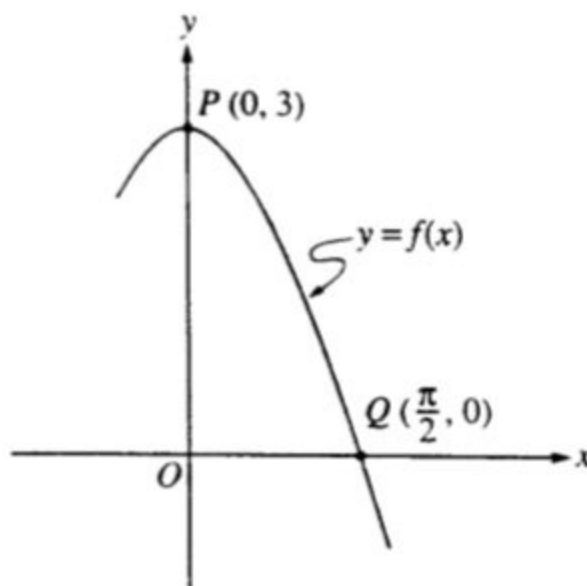
1 point is earned for the tangent line equation

$$g(\pi) = 0, g'(\pi) = -6$$

$$\text{Tangent line: } y = -6(x - \pi)$$

2-2-

79.



Let f be the function given by $f(x) = 3 \cos x$. As shown above, the graph of f crosses the y -axis at point P and the x -axis at point Q .

- Write an equation for the line passing through the points P and Q .
- Write an equation for the line tangent to the graph of f at point Q . Show the analysis that leads to your conclusion.
- Find the x -coordinate of the point on the graph of f , between points P and Q , at which the line tangent to the graph of f is parallel to line PQ .
- Let R be the region in the first quadrant bounded by the graph of f and line segment PQ . Write an integral expression for the volume of the solid generated by revolving the region R about the x -axis. Do not evaluate.

Part A

1 point is earned for the correct slope

$$\text{slope} = \frac{3-0}{0-\pi/2} = -\frac{6}{\pi}$$

1 point is earned for the correct equation

$$y - 3 = -\frac{6}{\pi}(x - 0)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct slope

2-2-

$$\text{slope} = \frac{3-0}{0-\pi/2} = -\frac{6}{\pi}$$

1 point is earned for the correct equation

$$y - 3 = -\frac{6}{\pi}(x - 0)$$

Part B

1 point is earned for the correct statement $f'(\pi/2) = -3$

1 point is earned for an equation using $f'(\pi/2)$ (or using a derivation) and $f(\pi/2)$

1/2 if equations only

$$f'(x) = -3 \sin x$$

$$f'(\pi/2) = -3 \sin(\pi/2) = -3$$

$$y - 0 = -3(x - \pi/2)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct statement $f'(\pi/2) = -3$

1 point is earned for an equation using $f'(\pi/2)$ (or using a derivation) and $f(\pi/2)$

1/2 if equations only

$$f'(x) = -3 \sin x$$

$$f'(\pi/2) = -3 \sin(\pi/2) = -3$$

$$y - 0 = -3(x - \pi/2)$$

Part C

1 point is earned for correctly equating the derivative to the slope

$$f'(x) = -3 \sin x = -\frac{6}{\pi}$$

$$\sin x = \frac{2}{\pi}$$

1 point is earned for the correct solution in $[0, \frac{\pi}{2}]$

1/2 if solution only

no credit for "no solution exists"

2-2-

$$x = 0.690$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly equating the derivative to the slope

$$f'(x) = -3 \sin x = -\frac{6}{\pi}$$

$$\sin x = \frac{2}{\pi}$$

1 point is earned for the correct solution in $[0, \frac{\pi}{2}]$

1/2 if solution only

no credit for "no solution exists"

$$x = 0.690$$

Part D

2 points are earned for the correct integrand

0/2 if not difference of 2 squares

1/2 if incorrect but of form $(a \cos x)^2 - (bx + c)^2$; $a, b \neq 0$

1/2 if reversed

1 point is earned for the correct constant and limits

0/3 for shell around y-axis

$$V = \pi \int_0^{\pi/2} \left[(3 \cos x)^2 - \left(-\frac{6}{\pi}x + 3 \right)^2 \right] dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct integrand

0/2 if not difference of 2 squares

2-2-

1/2 if incorrect but of form $(a \cos x)^2 - (bx + c)^2$; $a, b \neq 0$

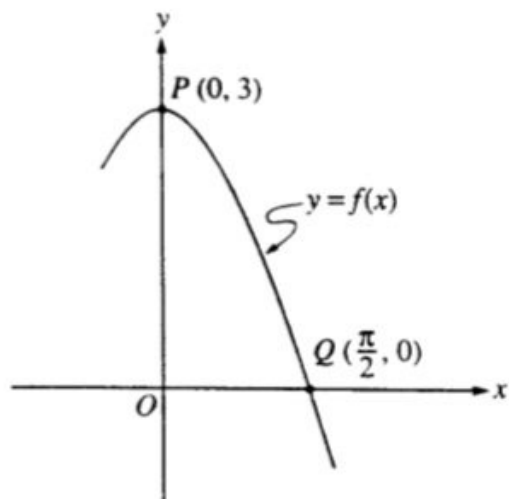
1/2 if reversed

1 point is earned for the correct constant and limits

0/3 for shell around y-axis

$$V = \pi \int_0^{\pi/2} \left[(3 \cos x)^2 - \left(-\frac{6}{\pi}x + 3 \right)^2 \right] dx$$

2-2-



Let f be the function given by $f(x) = 3 \cos x$. As shown above, the graph of f crosses the y -axis at point P and the x -axis at point Q .

80. Write an equation for the line tangent to the graph of f at point Q . Show the analysis that leads to your conclusion.

Part B

1 point is earned for the correct statement $f'(\frac{\pi}{2}) = -3$

1 point is earned for an equation using $f'(\frac{\pi}{2})$ (or using a derivation) and $f(\frac{\pi}{2})$

1/2 if equations only

$$f'(x) = -3 \sin x$$

$$f'(\frac{\pi}{2}) = -3 \sin(\frac{\pi}{2}) = -3$$

$$y - 0 = -3(x - \frac{\pi}{2})$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct statement $f'(\frac{\pi}{2}) = -3$

1 point is earned for an equation using $f'(\frac{\pi}{2})$ (or using a derivation) and $f(\frac{\pi}{2})$

1/2 if equations only

2-2-

$$f'(x) = -3 \sin x$$

$$f'\left(\frac{\pi}{2}\right) = -3 \sin\left(\frac{\pi}{2}\right) = -3$$

$$y - 0 = -3\left(x - \frac{\pi}{2}\right)$$

81. Find the x -coordinate of the point on the graph of f , between points P and Q , at which the line tangent to the graph of f is parallel to line PQ .

Part C

1 point is earned for correctly equating the derivative to the slope

$$f'(x) = -3 \sin x = -\frac{6}{\pi}$$

$$\sin x = \frac{2}{\pi}$$

1 point is earned for the correct solution in $\left[0, \frac{\pi}{2}\right]$

1/2 if solution only

no credit for "no solution exists"

$$x=0.690$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly equating the derivative to the slope

$$f'(x) = -3 \sin x = -\frac{6}{\pi}$$

$$\sin x = \frac{2}{\pi}$$

1 point is earned for the correct solution in $\left[0, \frac{\pi}{2}\right]$

1/2 if solution only

no credit for "no solution exists"

$$x=0.690$$

2-2-

82. If f is differentiable at $x = a$, which of the following could be false?

(A) f is continuous at $x = a$.

(B) $\lim_{x \rightarrow a} f(x)$ exists.

(C) $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists.

(D) $f'(a)$ is defined.

(E) $f''(a)$ is defined. ✓

83. At $x = 3$, the function given by $f(x) = \begin{cases} x^2, & x < 3 \\ 6x - 9, & x \geq 3 \end{cases}$ is

(A) undefined

(B) continuous but not differentiable

(C) differentiable but not continuous

(D) neither continuous nor differentiable

(E) both continuous and differentiable ✓

2-2-

84. Consider the curve defined by the equation $y + \cos y = x + 1$ for $0 \leq y \leq 2\pi$.

- (a) Find dy/dx in terms of y .
- (b) Write an equation for each vertical tangent to the curve
- (c) Find d^2y/dx^2 in terms of y .

Part A

2 points are earned for implicit differentiation

1 point is earned for solving dy/dx

$$\begin{aligned}\frac{dy}{dx} - \sin y \frac{dy}{dx} &= 1 \\ \frac{dy}{dx}(1 - \sin y) &= 1 \\ \frac{dy}{dx} &= \frac{1}{1 - \sin y}\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for implicit differentiation

1 point is earned for solving dy/dx

$$\begin{aligned}\frac{dy}{dx} - \sin y \frac{dy}{dx} &= 1 \\ \frac{dy}{dx}(1 - \sin y) &= 1 \\ \frac{dy}{dx} &= \frac{1}{1 - \sin y}\end{aligned}$$

Part B

1 point is earned for finding y where dy/dx does not exist

1 point is earned for solving that equation for y

1 point is earned for using that y solution to give equation of vertical line

dy/dx undefined when $\sin y = 1$

2-2-

$$y = \pi/2$$

$$\pi/2 + 0 = x + 1$$

$$x = \pi/2 - 1$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for finding y where dy/dx does not exist

1 point is earned for solving that equation for y

1 point is earned for using that y solution to give equation of vertical line

dy/dx undefined when $\sin y = 1$

$$y = \pi/2$$

$$\pi/2 + 0 = x + 1$$

$$x = \pi/2 - 1$$

Part C

2 points are earned for implicit differentiation

1 point is earned if substituted for student's dy/dx after differentiation and solves answer $\frac{d^2y}{dx^2}$

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d\left(\frac{1}{1-\sin y}\right)}{dx} \\ &= \frac{-\left(-\cos y \frac{dy}{dx}\right)}{(1-\sin y)^2} \\ &= \frac{\cos y \frac{1}{1-\sin y}}{(1-\sin y)^2} \\ &= \frac{\cos y}{(1-\sin y)^3} \end{aligned}$$

2-2-



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for implicit differentiation

1 point is earned if substituted for student's dy/dx after differentiation and solves answer $\frac{d^2y}{dx^2}$

$$\begin{aligned}
 \frac{d^2y}{dx^2} &= \frac{d\left(\frac{1}{1-\sin y}\right)}{dx} \\
 &= \frac{-\left(-\cos y \frac{dy}{dx}\right)}{(1-\sin y)^2} \\
 &= \frac{\cos y \frac{1}{1-\sin y}}{(1-\sin y)^2} \\
 &= \frac{\cos y}{(1-\sin y)^3}
 \end{aligned}$$

2-2-

85. Consider the curve defined by the equation $y + \cos y = x + 1$ for $0 \leq y \leq 2\pi$.

Write an equation for each vertical tangent to the curve

Part B

1 point is earned for equation of where dy/dx does not exist

1 point is earned for solving that equation for y

1 point is earned for using that y solution to give equation of vertical line

$\frac{dy}{dx}$ undefined when $\sin y = 1$

$$y = \frac{\pi}{2}$$

$$\frac{\pi}{2} + 0 = x + 1$$

$$x = \frac{\pi}{2} - 1$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for equation of where dy/dx does not exist

1 point is earned for solving that equation for y

1 point is earned for using that y solution to give equation of vertical line

$\frac{dy}{dx}$ undefined when $\sin y = 1$

$$y = \frac{\pi}{2}$$

$$\frac{\pi}{2} + 0 = x + 1$$

$$x = \frac{\pi}{2} - 1$$

86. $\lim_{h \rightarrow 0} \frac{e^{-1-h} - e^{-1}}{h}$ is

(A) -1

(B) $-1/e$

(C) 0

(D) $1/e$

(E) Nonexistent



2-2-

87. When $x = 2e$, $\lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln(x)}{h}$ is

(A) $\frac{1}{2e}$



(B) 1

(C) $\ln(2e)$

(D) nonexistent

88. $\lim_{h \rightarrow 0} \frac{\sin(\frac{\pi}{3} + h) - \sin(\frac{\pi}{3})}{h}$ is

(A) 0

(B) $\frac{1}{2}$



(C) 1

(D) $\frac{\sqrt{3}}{2}$

(E) nonexistent

89. The $\lim_{h \rightarrow 0} \frac{\tan 3(x+h) - \tan(3x)}{h}$ is

(A) 0

(B) $3\sec^2(3x)$



(C) $\sec^2(3x)$

(D) $3\cot(3x)$

(E) nonexistent

90. $\lim_{h \rightarrow 0} \frac{\ln(e+h) - 1}{h}$ is

(A) $f'(e)$, where $f(x) = \ln x$



(B) $f'(e)$, where $f(x) = \frac{\ln x}{x}$

(C) $f'(1)$, where $f(x) = \ln x$

(D) $f'(1)$, where $f(x) = \ln(x+e)$

(E) $f'(0)$, where $f(x) = \ln x$